FINANCIAL ECONOMETRICS

MODULE 3 | LESSON 2

SKEW NORMAL DISTRIBUTION AND SKEW-T DISTRIBUTION

Reading Time	60 minutes
Prior Knowledge	Basic Statistics
Keywords	Skewness, Kurtosis, Skew Normal Distribution, Skew-t Distribution

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In the last lesson, we visited the concepts of random variables and reviewed the basics of some common probability distributions. The most utilized distribution in many applications is the normal distribution. The normal distribution has a symmetric shape and properties that are easy to use for data analysis. However, in the real world, the observed data does not always have a bell-shaped distribution. The data may cluster on the right or the left of the whole data range. We usually call this kind of data distribution skewed to the right or to the left. In this lesson, we will look at the skewed data and learn how to measure the skewness. Apart from skewness, we will also learn what is a heavy tail or a light tail of a data distribution and how to measure it. Then, we will introduce skew normal distribution and its families. This is a distribution for observed data that exhibits some properties of a normal distribution but does not have the symmetric shape or similar tail pattern. Throughout this lesson, we will use the U.S. 10-Year Treasury Bond Yield daily return and the Dow Jones Industrial Index daily return for each technique we cover.

In [1]:
```
# Load libraries
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import scipy.stats as stats
from scipy.stats import kurtosis, skewnorm

plt.rcParams["figure.figsize"] = (16, 9)  # Figure size and width
```

In [2]:
```
# Download the dataset
data_set = pd.read_csv("M3. bond_and_stock_data.csv")
```

1. Skewness

A skew distribution means that the probability distribution is not symmetric. We usually draw a histogram from a dataset to inspect for skewness. In figure 1 below, the graph

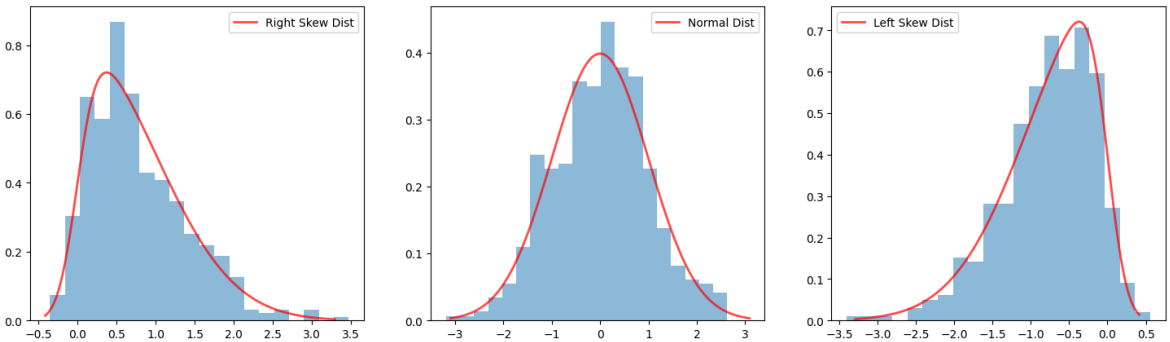
contains three histograms. The one in the middle is a histogram for a dataset sampled from a normal distribution. Normal distributions are symmetric distributions. We notice that the peak is located at the center of the distribution. The histogram on the lower left of the figure exhibits a right-skewed distribution. A right-skewed distribution has a longer tail, which extends to the right of the plot. The peak of a right-skewed distribution is located on the left region of the distribution. Finally, the histogram on the lower right of the figure is that of a left-skewed distribution. A left-skewed distribution has a longer tail, which extends to the left of the plot. The peak of a left-skewed distribution is found in the right region of the distribution.

Figure 1: Histograms of Normal Distribution, Right-Skew Distribution, and Left-Skew Distribution

In [3]:
```
# Histograms of Normal Distribution, Right-Skew and Left-Skew Distributions


# Helper HistogramPlot function
def HistogPlot(a, label):
    # np.random.seed(12345)
    r = skewnorm.rvs(a, size=500)
    x = np.linspace(skewnorm.ppf(0.001, a), skewnorm.ppf(0.999, a), 100)
    plt.plot(x, skewnorm.pdf(x, a), "r-", lw=2, alpha=0.7, label=label)  # Densi
    plt.hist(r, bins=20, density=True, histtype="stepfilled", alpha=0.5)  # Hist
    plt.legend()


# Plot
plt.figure(figsize=(18, 5))
ax1 = plt.subplot(131)
HistogPlot(5, "Right Skew Dist")  # Right Skew a>0
ax2 = plt.subplot(132)
HistogPlot(0, "Normal Dist")  # Normal Dist. a=0
ax3 = plt.subplot(133)
HistogPlot(-5, "Left Skew Dist")  # Left Skew a<0
plt.show()
```



One quick way to check if the data is skewed is to check the order of the mean, median, and mode of the data. The **mean** is the average of the data. The **median** is the value of the data point(s) at the 50th percentile of the data ordered from smallest to largest. The **mode** is the value of the data that shows up most frequently in the dataset. Figure 2 below shows different relationships among these three metrics for symmetric, right-skew and left-skew distributions.

Figure 2: Mean, Median, and Mode Relationships for Symmetric, Right-Skew and Left-Skew Distributions

In [4]:
```
# Mean, Median, and Mode Relationships for Symmetric, Right-Skew and Left-Skew D

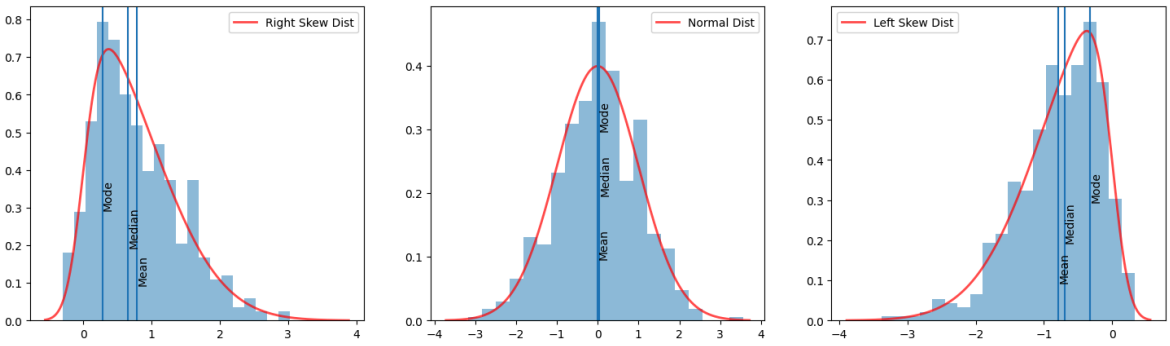

# Helper HistogramPlot function
def HistogPlot(a, label):
    np.random.seed(123)
    r = skewnorm.rvs(a, size=500)
    x = np.linspace(skewnorm.ppf(0.0001, a), skewnorm.ppf(0.9999, a), 500)
    plt.plot(x, skewnorm.pdf(x, a), "r-", lw=2, alpha=0.7, label=label)  # Densi
    n, bins, patches = plt.hist(
        r, bins=20, density=True, histtype="stepfilled", alpha=0.5
    )  # Histogram

    # Add Mean
    plt.axvline(x=np.mean(r))
    plt.text(np.mean(r), 0.1, "Mean", rotation=90)

    # Add Median
    plt.axvline(x=np.median(r))
    plt.text(np.median(r), 0.2, "Median", rotation=90)

    # Add Mode (by the most frequent bin)
    plt.axvline(x=(bins[n.argmax()] + bins[n.argmax() + 1]) / 2)
    plt.text((bins[n.argmax()] + bins[n.argmax() + 1]) / 2, 0.3, "Mode", rotatio
    plt.legend()


# Plot
plt.figure(figsize=(18, 5))
ax1 = plt.subplot(131)
HistogPlot(5, "Right Skew Dist")
ax2 = plt.subplot(132)
HistogPlot(0, "Normal Dist")
ax3 = plt.subplot(133)
HistogPlot(-5, "Left Skew Dist")
plt.show()
```



As we just saw, we can visually assess the skewness of a dataset via histograms. Alternatively, we can also use the relationship among the mean, median, and mode to decide if the data is skewed. However, is there a measurement to tell us if a dataset is skewed? Yes, there are measurements that can tell us if a dataset is skewed. Here's a popular one from many textbooks.

Fisher-Pearson Standardized Moment Coefficient (skewness):

$$\sum_{i=1}^{n} \frac{(X_i - \bar{X})^3/n}{s_X^3}$$

Where $\bar{X}$ is the mean of X and s is the standard deviation of X

The properties of this test coefficient are:

> 1. A symmetric distribution will have value = 0. A negative value indicates a left-skew distribution. A positive value indicates a right-skew distribution.
> 2. The coefficient compares the data with a normal distribution. The larger the value of the coefficient, the more the data deviates from normal distribution.
> 3. The metric can be adjusted for sample size by multiplying $\frac{\sqrt{n(n-1)}}{n-2}$. When the sample size is large, this adjustment factor approaches 1.

Most statistical software can calculate the skewness for us, and therefore, we do not have to calculate the number by hand.

2. Kurtosis

Kurtosis is another measure used to decide the shape of the distribution of a dataset. Kurtosis allows us to decide if the tails of a distribution are heavier or lighter than the tails of a normal distribution. What does it mean to have a heavy tail? In general, a distribution with heavy tails means its distribution has higher probability mass in the tails than a normal distribution. On the other hand, a distribution with thin tails means its distribution has lower probability density in the tails then a normal distribution. In figure 3, below, we will show four different distributions with different tail shapes.

Figure 3: Probability Functions with Different Tails Compared to a Normal Distribution

In [5]:
```
# Four distributions with different tail shapes
# Generate x values for plotting
x = np.linspace(-6, 6, 1000)

# 1. Normal-tailed distribution: Standard Normal
normal_tail = stats.norm(loc=0, scale=1)
normal_tail_pdf = normal_tail.pdf(x)

# 2. Heavy-tailed distribution 1: Student's t-distribution (df=2)
student_tail = stats.t(df=2)
student_tail_pdf = student_tail.pdf(x)

# 3. Heavy-tailed distribution 2: Laplace distribution
laplace_tail = stats.laplace(loc=0, scale=1)
laplace_tail_pdf = laplace_tail.pdf(x)

# 4. Thin-tailed distribution: Uniform distribution
```

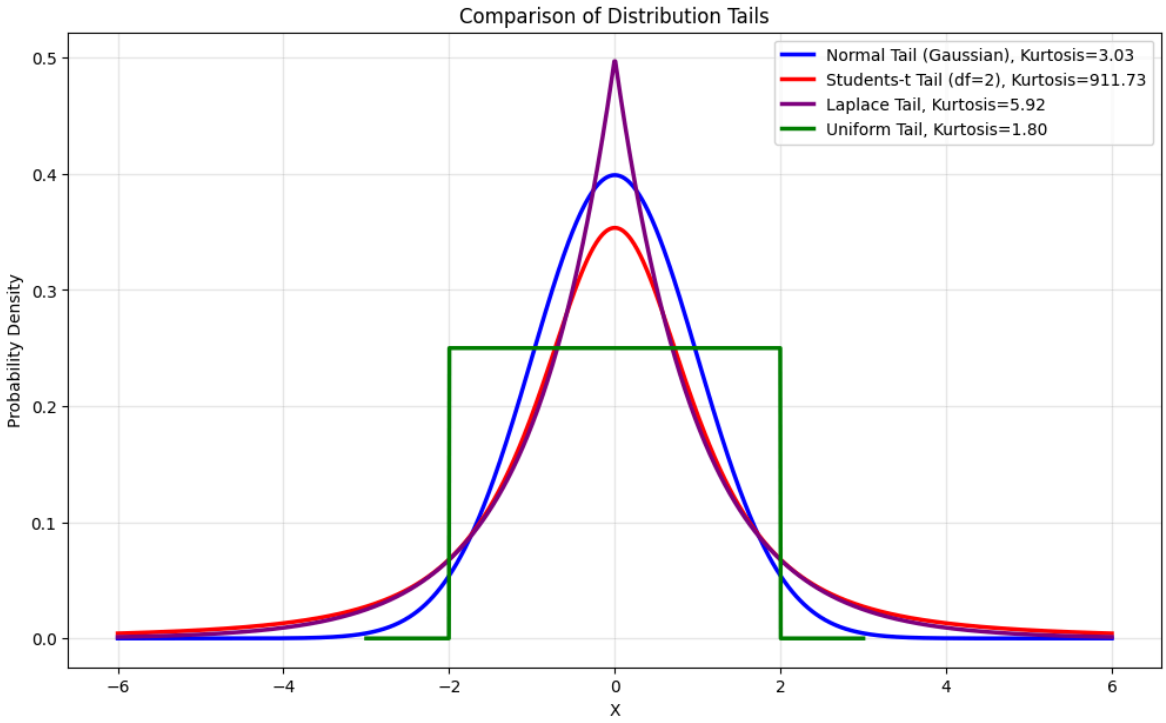
```
uniform_tail_x = np.linspace(-3, 3, 1000)
uniform_tail_pdf = np.where(np.abs(uniform_tail_x) <= 2, 0.25, 0)

# Calculate kurtosis for each distribution
# Generate samples to calculate empirical kurtosis
np.random.seed(42)  # For reproducibility
n_samples = 10000

normal_samples = stats.norm.rvs(size=n_samples)
student_samples = stats.t.rvs(df=2, size=n_samples)
laplace_samples = stats.laplace.rvs(size=n_samples)
uniform_samples = np.random.uniform(-2, 2, n_samples)

normal_kurtosis = stats.kurtosis(normal_samples)+3
student_kurtosis = stats.kurtosis(student_samples)+3
laplace_kurtosis = stats.kurtosis(laplace_samples)+3
uniform_kurtosis = stats.kurtosis(uniform_samples)+3

# Create the comparison plot
plt.figure(figsize=(12, 7))
plt.plot(x, normal_tail_pdf, label=f'Normal Tail (Gaussian), Kurtosis={normal_ku
plt.plot(x, student_tail_pdf, label=f'Students-t Tail (df=2), Kurtosis={student_
plt.plot(x, laplace_tail_pdf, label=f'Laplace Tail, Kurtosis={laplace_kurtosis:.
plt.plot(uniform_tail_x, uniform_tail_pdf, label=f'Uniform Tail, Kurtosis={unifo
plt.xlabel('X')
plt.ylabel('Probability Density')
plt.title('Comparison of Distribution Tails')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()
```



From figure 3, the blue curve is a normal distribution. The red curve is a student's-t distribution with degrees of freedom equals to 2. The purple curve is a Laplace (Double Exponential) distribution. The green curve is an uniform distribution with the range between -2 and 2.

We can see all four distributions are symmetric distributions centered around 0. However, compared to the normal distribution, the Student's-t distribution and Laplace distribution exhibit higher probability density in the tails than the normal distribution. The uniform distribution demonstrates a lower probability density in the tails relative to the normal distribution. Hence student's-t distribution and Laplace distribution are heavy-tailed(fat-tailed) distribution while uniform distribution is a thinner (lighter) tailed distribution.

One key point to be careful is that one shape of the tail pattern of a distribution does not mean having a higher peak or a lower peak compared to the normal distribution. In figure 3, we can see that both student's-t distribution and Laplace distribution have heavier tails compared to the normal distribution. But Laplace distribution has a taller peak than the normal distribution while student's-t distribution has a lower peak than the normal distribution.

By the tail shape of a symmetric distribution, a normal distribution is called **mesokurtic**. A fat tail distribution is called **leptokurtic**. A thin or light tail distribution is called **platykurtic**.

The formal Kurtosis formula is as follows:

$$\text{Kurtosis} = \sum_{i=1}^{n} \frac{(X_i - \bar{X})^4/N}{s^4}$$

A normal distribution has a kurtosis = 3. A fat tail distribution has a kurtosis > 3. A thin tail distribution has a kurtosis < 3. Any statistical application has a function to calculate kurtosis. Some statistical applications can mean excess kurtosis as kurtosis. **Excess kurtosis** is $kurtosis - 3$. When a distribution with excess kurtosis = 0, this distribution is a normal distribution. Be sure to check the selected statistical application's manual to verify the definition of kurtosis used in the application.

Both skewness and kurtosis are aimed to help us better understand the shape of a data distribution. If we want to check whether a dataset is normally distributed, we can draw the graph and check its skewness and kurtosis. However, the best way is to conduct a Shapiro–Wilk test, as described in the last lesson.

3. Skew-Normal Distribution and Skew-t Distribution

In the previous sections, we talked about skew data and heavy tail data. We learned from Module 1 that we can transform non-normally distributed data into normally distributed data. However, there are times we do not want to transform the data and want to keep its original scale for analysis. In this case, we need to find a method that we can use to directly analyze a non-normal dataset. As Azzalini and Capitanio summarized in their book (Azzalini et al, 2014), skew-normal distribution and skew-t distribution are used to handle skew and/or heavy tail data. Specifically, normal distribution is a special case for skew-normal distribution. In this section, we are going to give an overview of the skew-

normal distribution and skew-t distribution. Then, we will present an example to show how we can apply these distributions to regression analysis.

3.1 The Basics of Skew-Normal Distribution

A **skew-normal distribution** is a distribution that is not a symmetric distribution but has the shape of a normal distribution. When you check your data and it exhibits the above characteristics, you can try to use skew-normal distribution. Normal distribution is actually a special case for skew-normal distribution. We will talk about that later.

Figure 4: Simulated Skew-Normal Distributions with Different Shape Parameters

In [6]:
```
# Simulated Skew-Normal Distributions with Different Shape Parameters
fig, axes = plt.subplots(3, 1, figsize=(12, 10))
x = np.linspace(0, 60, 100)

# Function to covert mean to location
def calc_loc_for_mean(alpha, mean, scale):
    delta = alpha / np.sqrt(1 + alpha**2)
    return mean - scale * delta * np.sqrt(2/np.pi)

mean = 30
scale = 7

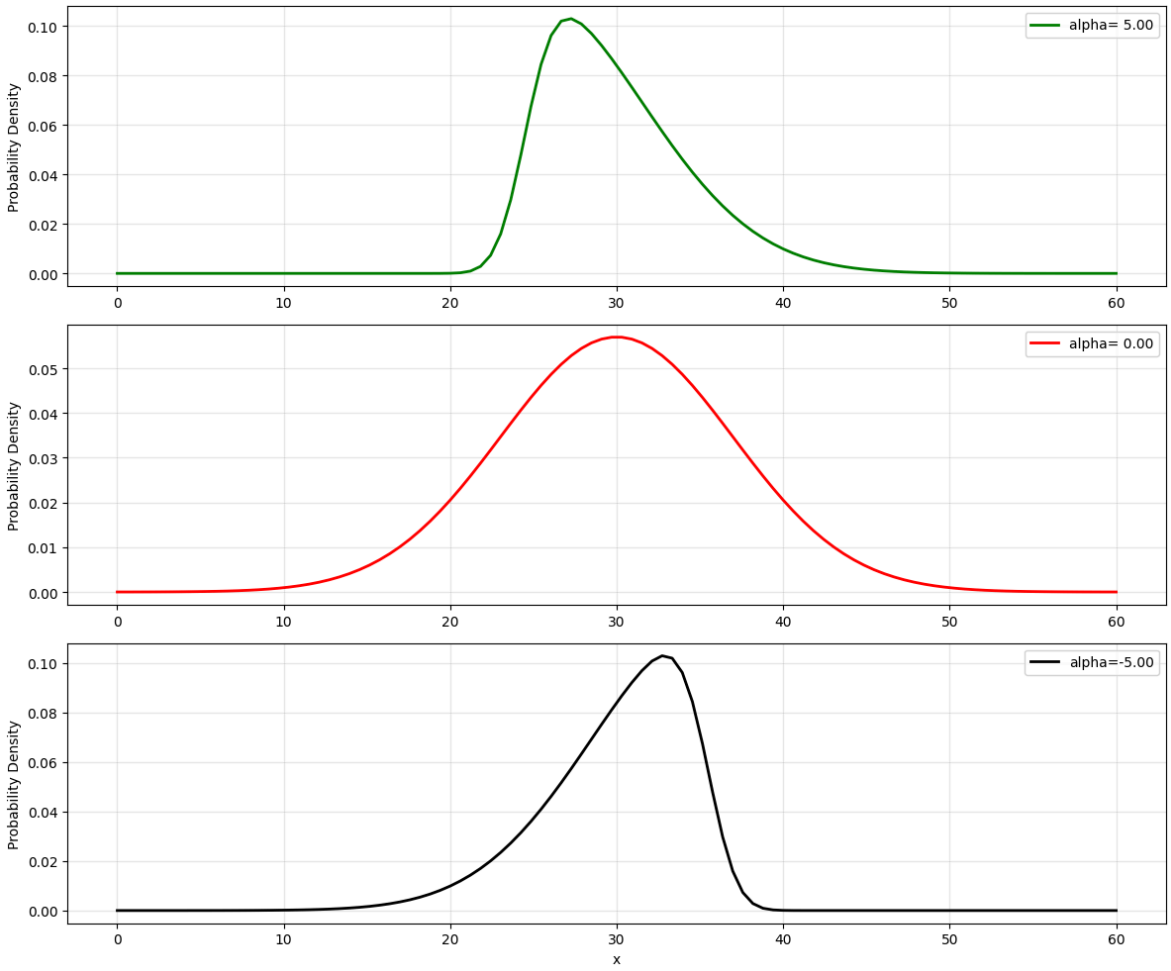
# Alpha = 5 (top subplot)
alpha = 5
loc = calc_loc_for_mean(alpha, mean, scale)
axes[0].plot(
    x,
    stats.skewnorm.pdf(x, alpha, loc=loc, scale=scale),
    c="g",
    linewidth=2,
    label="alpha= {:.2f}".format(alpha),
)
axes[0].legend()
axes[0].grid(True, alpha=0.3)
axes[0].set_ylabel('Probability Density')

# Alpha = 0 (middle subplot)
alpha = 0
loc = calc_loc_for_mean(alpha, mean, scale)
axes[1].plot(
    x,
    stats.skewnorm.pdf(x, alpha, loc=loc, scale=scale),
    c="r",
    linewidth=2,
    label="alpha= {:.2f}".format(alpha),
)
axes[1].legend()
axes[1].grid(True, alpha=0.3)
axes[1].set_ylabel('Probability Density')

# Alpha = -5 (bottom subplot)
alpha = -5
loc = calc_loc_for_mean(alpha, mean, scale)
axes[2].plot(
```

```
    x,
    stats.skewnorm.pdf(x, alpha, loc=loc, scale=scale),
    c="k",
    linewidth=2,
    label="alpha={:.2f}".format(alpha),
)
axes[2].legend()
axes[2].grid(True, alpha=0.3)
axes[2].set_ylabel('Probability Density')
axes[2].set_xlabel('x')

plt.tight_layout()
plt.show()
```



According to Azzalini and Capitanio, if X is a continuous random variable, its skew-normal density function is as follows:

$$f(x) = 2\phi(X)\Phi(\alpha x)$$

Where α is a predetermined parameter

And

$$\phi(x) = \frac{exp(-\frac{x^2}{2})}{\sqrt{2\pi}}$$

And

$$\Phi(\alpha x) = \int_{-\infty}^{\alpha x} \phi(t)dt$$

$\phi(x)$ is the standard normal density function and $\Phi(\alpha x)$ is the standard normal (cumulative) distribution function. α is a **shape parameter**. Figure 4 shows skew-normal distributions with different α's. Here are some properties about α:

> - When $\alpha = 0$, skew-normal distribution is standard normal distribution (red curve in figure 4).
> - When $\alpha > 0$, skew-normal distribution is a right-skew distribution (green curve in figure 4).
> - When $\alpha < 0$, skew-normal distribution is a left-skew distribution (black curve in figure 4).
> - When $|\alpha|$ increases, the peak of the distribution will move farther to either the right side or left side.

Skew-normal distribution and its related distribution families are heavy on math calculations. One tricky part is that skew-normal has two ways to express its parameters: **direct parameters** (DP) and **centered parameters** (CP). They exist due to maximum likelihood optimization calculations. We will not go into detail about these technicalities here. Let's talk about what they are and what we need to pay attention to when using them with skew-normal distribution.

Let's assume random variable X has a skew-normal distribution with mean = μ, standard deviation = σ, skewness = γ_1, we can denote X as follows:

Centered Parameters Method (CP):

$$X \sim SN(\mu, \sigma, \gamma_1)$$

Where $-0.995 < \gamma_1 < 0.995$

Direct Parameters Method (DP):

$$X \sim SN(\xi, \omega, \alpha)$$

Where ξ is location parameter, ω is scale parameter and α is shape parameter

There are functions to convert centered parameters to direct parameters and vice versa in most statistical applications. Therefore, you do not need to convert them by hand. We are more familiar with elements in the centered parameters method and their interpretation. You can always start from CP and then use the selected application to convert them to DP. Before you use any skew-normal distribution functions with your selected statistical application, make sure you understand what form of inputs (CP or DP) the function takes.

NOTE: Demonstration examples for the second part of this lesson are implemented in R software packages, which is why screenshots of R code and output are included here. A file with all the R code for this lesson is also provided.

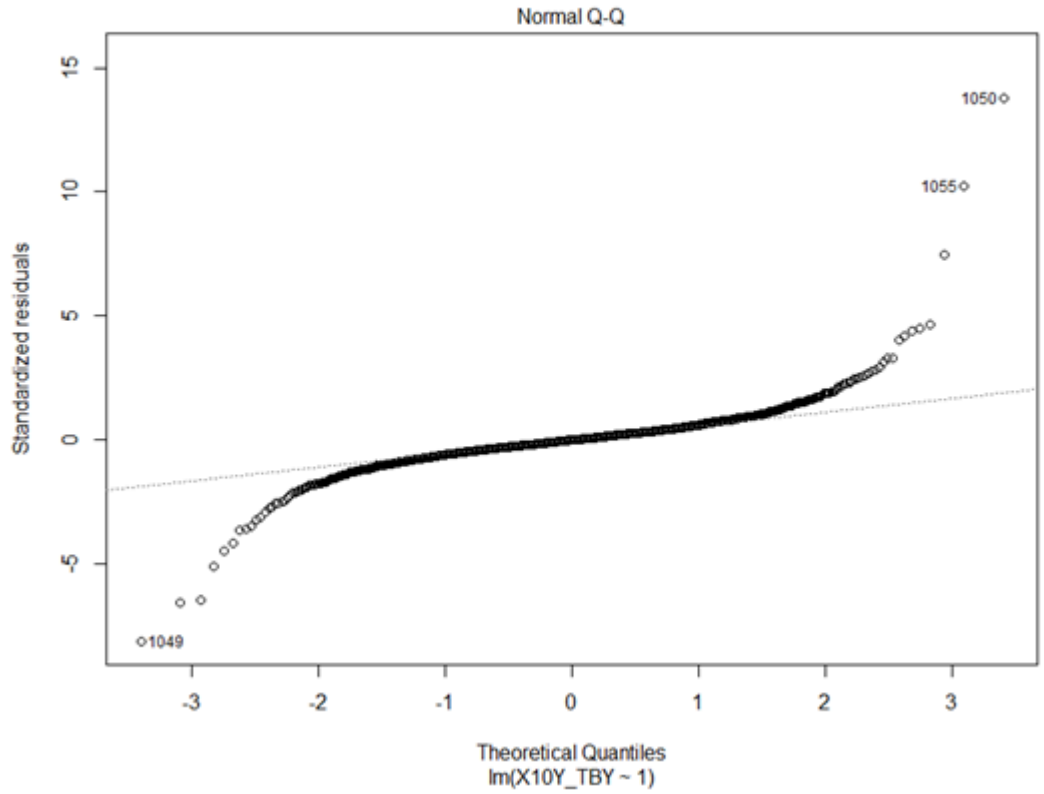
3.2 U.S. 10-Year Treasury Bond Yield Daily Return and Skew-Normal Distribution

We talked about the basics of skew-normal distribution in the last section. Now let's apply this method to our U.S. 10-Year Treasury Bond Yield Daily Return data. The data spans from the beginning of 2016 to the end of 2021. Here are some statistics from the data.

Figure 5: Key Statistics for U.S. 10-Year Treasury Bond Yield Daily Return

UST 10Y Yield Daily Return	
Mean	0.0004
Standard Deviation	0.0362
Skewness	2.0381
Kurtosis	42.7225

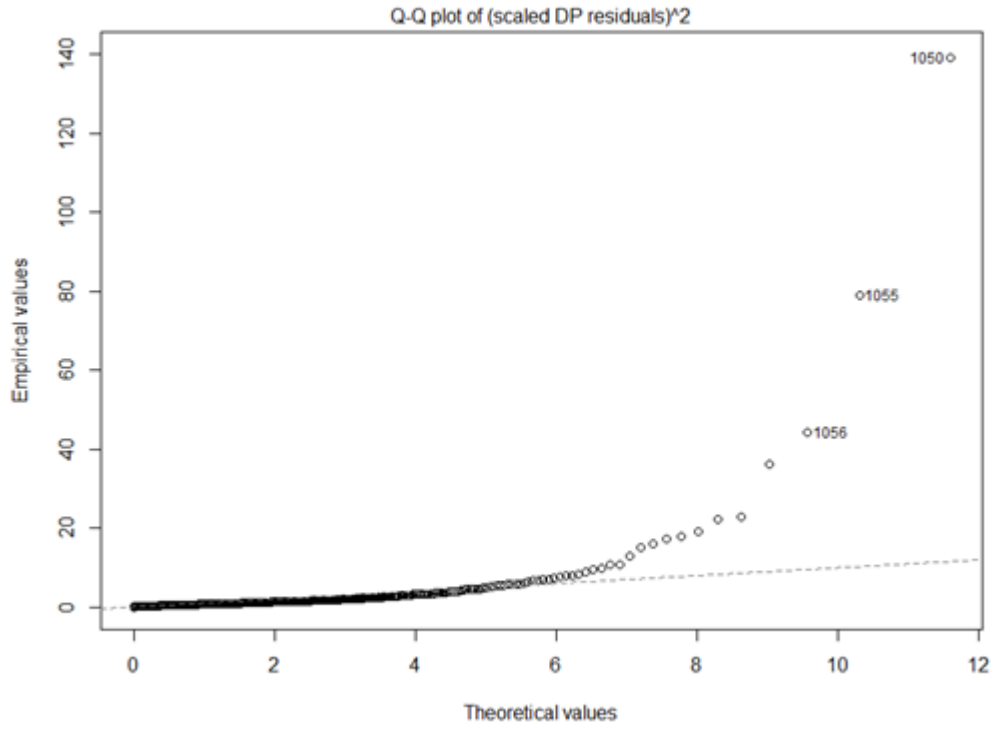
Figure 6: QQ Plot Normal Distribution vs U.S. 10-Year Treasury Bond Yield Daily Return



From figure 5, we can see that the U.S. Treasury 10-Year Bond Yield daily return is skewed to the right and has a heavy tail. Figure 6 also shows the U.S. Treasury 10-Year Bond Yield daily return has heavy tails on both ends compared to a normal distribution. Hence, normal distribution is not a good choice to describe the variable.

Now let's try skew-normal distribution on the U.S. Treasury 10-Year Bond Yield daily return to see if the distribution fits the data. Figure 7 show the result.

Figure 7: QQ Plot Skew-Normal Distribution vs U.S. Treasury 10-Year Bond Yield Daily Return



From figure 7, we can see that skew-normal distribution fits the data better on the left side of the graph. However, there are still some large outliers on the right side that skew-normal distribution cannot catch.

Now let's proceed to run a simple regression to understand the impact of the Dow Jones Industrial Average Index daily return on the U.S. Treasury 10-Year Bond Yield daily return. Our dependent variable will be the U.S. Treasury 10-Year Bond Yield daily return and our independent variable will be the Dow Jones Industrial Average Index daily return. We will run one simple regression model with a normal dependent variable setting and one with a skew-normal dependent variable setting. We then compare the QQ plots for the respective residuals. Figure 8 shows the normal least square model result.

Figure 8: Least Square Model Result

```
Call:
lm(formula = X10Y_TBY ~ DWJ, data = data_set)

Residuals:
     Min       1Q   Median       3Q      Max
-0.22532 -0.01422 -0.00128  0.01208  0.43982

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0003206  0.0008533  -0.376    0.707
DWJ          1.2157799  0.0709167  17.144   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.03309 on 1505 degrees of freedom
Multiple R-squared:  0.1634,    Adjusted R-squared:  0.1628
F-statistic: 293.9 on 1 and 1505 DF,  p-value: < 2.2e-16
```

Figure 9: Skew-Normal Regression Result

```
Call: selm(formula = X10Y_TBY ~ DWJ, family = "SN", data = data_set)
Number of observations: 1507
Family: SN
Estimation method: MLE
Log-likelihood: 3053.266
Parameter type: DP

DP residuals:
     Min       1Q   Median       3Q      Max
-0.19895  0.01181  0.02482  0.03813  0.46436

Regression coefficients
                estimate    std.err    z-ratio Pr{>|z|}
(Intercept.DP) -0.026387   0.001335 -19.768647        0
DWJ             1.247159   0.066077  18.874188        0

Parameters of the SEC random component
      estimate std.err
omega   0.0421   0.001
alpha   1.3602   0.091
```

From figures 8 and 9, we can see the coefficient for the Dow Jones Industrial Average Index daily return from our least square regression is 1.216, and from our skew-normal regression, it is 1.247. The alpha from skew-normal regression is 1.36, which means that the dependent variable is skewed to the right. Now, let's compare the QQ plots from our least square and skew-normal regressions.

Figure 10: QQ Plots for Residuals from Normal Regression and for Residuals from Skew-Normal Regression

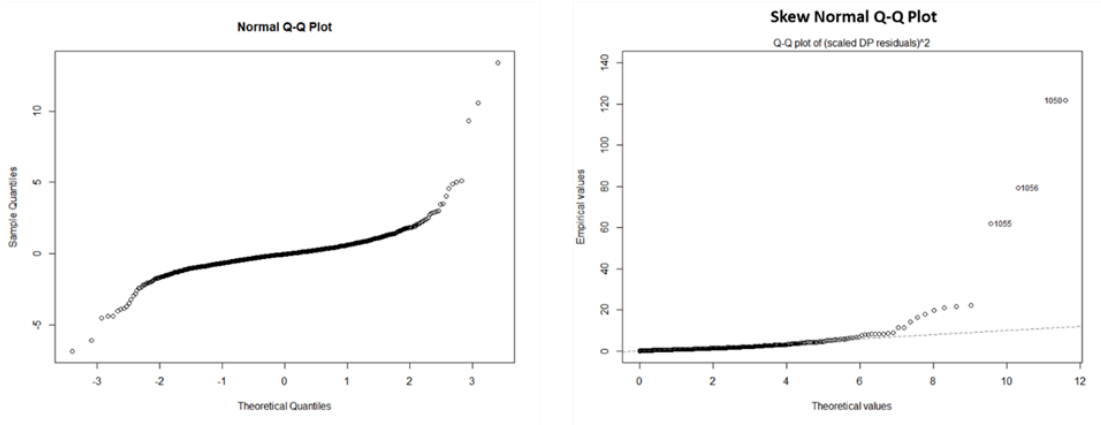
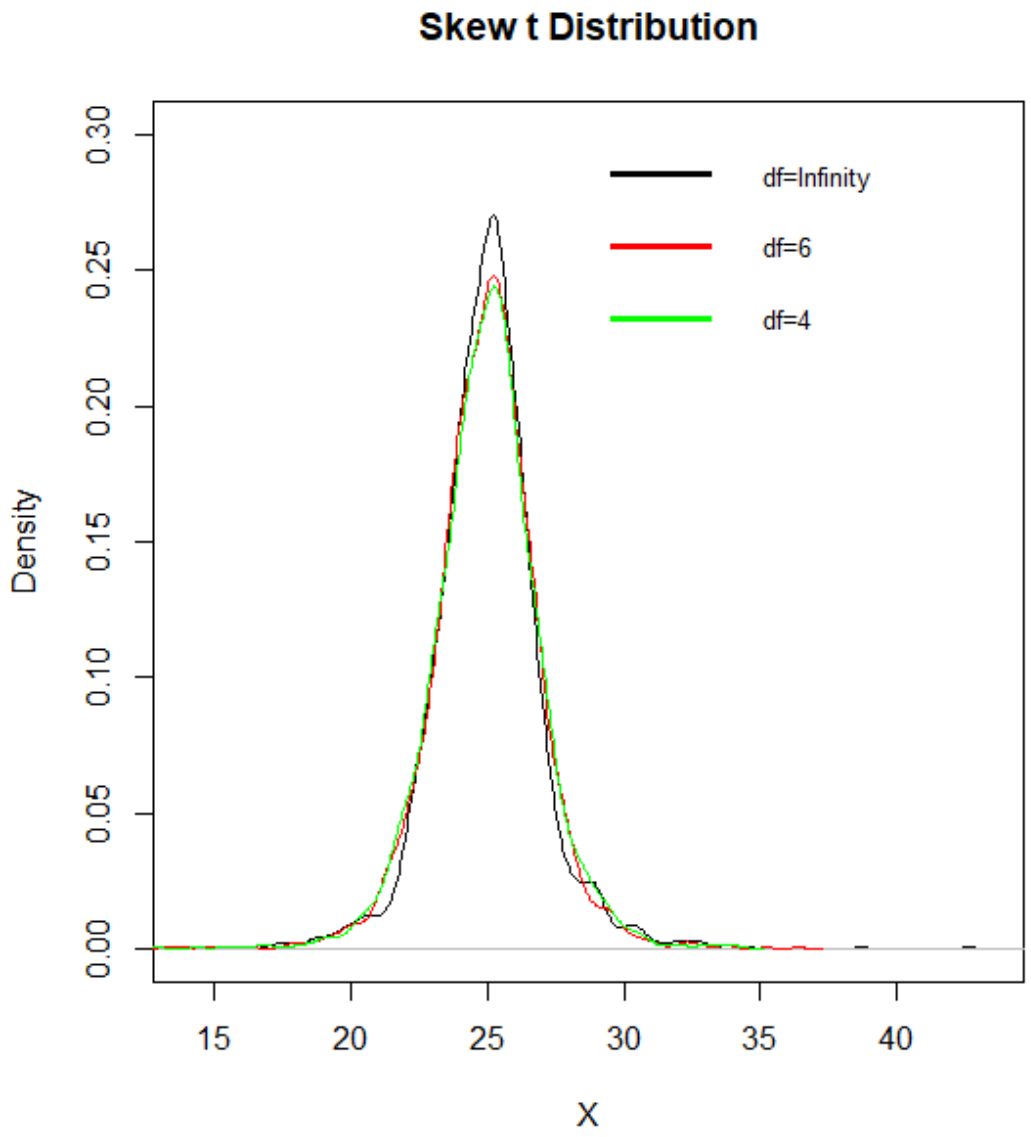


Figure 10 shows the QQ plots for residuals for two models. The scales of the two QQ plots are different because of the CP-DP adjustment in skew-normal distribution. The interpretation is still the same. From the normal (least square) QQ plot on the left, we can see the data are all curvy on both ends of the graph. Only the middle part follows a straight line. However, the skew-normal QQ plot on the right seems to have a better fit. The left end of the data also fits well with the straight line. Only to the right of the graph do we see that the data curves up. We see this phenomenon because the actual data has a thicker right tail than a normal distribution. In the next section, we will introduce skew-t distribution to be able to handle heavier tails.

3.3 The Basics of Skew-t Distribution

A skew-t distribution is like a Student's t-distribution with the flexibility to adjust for the heaviness of the tails. Student's t-distribution is a symmetric distribution that has heavier tails than normal distribution. Usually Student's t-distribution is used when the data distribution is symmetric but the tails are thicker or longer than Normal distribution. In this section, we are going to introduce a distribution that is like Student's t-distribution but can both handle skewness and kurtosis. This is skew-t distribution.

Figure 11: Simulated Skew-t Distributions with Different Degrees of Freedom



If X is a continuous random variable, it's skew-t density function is as follows:

$2t(X;\nu)T(\alpha X;\nu)$

Where ν is degree of freedom

And $t(X;\nu)$ is the probability density function for Student's-t distribution

And $T(\alpha X;\nu)$ is the cumulative probability distribution function for Student's t-distribution.

Figure 11 shows three skew-t distributions with different degrees of freedom. Both ν and α are parameters to decide the shape and tail pattern of the skwe-t distribution. Here are some properties of skew-t distribution:

When α =0, skew-t distribution reduces to Student's t-distribution.

When $\nu \to \infty$, skew-t distribution reduces to skew normal distribution

When α =0 and $\nu \to \infty$, skew-t distribution reduces to standard normal distribution.

Similar to skew-normal distribution, skew-t distribution also has a CP-DP conversion adjustment. Let's look at skew-t distribution's CP and DP. Assume random variable X has a skew-t distribution with mean = μ, standard deviation = σ, skewness = γ_1 and kurtosis = γ_2, and we can denote X as follows:

Centered Parameters Method (CP): $X \sim ST(\mu, \sigma, \gamma_1, \gamma_2)$

Where γ_1 < 0.995

Direct Parameters Method (DP): $X \sim ST(\xi, \omega, \alpha, \nu)$

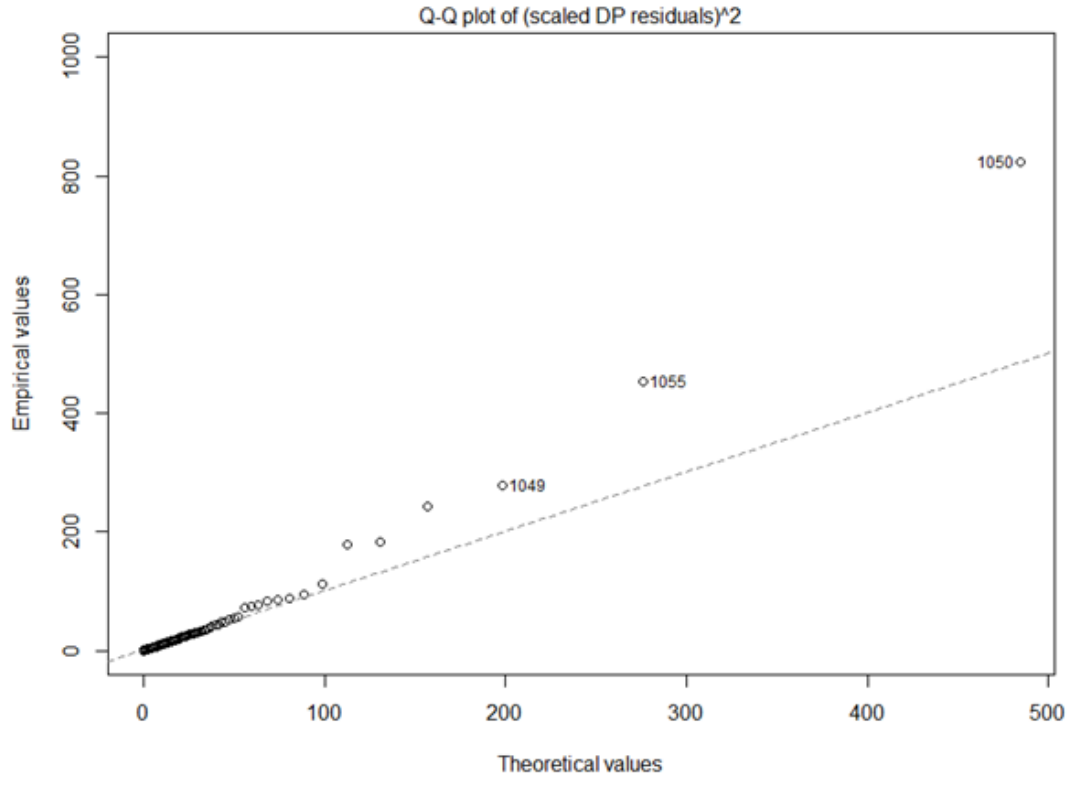
Where ξ is location parameter, ω is scale parameter, α is shape parameter and ν is degree of freedom

Any statistical application can convert CP to DP and vice versa for skew-t distribution.

3.4 U.S. 10-Year Treasury Bond Yield Daily Return and Skew-t Distribution

We tried to use skew-normal to fit the U.S. 10-Year Treasury Bond Yield Daily Return data in the last section. However, skew-normal distribution cannot describe well the heavy right tail nature of the U.S. 10-Year Treasury Bond Yield Daily Return data. In this section, let's try skew-t distribution to see if this distribution can do a better job.

Figure 12: QQ Plot Skew-t Distribution vs. U.S. Treasury 10-Year Bond Yield Daily Return



From figure 12, we can see more data points rest on the straight line for the skew-t distribution QQ plot than the skew-normal distribution QQ plot in Figure 7. Skew-t distribution fits the U.S. Treasury 10-Year Bond Yield daily return data better than skew-normal distribution. There are fewer data points not on the straight line for skew-t distribution.

Now let's run skew-t distribution regression of the U.S. Treasury 10-Year Bond Yield daily return on the Dow Jones Industrial Average daily return. Figure 13 shows the result from skew-t distribution regression.

Figure 13: Skew-t Distribution Regression Result

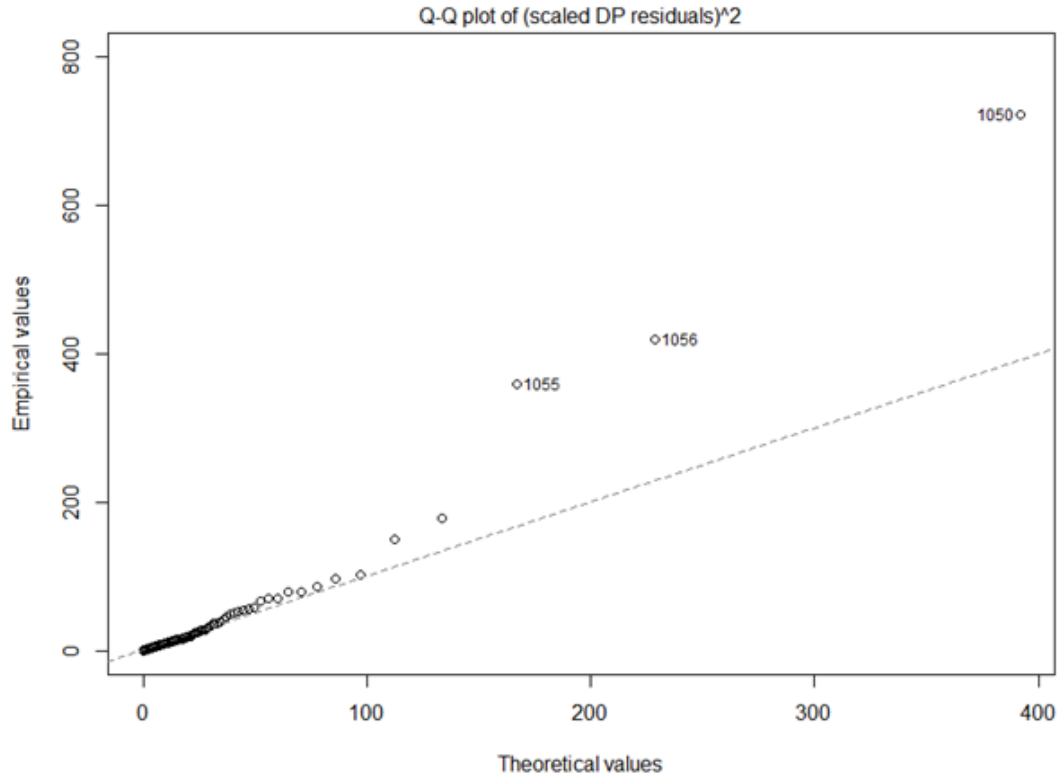
```
Call: selm(formula = X10Y_TBY ~ DWJ, family = "ST", data = data_set)
Number of observations: 1507
Family: ST
Estimation method: MLE
Log-likelihood: 3419.031
Parameter type: DP

DP residuals:
      Min        1Q    Median        3Q       Max
-0.222990 -0.010489  0.002294  0.015836  0.449492

Regression coefficients
                estimate   std.err   z-ratio Pr{>|z|}
(Intercept.DP) -0.003879  0.001727 -2.245821    0.025
DWJ             1.090937  0.078792 13.845841    0.000

Parameters of the SEC random component
      estimate std.err
omega  0.01671   0.001
alpha  0.17386   0.114
nu     2.59760   0.202
```

Figure 14: QQ Plots for Residuals from Skew-Normal Regression and for Residuals from Skew-t Regression



From Figure 14, we can see skew-t distribution has a better regression fit for the right side of the regression residuals QQ plot than skew-normal distribution regression does. Also, the coefficient for the Dow Jones Industrial Average index daily return is much smaller (1.091 from figure 13) than in skew-normal distribution. It is more evidence that we should properly model the thicker tail for underlying data.

4. Conclusion

In this lesson, we discussed what skewness and kurtosis are. We talked about the patterns of skewness and its measurement. We also talked about the patterns of kurtosis and its measurement. Built on the knowledge of skewness and kurtosis, we introduced skew normal distribution and skew-t distribution to handle normal distribution like a dataset but with skewness and thick tails. We went through the basics of skew normal distribution and skew-t distribution. We then applied these distributions to our U.S. Treasury Bond Yield and U.S. stock index example. In the next lesson, we will explore the metrics to calculate relationships among random variables.

References

- Azzalini, Adelchi, and Antonella Capitanio. The Skew-Normal and Related Families. Cambridge University Press, IMS Monographs Series, 2014.

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14.30 Introduction to Statistical Methods in Economics

Spring 2009

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14.30 Introduction to Statistical Methods in Economics
Lecture Notes 9

Konrad Menzel

March 10, 2009

1 Functions of Random Variables

In this part of the lecture we are going to look at functions of random variables, $Y = u(X)$. Note that Y is again a random variable: since X is a mapping from the sample space S into the real numbers,

$$X : S \to \mathbb{R}$$

and $u : \mathbb{R} \to \mathbb{R}$, the composition of u and X is also a mapping from S into the real numbers:

$$Y = u \circ X : S \to \mathbb{R}$$

Example 1 *If X is the life of the first spark plug in a lawnmower, and Y the life of the second, we may be interested in the sum of the two, $Z = X + Y$.*

Example 2 *Before coming to MIT, I applied for several German fellowships, so I would receive a monthly stipend of X Euros, depending on which fellowship I was going to get, and the exchange rate in, say, September 2005 was going to be Y Dollars per Euro. Each quantity was uncertain at the time I was applying, but since I was going to spend the money in the US, the main quantity of interest was the dollar amount $Z = X * Y$ I was going to receive (at least in terms of the Dollar exchange rate, I could not complain).*

We now want to know how to obtain the density and c.d.f. for the transformed random variable $u(X)$, so we can treat each problem involving a *function* of a random variable in the same way as a question involving only the random variable itself with a known p.d.f.

We'll consider three cases:

1. underlying variable X discrete
2. underlying variable X continuous
3. X continuous *and* $u(X)$ strictly increasing

The last case is of course a special case of the second, but we'll see that it's much easier to work with.

1.1 Discrete Case - "2-Step" Method

If X is a discrete random variable with p.d.f. $f_X(x)$, and $Y = u(X)$, where $u(\cdot)$ is a known deterministic function. Then

$$f_Y(y) = P(Y = y) = P(u(X) = y) = \sum_{x:u(x)=y} f_X(x)$$

Example 3

$$f_X(x) = \begin{cases} \frac{1}{5} & \text{if } x \in \{-2, -1, 0, 1, 2\} \\ 0 & \text{otherwise} \end{cases}$$

Then if $Y = g(X) = |X|$,

$$f_Y(y) = \begin{cases} f_X(0) = \frac{1}{5} & \text{if } y = 0 \\ f_X(-1) + f_X(1) = \frac{2}{5} & \text{if } y = 1 \\ f_X(-2) + f_X(2) = \frac{2}{5} & \text{if } y = 2 \\ 0 & \text{otherwise} \end{cases}$$

Note that if X is discrete, then Y is also discrete.

1.2 Continuous Case - "2-Step" Method

If X is a continuous random variable with p.d.f. $f_X(x)$, and $Y = u(X)$, then the c.d.f. of Y is given by

$$F_Y(y) = P(Y \leq y) = P(u(X) \leq y) = \int_{x:u(x)\leq y} f_X(x)dx$$

If Y is also continuous, then

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Note that even if X is continuous, Y need not be continuous.

Example 4 $Y = \lceil X \rceil$*, the largest integer smaller than* X*, is discrete regardless whether* X *is continuous or discrete.*

Example 5

$$f_X(x) = \begin{cases} \frac{1}{2} & \text{if } -1 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Let's look at

$$Y = X^2$$

From $X \in [-1, 1]$*, it follows that* $Y = [0.1]$*. How do we get the density of* Y*? For* $y \in [0, 1]$*, the c.d.f. is*

$$F_Y(y) = P(Y \leq y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} dx = \sqrt{y}$$

In sum

$$F_Y(y) = \begin{cases} 0 & \text{if } y < 0 \\ \sqrt{y} & \text{if } y \in [0, 1) \\ 1 & \text{if } y \geq 1 \end{cases}$$

Since Y *is continuous, we can derive the density*

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \begin{cases} \frac{1}{2\sqrt{y}} & \text{if } y \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

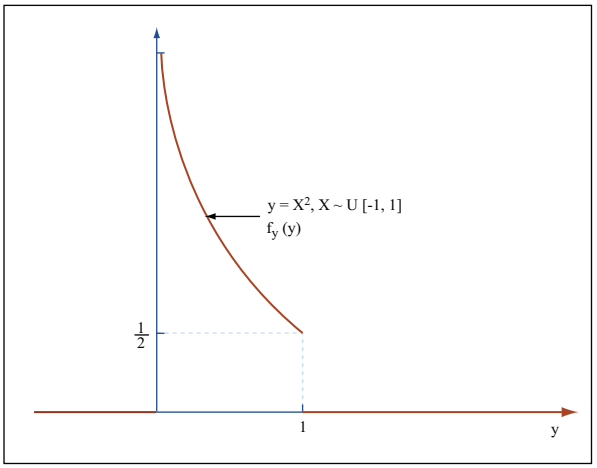


Image by MIT OpenCourseWare.

1.3 Change of Variables Formula for One-to-One Transformations

It is in general not very convenient to derive the density of Y from the density $f_X(x)$ of X through the c.d.f.s, in particular because this involves one integration and one differentiation. So one might wonder whether there is a more direct connection between the p.d.f.s.
Before going to the more general case, suppose $u(x) = ax$ for some constant $a > 0$. Then the c.d.f. of $Y = u(X) = aX$ is given by

$$F_Y(y) = \int_{ax \leq y} f_X(x)dx = \int_{x \leq \frac{y}{a}} f_X(x)dx = F_X\left(\frac{y}{a}\right)$$

Using the chain rule, we can derive the p.d.f. of Y

$$f_Y(y) = \frac{d}{dy}F_Y(y) = \frac{d}{dy}F_X\left(\frac{y}{a}\right) = \frac{1}{a}\frac{d}{dx}F_X(x) = \frac{1}{a}f_X(x)$$

What is a good intuition for this? - if $a > 1$, we could think of the transformation as stretching the axis on which the random variable falls. This moves any pair of points on the axis apart by a factor of a, but leaves constant the probability that the variable falls between the points. Therefore, the distribution of Y is "thinned out" by a factor of $\frac{1}{a}$ compared to the distribution of X. One could visualize this by thinking about a lump of dough containing a number of raisins - the more we spread out the dough, the sparser the distribution of the raisins in the dough will be with respect to the surface of the table.

For differentiable monotone transformations $u(\cdot)$ of X we have the following result

Proposition 1 *Let X be a continuous random variable with known density $f_X(x)$ such that $P(a \leq X \leq b) = 1$, and $Y = u(X)$. If $u(\cdot)$ is strictly increasing and differentiable on an interval $[a, b]$, and has an inverse $s(y) = u^{-1}(y)$, then the density of Y is given by*

$$f_Y(y) = \begin{cases} f_X(s(y))\left|\frac{d}{dy}s(y)\right| & \text{if } u(a) \leq y \leq u(b) \\ 0 & \text{otherwise} \end{cases}$$

Notice that an analogous result is true if $u(x)$ is strictly *decreasing* on $[a, b]$.

Example 6 *Let X be uniform on $[0, 1]$, so it has p.d.f.*

$$f_X(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

What is the p.d.f. of $Y = X^2$*? We can see that* on the support of X, $u(x) = x^2$ *is strictly increasing and differentiable, so that we can use the inverse of* $u(\cdot)$, $s(y) = \sqrt{y}$ *to obtain the p.d.f. of* Y,

$$f_Y(y) = f_X(s(y)) \left| \frac{d}{dy} s(y) \right| = f_X(\sqrt{y}) \frac{1}{2\sqrt{y}} = \begin{cases} \frac{1}{2\sqrt{y}} & \text{if } 0 \leq \sqrt{y} \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

This is similar to one example we did above, except that in the previous case, the support of X *was* $[-1, 1]$*, so that* $u(x) = x^2$ *was not monotone on the support of* X*.*

It is very important to note that this formula *works only for differentiable one-to-one - i.e. monotone - transformations*. In other cases, we have to stick to the more cumbersome "2-step" methods for the discrete and continuous case, respectively.

1.4 Probability Integral / Quantile Transformation

For continuous random variables, there is an interesting - and also very useful - result: the "c.d.f. of a c.d.f." is that of a uniform variable in the following sense:

Proposition 2 *Let* X *be a continuous random variable with c.d.f.* $F_X(x)$*. Then the c.d.f. evaluated at a random draw of* X, $F_X(X)$ *is uniformly distributed, i.e.*

$$F_X(X) \sim U[0, 1]$$

You should notice that a function of a random variable is itself a random variable (we'll discuss this in more detail later on).
PROOF: Since the c.d.f. takes values only between zero and 1, we can already see that the c.d.f. $G(\cdot)$ of $F(X)$ satisfies

$$\begin{aligned} G(F(X)) = P(F(X) \leq x) &= 0 \quad \text{if } x < 0 \\ G(F(X)) = P(F(X) \leq x) &= 1 \quad \text{if } x > 1 \end{aligned}$$

Without loss of generality (i.e. only avoiding a few uninteresting extra definitions or case distinctions), suppose $F(\cdot)$ is *strictly* monotonic - keeping in mind that *all* c.d.f.s are nondecreasing. This means that there is an inverse function $F^{-1}(\cdot)$, i.e. a function such that $F^{-1}(F(x)) = x$.

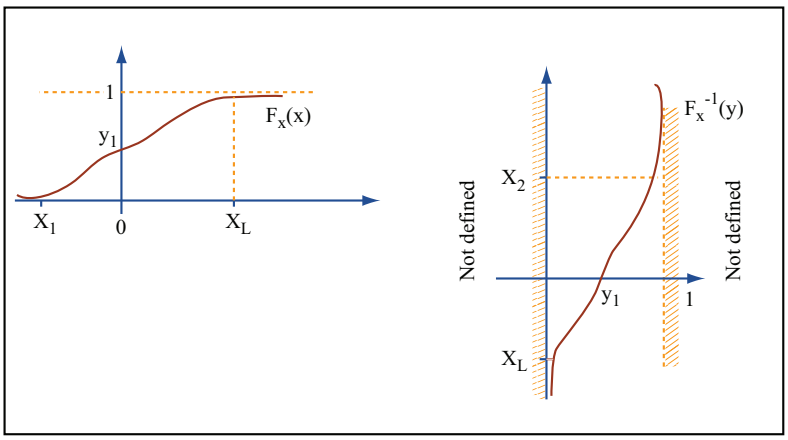


Image by MIT OpenCourseWare.

The inverse function will also be strictly monotonic, so that for $0 \leq x \leq 1$, the c.d.f. of the random variable $F_X(X)$ is given by

$$P\left(F_X(X) \leq x\right) = P\left(F_X^{-1}(F_X(X)) \leq F_X^{-1}(x)\right) = P\left(X \leq F_X^{-1}(x)\right) = F_X(F_X^{-1}(x)) = x$$

(the first equality uses monotonicity of $F^{-1}(\cdot)$, and the third the definition of a c.d.f.).
Summarizing, the c.d.f. $G(\cdot)$ of the random variable $F(X)$ is given by

$$G(F(x)) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

We can easily check that this is also the c.d.f. of a uniform random variable on the interval $[0, 1]$, so that $F(X)$ has the same probability distribution as $U[0, 1]$ $\square$

What is this result useful for? As an example, there are very efficient ways of generating *uniform* random numbers with a computer. If you want to get a sample of n draws from a random variable with c.d.f. $F_X(\cdot)$, you can

- draw $U_1, \ldots, U_n \sim U[0, 1]$
- transform each uniform draw according to

$$X_i = F_X^{-1}(U_i)$$

By the argument we made before, $X_1, \cdot, X_n$ behave like a random variable with c.d.f. $F_X(\cdot)$. This method is known as *integral (or quantile) transformation.*

Example 7 *Say, we have a computer program which allows us to draw a random variable U from a uniform distribution, but we actually want to obtain a random draw of X with p.d.f.*

$$f_X(x) = \begin{cases} \frac{1}{2}e^{-\frac{1}{2}x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

We can obtain the c.d.f. of X by integration:

$$F_X(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 - e^{-\frac{1}{2}x} & \text{if } x \geq 0 \end{cases}$$

so that the inverse of the c.d.f. is given by

$$F_X^{-1}(u) = -2\log(1-u) \quad \text{for } u \in [0, 1]$$

If we try this using some statistics software or Excel, a histogram of the draws will look like this:

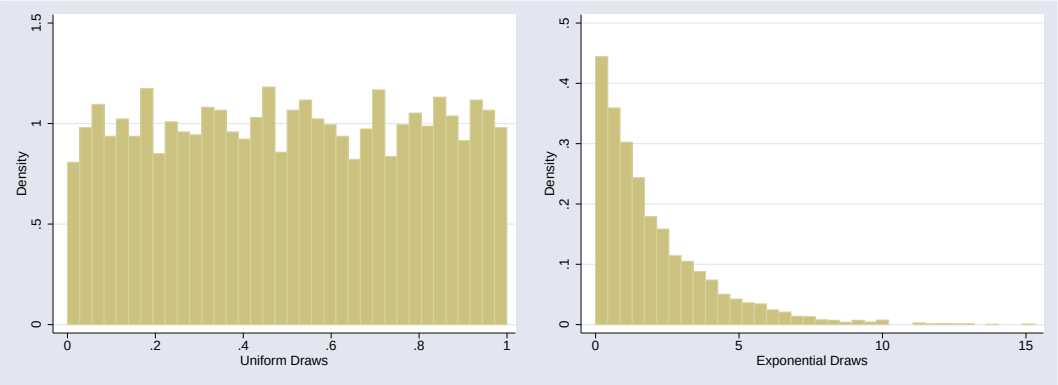


Figure 1: Histogram of 5,000 draws U_i from a Uniform (left), and the transformation $X_i = -2\log(1-U_i)$ (right)

If you want to try a few examples on your own in Excel, you can create uniform random draws using the RAND() function. You can then plot histograms by clicking yourself through the menus ("Tools">"Data Analysis">"Analysis Tools">"Histogram").

Gallery of Continuous Random Variables
Class 5, 18.05
Jeremy Orloff and Jonathan Bloom

1 Learning Goals

1. Be able to give examples of what uniform, exponential and normal distributions are used to model.
2. Be able to give the range and pdf's of uniform, exponential and normal distributions.

2 Introduction

Here we introduce a few fundamental continuous distributions. These will play important roles in the statistics part of the class. For each distribution, we give the range, the pdf, the cdf, and a short description of situations that it models. These distributions all depend on parameters, which we specify.

As you look through each distribution do not try to memorize all the details; you can always look those up. Rather, focus on the shape of each distribution and what it models.

Although it comes towards the end, we call your attention to the normal distribution. It is easily the most important distribution defined here.

2.1 Parametrized distributions

When we studied discrete random variables we learned, for example, about the Bernoulli(p) distribution. The probability p used to define the distribution is called a parameter and Bernoulli(p) is called a parametrized distribution. For example, tosses of fair coin follow a Bernoulli distribution where the parameter $p = 0.5$. When we study statistics one of the key questions will be to estimate the parameters of a distribution. For example, if I have a coin that may or may not be fair then I know it follows a Bernoulli(p) distribution, but I don't know the value of the parameter p. I might run experiments and use the data to estimate the value of p.

As another example, the binomial distribution Binomial(n, p) depends on two parameters n and p.

In the following sections we will look at specific parametrized continuous distributions. The applet https://mathlets.org/mathlets/probability-distributions/ allows you to visualize the pdf and cdf of these distributions and to dynamically change the parameters.

3 Uniform distribution

1. Parameters: a, b.
2. Range: $[a, b]$.

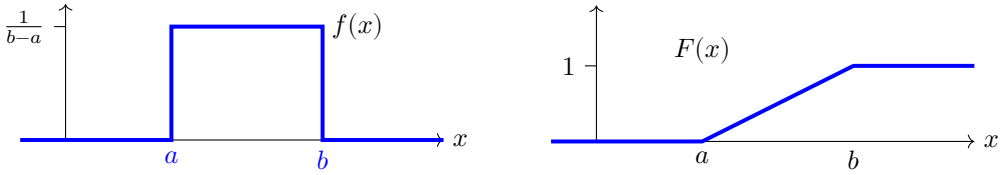
3. Notation: uniform(a, b) or U(a, b).

4. Probability density function (pdf): $f(x) = \dfrac{1}{b-a}$ for $a \le x \le b$.

5. Cumulative distribution function (cdf): $F(x) = (x-a)/(b-a)$ for $a \le x \le b$.

6. Models: Situtations where all outcomes in the range have equal probability (more precisely all outcomes have the same probability density).

Graphs:



pdf and cdf for uniform(a,b) distribution.

Example 1. 1. Suppose we have a tape measure with markings at each millimeter. If we measure (to the nearest marking) the length of items that are roughly a meter long, the rounding error will be uniformly distributed between -0.5 and 0.5 millimeters.

2. Many board games use spinning arrows (spinners) to introduce randomness. When spun, the arrow stops at an angle that is uniformly distributed between 0 and 2π radians.

3. In most pseudo-random number generators, the basic generator simulates a uniform distribution and all other distributions are constructed by transforming the basic generator.

4 Exponential distribution

1. Parameter: λ.

2. Range: $[0, \infty)$.

3. Notation: exponential(λ) or exp(λ).

4. Probability density function (pdf): $f(x) = \lambda e^{-\lambda x}$ for $0 \le x$.

5. Cumulative distribution function (cdf): (This is an easy integral.)

$$F(x) = 1 - e^{-\lambda x} \text{ for } x \ge 0$$

6. Right tail distribution: $P(X > x) = 1 - F(x) = e^{-\lambda x}$. (Note: this is defined as $P(X > x)$, i.e. that X is to the right of x on the number line.)

7. Models: The waiting time for a continuous process to change state.

Example 2. If I step out to 77 Mass Ave after class and wait for the next taxi, my waiting time in minutes is exponentially distributed. We will see that in this case λ is given by 1/(average number of taxis that pass per minute).

Example 3. The exponential distribution models the waiting time until an unstable isotope undergoes nuclear decay. In this case, the value of λ is related to the half-life of the isotope.

Memorylessness: There are other distributions that also model waiting times, but the exponential distribution has the additional property that it is memoryless. Here's what this means in the context of Example 2: suppose that the probability that a taxi arrives within the first five minutes is p. If I wait five minutes and, in this case, no taxi arrives, then the probability that a taxi arrives within the next five minutes is still p. That is, my previous wait of 5 minutes has no impact on the length of my future wait!

By contrast, suppose I were to instead go to Kendall Square subway station and wait for the next inbound train. Since the trains are coordinated to follow a schedule (e.g., roughly 12 minutes between trains), if I wait five minutes without seeing a train then there is a far greater probability that a train will arrive in the next five minutes. In particular, waiting time for the subway is not memoryless, and a better model would be the uniform distribution on the range [0,12].

The memorylessness of the exponential distribution is analogous to the memorylessness of the (discrete) geometric distribution, where having flipped 5 tails in a row gives no information about the next 5 flips. Indeed, the exponential distribution is precisely the continuous counterpart of the geometric distribution, which models the waiting time for a discrete process to change state. More formally, memoryless means that the probability of waiting t more minutes is independent of the amount of time already waited. In symbols,

$$P(X > s + t \,|\, X > s) = P(X > t).$$

Proof of memorylessness: We know that

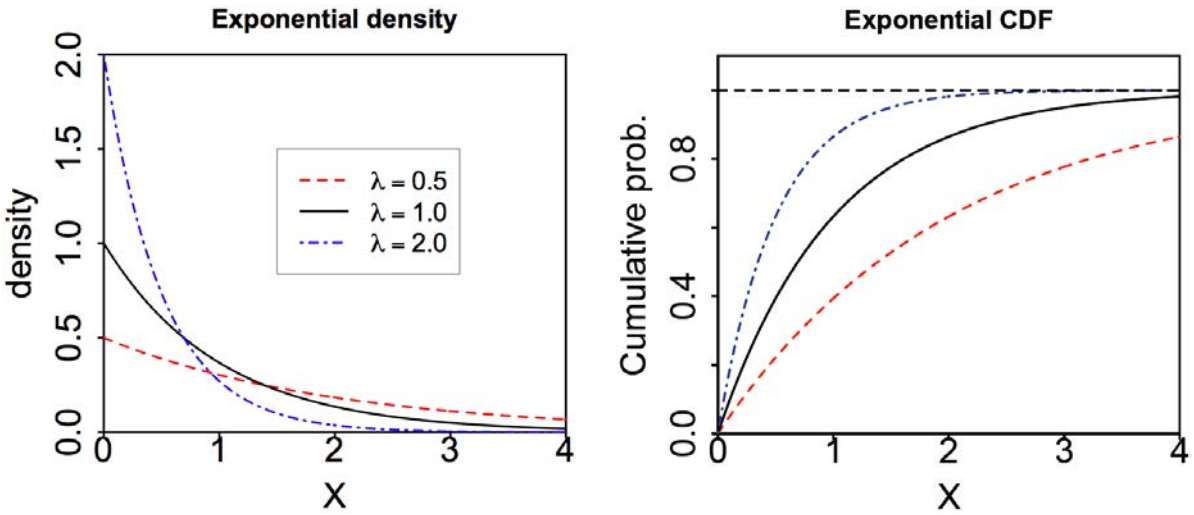
$$(X > s + t) \cap (X > s) = (X > s + t),$$

since the event 'waited at least s minutes' contains the event 'waited at least $s + t$ minutes'. Therefore the formula for conditional probability gives

$$P(X > s + t \,|\, X > s) = \frac{P(X > s + t)}{P(X > s)} = \frac{\mathrm{e}^{-\lambda(s+t)}}{\mathrm{e}^{-\lambda s}} = \mathrm{e}^{-\lambda t} = P(X > t).$$

The probability $P(X > s + t) = \mathrm{e}^{-\lambda(s+t)}$ is the formula for the right tail probability given above.

Graphs:



5 Normal distribution

In 1809, Carl Friedrich Gauss published a monograph introducing several notions that have become fundamental to statistics: the normal distribution, maximum likelihood estimation, and the method of least squares (we will cover all three in this course). For this reason, the normal distribution is also called the Gaussian distribution, and it is by far the most important continuous distribution.

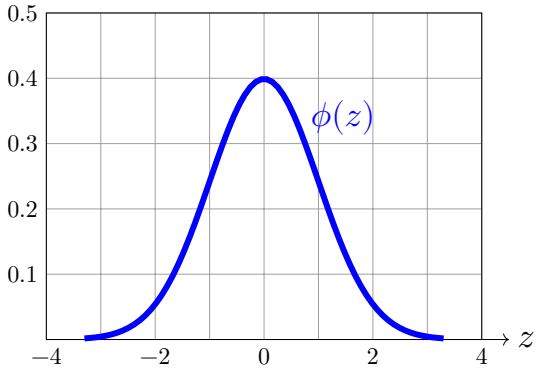
1. Parameters: μ, σ.
2. Range: $(-\infty, \infty)$.
3. Notation: Normal(μ, σ^2) or $N(\mu, \sigma^2)$.
4. Probability density function (pdf): $f(x) = \dfrac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$.
5. Cumulative distribution function (cdf): $F(x)$ has no formula, so use tables or software such as `pnorm` in R to compute $F(x)$.
6. Models: Measurement error, intelligence/ability, height, averages of lots of data.

The standard normal distribution $N(0, 1)$ has mean 0 and variance 1. We reserve Z for a standard normal random variable, $\phi(z) = \dfrac{1}{\sqrt{2\pi}} e^{-z^2/2}$ for the standard normal density, and $\Phi(z)$ for the standard normal distribution.

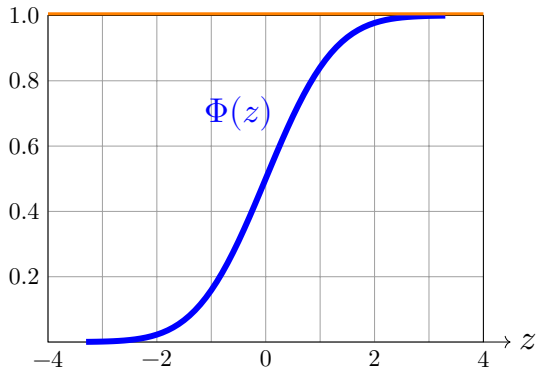
Note: we will define mean and variance for continuous random variables next time. They have the same interpretations as in the discrete case. As you might guess, the normal distribution $N(\mu, \sigma^2)$ has mean μ, variance σ^2, and standard deviation σ.

Here are some graphs of normal distributions. Note that they are shaped like a bell curve. Note also that as σ increases they become more spread out.

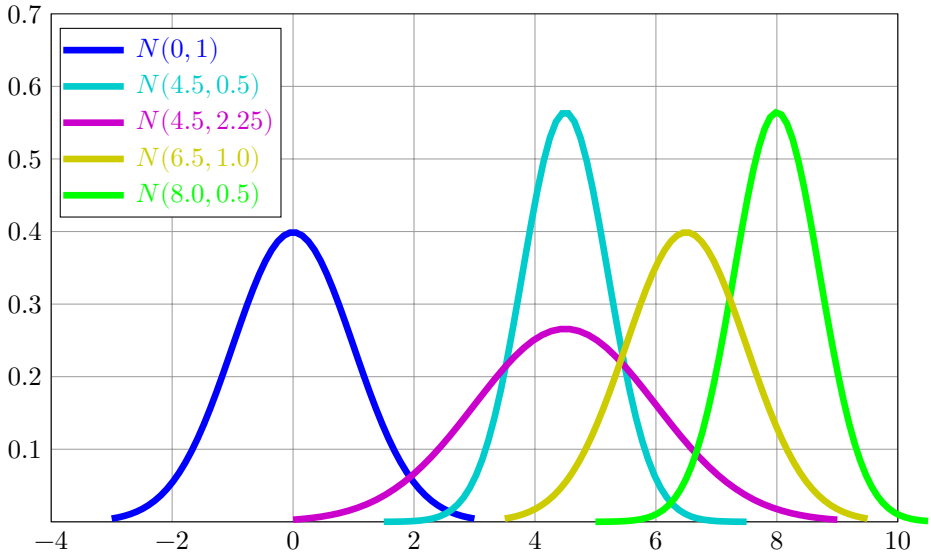
The bell curve: First we show the standard normal probability density and cumulative distribution functions. Below that is a selection of normal densities. Notice that the graph is centered on the mean and the bigger the variance the more spread out the curve.



Standard normal pdf



Standard normal cdf



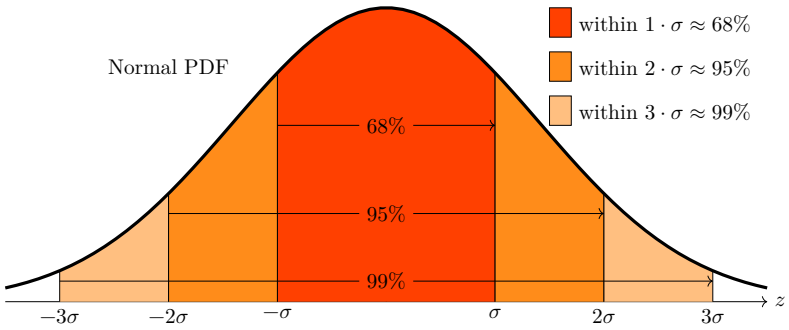
Notation note. In the figure above we use our notation $N(\mu, \sigma^2)$. So, for example, $N(8, 0.5)$ has variance 0.5 and standard deviation $\sigma = \sqrt{0.5} \approx 0.7071$.

5.1 Normal probabilities

To make approximations it is useful to remember the following rule of thumb for three approximate probabilities from the standard normal distribution:

$$P(-1 \le Z \le 1) \approx 0.68, \qquad P(-2 \le Z \le 2) \approx 0.95, \qquad P(-3 \le Z \le 3) \approx 0.99.$$

The figure below shows these probabilities as areas under the graph of the standard normal pdf $\phi(z)$.

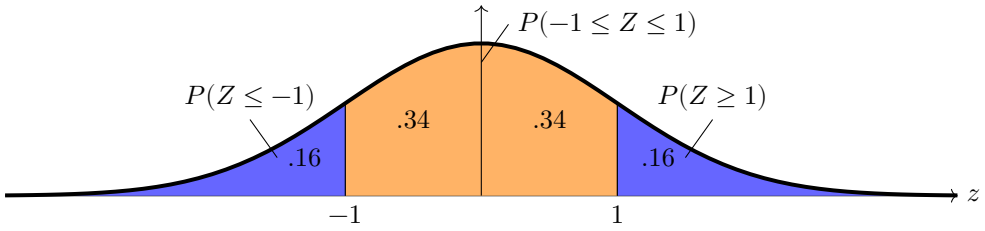


Symmetry calculations

We can use the symmetry of the standard normal distribution about $z = 0$ to make some calculations.

Example 4. The rule of thumb says $P(-1 \le Z \le 1) \approx 0.68$. Use this to estimate $\Phi(1)$.

Solution: $\Phi(1) = P(Z \le 1)$. In the figure, the two tails (in blue) have combined area $1 - 0.68 = 0.32$. By symmetry the left tail has area 0.16 (half of 0.32), so $P(Z \le 1) \approx 0.68 + 0.16 = 0.84$.



5.2 Using R to compute the standard normal cdf

```
# Use the R function pnorm(x,μ,σ) to compute F(x) for N(μ,σ^2)
pnorm(1,0,1)
[1] 0.8413447

pnorm(0,0,1)
[1] 0.5

pnorm(1,0,2)
[1] 0.6914625

pnorm(1,0,1) - pnorm(-1,0,1)
[1] 0.6826895

pnorm(5,0,5) - pnorm(-5,0,5)
[1] 0.6826895

# Of course z can be a vector of values
pnorm(c(-3,-2,-1,0,1,2,3),0,1)
[1] 0.001349898 0.022750132 0.158655254 0.500000000 0.841344746 0.977249868 0.998650102
```

Note: The R function pnorm(x, μ, σ) uses σ whereas our notation for the normal distribution $\mathrm{N}(\mu, \sigma^2)$ uses σ^2.

Here's a table of values with fewer decimal points of accuracy

z:	-2	-1	0	0.3	0.5	1	2	3
$\Phi(z)$:	0.0228	0.1587	0.5000	0.6179	0.6915	0.8413	0.9772	0.9987

Example 5. Use R to compute $P(-1.5 \le Z \le 2)$.

Solution: This is $\Phi(2) - \Phi(-1.5) =$ `pnorm(2,0,1) - pnorm(-1.5,0,1) = 0.91044`

6 Pareto and other distributions

In 18.05, we only have time to work with a few of the many wonderful distributions that are used in probability and statistics. We hope that after this course you will feel comfortable learning about new distributions and their properties when you need them. Wikipedia is often a great starting point.

The Pareto distribution is one common, beautiful distribution that we will not have time to cover in depth.

1. Parameters: $m > 0$ and $\alpha > 0$.
2. Range: $[m, \infty)$.
3. Notation: Pareto(m, α).
4. Density: $f(x) = \dfrac{\alpha\, m^\alpha}{x^{\alpha+1}}$.
5. Distribution: (easy integral)

$$F(x) = 1 - \frac{m^\alpha}{x^\alpha}, \text{ for } x \ge m$$

6. Tail distribution: $P(X > x) = m^\alpha / x^\alpha$, for $x \ge m$.
7. Models: The Pareto distribution models a power law, where the probability that an event occurs varies as a power of some attribute of the event. Many phenomena follow a power law, such as the size of meteors, income levels across a population, and population levels across cities. See Wikipedia for loads of examples:

   https://en.wikipedia.org/wiki/Pareto_distribution#Applications

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Joint Distributions, Independence
Class 7, 18.05
Jeremy Orloff and Jonathan Bloom

1 Learning Goals

1. Understand what is meant by a joint pmf, pdf and cdf of two random variables.
2. Be able to compute probabilities and marginals from a joint pmf or pdf.
3. Be able to test whether two random variables are independent.

2 Introduction

In science and in real life, we are often interested in two (or more) random variables at the same time. For example, we might measure the height and weight of giraffes, or the IQ and birthweight of children, or the frequency of exercise and the rate of heart disease in adults, or the level of air pollution and rate of respiratory illness in cities, or the number of Facebook friends and the age of Facebook members.

Think: What relationship would you expect in each of the five examples above? Why?

In such situations the random variables have a joint distribution that allows us to compute probabilities of events involving both variables and understand the relationship between the variables. This is simplest when the variables are independent. When they are not, we use covariance and correlation as measures of the nature of the dependence between them.

3 Joint Distribution

3.1 Discrete case

Suppose X and Y are two discrete random variables and that X takes values $\{x_1, x_2, \dots, x_n\}$ and Y takes values $\{y_1, y_2, \dots, y_m\}$. The ordered pair (X, Y) take values in the product $\{(x_1, y_1), (x_1, y_2), \dots (x_n, y_m)\}$. The joint probability mass function (joint pmf) of X and Y is the function $p(x_i, y_j)$ giving the probability of the joint outcome $X = x_i$, $Y = y_j$.

We organize this in a joint probability table as shown:

$X\backslash Y$	y_1	y_2	...	y_j	...	y_m
x_1	$p(x_1,y_1)$	$p(x_1,y_2)$	$\cdots$	$p(x_1,y_j)$	$\cdots$	$p(x_1,y_m)$
x_2	$p(x_2,y_1)$	$p(x_2,y_2)$	$\cdots$	$p(x_2,y_j)$	$\cdots$	$p(x_2,y_m)$
...	...	...	...	...	...	...
...	...	...	...	...	...	...
x_i	$p(x_i,y_1)$	$p(x_i,y_2)$	$\cdots$	$p(x_i,y_j)$	$\cdots$	$p(x_i,y_m)$
...	...	...	...	...	...	
x_n	$p(x_n,y_1)$	$p(x_n,y_2)$	$\cdots$	$p(x_n,y_j)$	$\cdots$	$p(x_n,y_m)$

Example 1. Roll two dice. Let X be the value on the first die and let Y be the value on the second die. Then both X and Y take values 1 to 6 and the joint pmf is $p(i,j) = 1/36$ for all i and j between 1 and 6. Here is the joint probability table:

$X\backslash Y$	1	2	3	4	5	6
1	1/36	1/36	1/36	1/36	1/36	1/36
2	1/36	1/36	1/36	1/36	1/36	1/36
3	1/36	1/36	1/36	1/36	1/36	1/36
4	1/36	1/36	1/36	1/36	1/36	1/36
5	1/36	1/36	1/36	1/36	1/36	1/36
6	1/36	1/36	1/36	1/36	1/36	1/36

Example 2. Roll two dice. Let X be the value on the first die and let T be the total on both dice. Here is the joint probability table:

$X\backslash T$	2	3	4	5	6	7	8	9	10	11	12
1	1/36	1/36	1/36	1/36	1/36	1/36	0	0	0	0	0
2	0	1/36	1/36	1/36	1/36	1/36	1/36	0	0	0	0
3	0	0	1/36	1/36	1/36	1/36	1/36	1/36	0	0	0
4	0	0	0	1/36	1/36	1/36	1/36	1/36	1/36	0	0
5	0	0	0	0	1/36	1/36	1/36	1/36	1/36	1/36	0
6	0	0	0	0	0	1/36	1/36	1/36	1/36	1/36	1/36

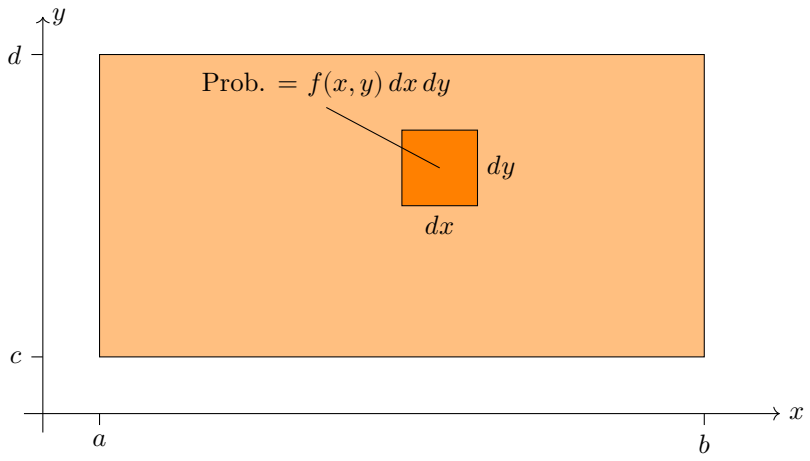
A joint probability mass function must satisfy two properties:
1. $0 \le p(x_i, y_j) \le 1$
2. The total probability is 1. We can express this as a double sum:

$$\sum_{i=1}^{n}\sum_{j=1}^{m} p(x_i,y_j) = 1$$

3.2 Continuous case

The continuous case is essentially the same as the discrete case: we just replace discrete sets of values by continuous intervals, the joint probability mass function by a joint probability density function, and the sums by integrals.

If X takes values in $[a, b]$ and Y takes values in $[c, d]$ then the pair (X, Y) takes values in the product $[a, b] \times [c, d]$. The joint probability density function (joint pdf) of X and Y is a function $f(x, y)$ giving the probability density at (x, y). That is, the probability that (X, Y) is in a small rectangle of width dx and height dy around (x, y) is $f(x, y)\, dx\, dy$.



A joint probability density function must satisfy two properties:
1. $0 \le f(x, y)$
2. The total probability is 1. We now express this as a double integral:

$$\int_c^d \int_a^b f(x, y)\, dx\, dy = 1$$

Note: as with the pdf of a single random variable, the joint pdf $f(x, y)$ can take values greater than 1; it is a probability density, **not** a probability.

In 18.05 we won't expect you to be experts at double integration. Here's what we will expect.

- You should understand double integrals conceptually as double sums.
- You should be able to compute double integrals over rectangles.
- For a non-rectangular region, when $f(x, y) = c$ is constant, you should know that the double integral is the same as the $c \times$ (the area of the region).

3.3 Events

Random variables are useful for describing events. Recall that an event is a set of outcomes and that random variables assign numbers to outcomes. For example, the event '$X > 1$' is the set of all outcomes for which X is greater than 1. These concepts readily extend to pairs of random variables and joint outcomes.

Example 3. In Example 1, describe the event $B =$ '$Y - X \geq 2$' and find its probability.

Solution: We can describe B as a set of (X, Y) pairs:

$$B = \{(1,3), (1,4), (1,5), (1,6), (2,4), (2,5), (2,6), (3,5), (3,6), (4,6)\}.$$

We can also describe it visually

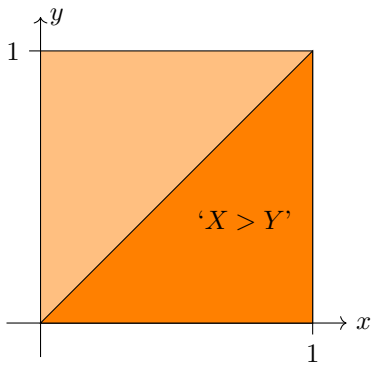
$X\backslash Y$	1	2	3	4	5	6
1	1/36	1/36	1/36	1/36	1/36	1/36
2	1/36	1/36	1/36	1/36	1/36	1/36
3	1/36	1/36	1/36	1/36	1/36	1/36
4	1/36	1/36	1/36	1/36	1/36	1/36
5	1/36	1/36	1/36	1/36	1/36	1/36
6	1/36	1/36	1/36	1/36	1/36	1/36

The event B consists of the outcomes in the shaded squares.

The probability of B is the sum of the probabilities in the orange shaded squares, so $P(B) = 10/36$.

Example 4. Suppose X and Y both take values in [0,1] with uniform density $f(x, y) = 1$. Visualize the event '$X > Y$' and find its probability.

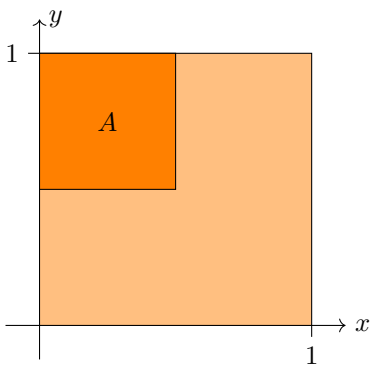
Solution: Jointly X and Y take values in the unit square. The event '$X > Y$' corresponds to the shaded lower-right triangle below. Since the density is constant, the probability is just the fraction of the total area taken up by the event. In this case, it is clearly 0.5.



The event '$X > Y$' in the unit square.

Example 5. Suppose X and Y both take values in [0,1] with density $f(x, y) = 4xy$. Show $f(x, y)$ is a valid joint pdf, visualize the event $A =$ '$X < 0.5$ and $Y > 0.5$' and find its probability.

Solution: Jointly X and Y take values in the unit square.



The event A in the unit square.

To show $f(x,y)$ is a valid joint pdf we must check that it is positive (which it clearly is) and that the total probability is 1.

$$\text{Total probability} = \int_0^1 \int_0^1 4xy\,dx\,dy = \int_0^1 \left[2x^2y\right]_0^1 dy = \int_0^1 2y\,dy = 1. \quad \text{QED}$$

The event A is just the upper-left-hand quadrant. Because the density is not constant we must compute an integral to find the probability.

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy\,dy\,dx = \int_0^{0.5} \left[2xy^2\right]_{0.5}^1 dx = \int_0^{0.5} \frac{3x}{2}\,dx = \boxed{\frac{3}{16}}.$$

3.4 Joint cumulative distribution function

Suppose X and Y are jointly-distributed random variables. We will use the notation '$X \le x$, $Y \le y$' to mean the event '$X \le x$ and $Y \le y$'. The joint cumulative distribution function (joint cdf) is defined as

$$F(x,y) = P(X \le x,\, Y \le y)$$

Continuous case: If X and Y are continuous random variables with joint density $f(x,y)$ over the range $[a,b] \times [c,d]$ then the joint cdf is given by the double integral

$$F(x,y) = \int_c^y \int_a^x f(u,v)\,du\,dv.$$

To recover the joint pdf, we differentiate the joint cdf. Because there are two variables we need to use partial derivatives:

$$f(x,y) = \frac{\partial^2 F}{\partial x \partial y}(x,y).$$

Discrete case: If X and Y are discrete random variables with joint pmf $p(x_i, y_j)$ then the joint cdf is give by the double sum

$$F(x,y) = \sum_{x_i \le x} \sum_{y_j \le y} p(x_i, y_j).$$

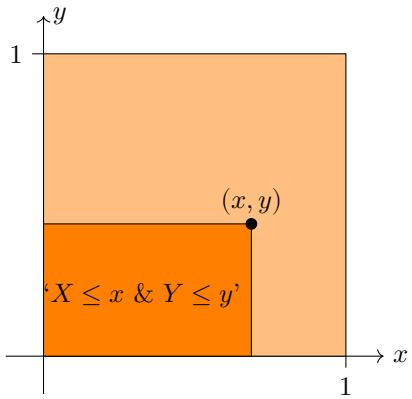
3.5 Properties of the joint cdf

The joint cdf $F(x, y)$ of X and Y must satisfy several properties:

1. $F(x, y)$ is non-decreasing: i.e. if x or y increase then $F(x, y)$ must stay constant or increase.
2. $F(x, y) = 0$ at the lower-left of the joint range.
   If the lower left is $(-\infty, -\infty)$ then this means $\lim_{(x,y)\to(-\infty,-\infty)} F(x, y) = 0$.
3. $F(x, y) = 1$ at the upper-right of the joint range.
   If the upper-right is (∞, ∞) then this means $\lim_{(x,y)\to(\infty,\infty)} F(x, y) = 1$.

Example 6. Find the joint cdf for the random variables in Example 5.

Solution: The event '$X \le x$ and $Y \le y$' is a rectangle in the unit square.



To find the cdf $F(x, y)$ we compute a double integral:

$$F(x, y) = \int_0^y \int_0^x 4uv \, du \, dv = \boxed{x^2y^2}.$$

Example 7. In Example 1, compute $F(3.5, 4)$.

Solution: We redraw the joint probability table. Notice how similar the picture is to the one in the previous example.

$F(3.5, 4)$ is the probability of the event '$X \le 3.5$ and $Y \le 4$'. We can visualize this event as the shaded rectangles in the table:

$X\backslash Y$	1	2	3	4	5	6
1	1/36	1/36	1/36	1/36	1/36	1/36
2	1/36	1/36	1/36	1/36	1/36	1/36
3	1/36	1/36	1/36	1/36	1/36	1/36
4	1/36	1/36	1/36	1/36	1/36	1/36
5	1/36	1/36	1/36	1/36	1/36	1/36
6	1/36	1/36	1/36	1/36	1/36	1/36

The event '$X \le 3.5$ and $Y \le 4$'.

Adding up the probability in the shaded squares we get $F(3.5, 4) = 12/36 = 1/3$.

Note. One unfortunate difference between the continuous and discrete visualizations is that for continuous variables the value increases as we go up in the vertical direction while the opposite is true for the discrete case. We have experimented with changing the discrete tables to match the continuous graphs, but it causes too much confusion. We will just have to live with the difference!

3.6 Marginal distributions

When X and Y are jointly-distributed random variables, we may want to consider only one of them, say X. In that case we need to find the pmf (or pdf or cdf) of X without Y. This is called a marginal pmf of the joint pmf (or pdf or cdf). The next example illustrates the way to compute this and the reason for the term 'marginal'.

3.7 Marginal pmf

Example 8. In Example 2 we rolled two dice and let X be the value on the first die and T be the total on both dice. Compute the marginal pmf for X and for T.

Solution: In the table each row represents a single value of X. So the event '$X = 3$' is the third row of the table. To find $P(X = 3)$ we simply have to sum up the probabilities in this row. We put the sum in the right-hand margin of the table. Likewise $P(T = 5)$ is just the sum of the column with $T = 5$. We put the sum in the bottom margin of the table.

$X\backslash T$	2	3	4	5	6	7	8	9	10	11	12	$p(x_i)$
1	1/36	1/36	1/36	1/36	1/36	1/36	0	0	0	0	0	1/6
2	0	1/36	1/36	1/36	1/36	1/36	1/36	0	0	0	0	1/6
3	0	0	1/36	1/36	1/36	1/36	1/36	1/36	0	0	0	1/6
4	0	0	0	1/36	1/36	1/36	1/36	1/36	1/36	0	0	1/6
5	0	0	0	0	1/36	1/36	1/36	1/36	1/36	1/36	0	1/6
6	0	0	0	0	0	1/36	1/36	1/36	1/36	1/36	1/36	1/6
$p(t_j)$	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36	1

Computing the marginal probabilities $P(X = 3) = 1/6$ and $P(T = 5) = 4/36$.

Note: Of course in this case we already knew the pmf of X and of T. It is good to see that our computation here is in agreement!

As motivated by this example, marginal pmfs are obtained from the joint pmf by summing:

$$p_X(x_i) = \sum_j p(x_i, y_j), \qquad p_Y(y_j) = \sum_i p(x_i, y_j)$$

The term marginal refers to the fact that the values are written in the margins of the table.

3.8 Marginal pdf

For a continous joint density $f(x, y)$ with range $[a, b] \times [c, d]$, the marginal pdfs are:

$$f_X(x) = \int_c^d f(x, y)\, dy, \qquad f_Y(y) = \int_a^b f(x, y)\, dx.$$

Compare these with the marginal pmfs above; as usual the sums are replaced by integrals. We say that to obtain the marginal for X, we integrate out Y from the joint pdf and vice versa.

Example 9. Suppose (X, Y) takes values on the square $[0, 1] \times [1, 2]$ with joint pdf $f(x, y) = \frac{8}{3}x^3 y$. Find the marginal pdfs $f_X(x)$ and $f_Y(y)$.

Solution: To find $f_X(x)$ we integrate out y and to find $f_Y(y)$ we integrate out x.

$$f_X(x) = \int_1^2 \frac{8}{3}x^3 y\, dy = \left[\frac{4}{3}x^3 y^2\right]_1^2 = \boxed{4x^3}$$

$$f_Y(y) = \int_0^1 \frac{8}{3}x^3 y\, dx = \left[\frac{2}{3}x^4 y^1\right]_0^1 = \boxed{\frac{2}{3}y}.$$

Example 10. Suppose (X, Y) takes values on the unit square $[0, 1] \times [0, 1]$ with joint pdf $f(x, y) = \frac{3}{2}(x^2 + y^2)$. Find the marginal pdf $f_X(x)$ and use it to find $P(X < 0.5)$.

Solution:

$$f_X(x) = \int_0^1 \frac{3}{2}(x^2 + y^2)\, dy = \left[\frac{3}{2}x^2 y + \frac{y^3}{2}\right]_0^1 = \boxed{\frac{3}{2}x^2 + \frac{1}{2}}.$$

$$P(X < 0.5) = \int_0^{0.5} f_X(x)\, dx = \int_0^{0.5} \frac{3}{2}x^2 + \frac{1}{2}\, dx = \left[\frac{1}{2}x^3 + \frac{1}{2}x\right]_0^{0.5} = \boxed{\frac{5}{16}}.$$

3.9 Marginal cdf

Finding the marginal cdf from the joint cdf is easy. If X and Y jointly take values on $[a, b] \times [c, d]$ then

$$F_X(x) = F(x, d), \qquad F_Y(y) = F(b, y).$$

If d is ∞ then this becomes a limit $F_X(x) = \lim_{y \to \infty} F(x, y)$. Likewise for $F_Y(y)$.

Example 11. The joint cdf in the last example was $F(x, y) = \frac{1}{2}(x^3 y + x y^3)$ on $[0, 1] \times [0, 1]$. Find the marginal cdfs and use $F_X(x)$ to compute $P(X < 0.5)$.

Solution: We have $F_X(x) = F(x, 1) = \frac{1}{2}(x^3 + x)$ and $F_Y(y) = F(1, y) = \frac{1}{2}(y + y^3)$. So $P(X < 0.5) = F_X(0.5) = \frac{1}{2}(0.5^3 + 0.5) = \frac{5}{16}$: exactly the same as before.

3.10 3D visualization

We visualized $P(a < X < b)$ as the area under the pdf f(x) over the interval $[a, b]$. Since the range of values of (X, Y) is already a two dimensional region in the plane, the graph of

$f(x, y)$ is a surface over that region. We can then visualize probability as volume under the surface.

Think: Summoning your inner artist, sketch the graph of the joint pdf $f(x, y) = 4xy$ and visualize the probability $P(A)$ as a volume for Example 5.

4 Independence

We are now ready to give a careful mathematical definition of independence. Of course, it will simply capture the notion of independence we have been using up to now. But, it is nice to finally have a solid definition that can support complicated probabilistic and statistical investigations.

Recall that events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

Random variables X and Y define events like '$X \leq 2$' and '$Y > 5$'. So, X and Y are independent if any event defined by X is independent of any event defined by Y. The formal definition that guarantees this is the following.

Definition: Jointly-distributed random variables X and Y are independent if their joint cdf is the product of the marginal cdfs:

$$F(X, Y) = F_X(x)F_Y(y).$$

For discrete variables this is equivalent to the joint pmf being the product of the marginal pmfs.:

$$p(x_i, y_j) = p_X(x_i)p_Y(y_j).$$

For continous variables this is equivalent to the joint pdf being the product of the marginal pdfs.:

$$f(x, y) = f_X(x)f_Y(y).$$

Once you have the joint distribution, checking for independence is usually straightforward although it can be tedious.

Example 12. For discrete variables independence means the probability in a cell must be the product of the marginal probabilities of its row and column. In the first table below this is true: every marginal probability is 1/6 and every cell contains 1/36, i.e. the product of the marginals. Therefore X and Y are independent.

In the second table below most of the cell probabilities are not the product of the marginal probabilities. For example, none of marginal probabilities are 0, so none of the cells with 0 probability can be the product of the marginals.

$X\backslash Y$	1	2	3	4	5	6	$p(x_i)$
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	1/36	1/36	1/36	1/36	1/36	1/6
3	1/36	1/36	1/36	1/36	1/36	1/36	1/6
4	1/36	1/36	1/36	1/36	1/36	1/36	1/6
5	1/36	1/36	1/36	1/36	1/36	1/36	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
$p(y_j)$	1/6	1/6	1/6	1/6	1/6	1/6	1

$X\backslash T$	2	3	4	5	6	7	8	9	10	11	12	$p(x_i)$
1	1/36	1/36	1/36	1/36	1/36	1/36	0	0	0	0	0	1/6
2	0	1/36	1/36	1/36	1/36	1/36	1/36	0	0	0	0	1/6
3	0	0	1/36	1/36	1/36	1/36	1/36	1/36	0	0	0	1/6
4	0	0	0	1/36	1/36	1/36	1/36	1/36	1/36	0	0	1/6
5	0	0	0	0	1/36	1/36	1/36	1/36	1/36	1/36	0	1/6
6	0	0	0	0	0	1/36	1/36	1/36	1/36	1/36	1/36	1/6
$p(y_j)$	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36	1

Example 13. For continuous variables independence means you can factor the joint pdf or cdf as the product of a function of x and a function of y.

(i) Suppose X has range $[0, 1/2]$, Y has range $[0, 1]$ and $f(x, y) = 96x^2y^3$ then X and Y are independent. The marginal densities are $f_X(x) = 24x^2$ and $f_Y(y) = 4y^3$.

(ii) If $f(x, y) = 1.5(x^2+y^2)$ over the unit square then X and Y are not independent because there is no way to factor $f(x, y)$ into a product $f_X(x)f_Y(y)$.

(iii) If $F(x, y) = \frac{1}{2}(x^3y + xy^3)$ over the unit square then X and Y are not independent because the cdf does not factor into a product $F_X(x)F_Y(y)$.

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The Binomial Distribution



Summer 2003

Internet Bubble

"Several industry experts believe that 30% of internet companies will run out of cash in 6 months and that these companies will find it very hard to secure additional funding" Association for Computing Machinery, August, 2001

- Pick 10 companies at random. How many would we expect to run out of cash in 6 months?
- What is p(4 run out of cash in 6 months)?
- What is p(2 run out…)?
- What is p(all 10 run out…)?

Binomial Distribution

Derived from theory, not from experience

- An experiment consists of *n "trials"*
- Each trial results in : *yes* or *no* (*"binomial"* means *"2 names" or "2 labels")*
- Trials are independent of each other
- Each trial has same probability: success p, failure 1-p

r.v. X = # successes in n trials

Random variable X has binomial distribution with parameters n and p
"Binomial(n,p)"

An Example "Experiment"

Select, randomly, one team of 3 students from a large pool of students composed of 70% US, and 30% International.

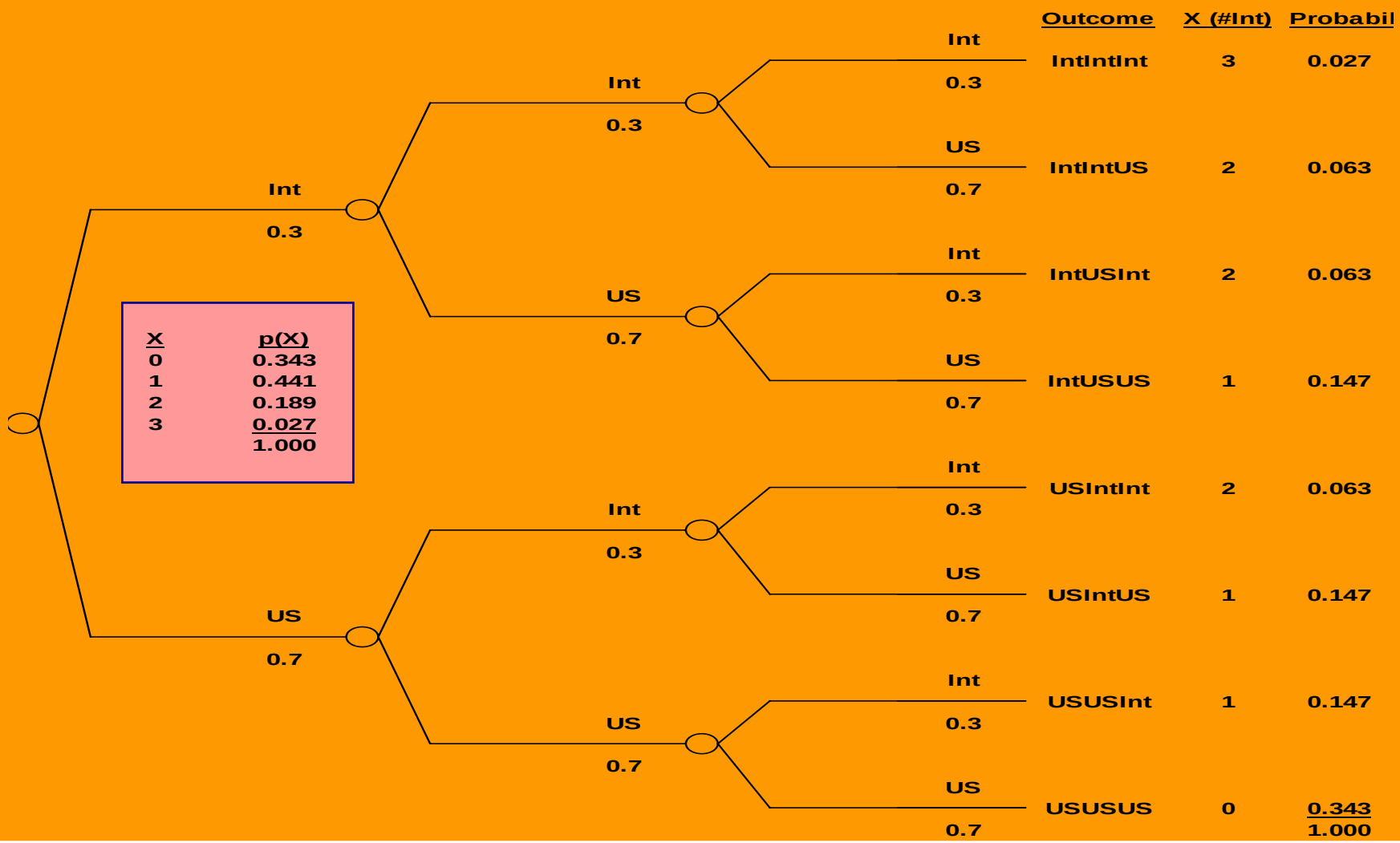
r.v. X = number of international students on the three student team selected randomly.

- n = 3 trials
- define: "yes": picking an international student
  "no": picking a US student
- P(yes) = p = 0.30; P(no) = 1 - p = 0.70

X is a Binomial (n=3 , p=0.30)

(compare to experiment 2 in which we flipped three coins with p(H) = 0.3 and looked at the total # of heads)

Probability tree and probability distribution for r.v. X (total # Int students)



Binomial Probability Distribution Formula

Let X be Binomial(n, p):

The probability of having x "yes" events (also called "successes" in n trials is

$$P(X=x) = \frac{n!}{x!(n-x)!} \; p^x (1-p)^{n-x}$$

(where x! = x(x-1)(x-2)...1, and 0! = 1)

Check in Example 1: n=3 , p=0.3

$$P(X=1) = \frac{3!}{1!/(3-1)!} (0.3)(0.7)^2 = 0.441$$

Internet Bubble Again

- What is the p(4 of 10 run out of cash)?
- $P(X=x) = \frac{n!}{x!(n-x)!} p^x(1-p)^{n-x}$
- $P(X=4) = \frac{10!}{4!6!} .3^4 (.7^6) = .20$
- P(X=10) = ?

Excel Calculates Binomial Probabilities

Use the function wizard icon or the Insert_Function Command to choose the Statistical function BINOMDIST. This function calculates either the individual binomial probability P(X=Value) or the cumulative binomial probability P(X<=Value).

Mean, Variance and SD for Binomial Probability Distribution (r.v.)

Let X be Binomial(n, p):

The probability of having x successes in n trials is:

$$P(X=x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}$$

(where x! = x(x-1)(x-2)...1, and 0! = 1)

$$E(X) = np = 3 * 0.3 = 0.9$$

$$VAR(X) = np(1-p) = 3 * 0.3 * 0.7 = 0.63$$

$$SD(X) = \sqrt{n\ p\ (1\ -\ p)} = 0.794$$

Calculations shown for Binomial (n=3, p=0.3)

Note: this is equivalent to counting success = 1 and failure = 0

Another example:

A laser production facility is known to have a 75% yield; that is, 75% of the lasers manufactured by the facility pass the quality test. Suppose that today the facility is scheduled to produce 15 lasers.

a) What is the expected number of lasers to pass the test (the mean)?
b) What is the variance?
c) What is the standard deviation?
d) What is the probability that at least 14 lasers will pass the quality test?

Laser Production

- What is the mean?

  E(X) = np = 15*.75 = 11.25 lasers
- What is the variance?

  Var(X) = np(1-p) = 15*.75*.25 = 2.812
- What is the standard deviation?

  SD(X) = sqrt (Variance) = 1.677 lasers
- What is the p of 14 or more passing?

  P(14 or more) = p(14) + p(15) = .067 + .013 = .08

Acquiring A Company Decision

- You represent Company A (the Acquirer), which is considering acquiring Company T (the Target) by means of a cash offer for 100% of Company T shares. You are unsure what price to offer because the value of Company T depends on the outcome of a major oil exploration project. If the project fails, Company T will be worth nothing, but if it succeeds, shares of Company T could be worth up to $100. All share values between $0 and $100 are considered equally likely. By all estimates, whatever the value of Company T under current management, it will be worth 50% more under the management of Company A. The Board of Directors wants your recommendation today for how much to offer per share for Company T. You do not know the outcome of the exploration project, however, you expect Company T to delay their decision about accepting your offer until after they know the results of the project. Company T will accept any offer above their true value and reject any offer below. There will be no negotiation or opportunity to make a second offer. What do you recommend as a take-it-or-leave-it price for Company T? $ _____ per share

Setting Up the Problem

- What is our goal? To maximize expected profit.
- What are the options? Make an offer of somewhere between $0 and $100/share.

If you have trouble setting up the continuous problem, try making it discrete, e.g., what if I offered $20, $40, $60, $80?

- What are the outcomes? Profit = Revenue – Costs. Revenue = 1.5 * Book Value. Costs = Offer. So Profit = 1.5 BV – O.
- What is E(BV)? And what are the probabilities?

Probabilities

- We are told there is a uniform distribution between $0 and $100. Therefore, is the expected value = (0 + 100)/2 = $50/share?
- No. The seller will not accept any offer less than the book value. Therefore, if my offer is accepted, the book value must be uniform between $0 and Offer.

  E(BV) is therefore (0 + Offer)/2 = .5 Offer !!
- And, Profit = 1.5 BV – Offer

  = 1.5*.5 Offer – Offer = -.25 Offer

Uniform Distributions



Conditional Probabilities

- The key to the Acquiring A Company decision is to realize that the seller is making a decision conditional on receiving an acceptable offer
- If you offer $60/share, the offer will be accepted only if the value is <= $60/share
- Your expected book value is therefore conditional on your offer! If you offer $60/share, the expected book value is not $50/share, but rather $30/share (all values between $0 and $60 equally likely)
- Your E(profit) is always <0, so don't bid !

Your Answers

- 37% answered $50/share, which is the EMV of T; ignores the conditional situation
- 19% bid between $51-$75 which tries to give each side some profit
- 9% gave the correct answer ($0, or don't make a bid)
- Even $1/share bid will lose an expected 25 cents per share (times millions of shares)

Next Lecture

Question: Given the choice to invest $1 million in these three firms, which one(s) would you choose?

Asset	Expected Return (% per year)	Risk (% per year)	Correlation		
			Exxon	Boeing	GM
Exxon	11.2%	13.9%	-----	0.41	0.13
Boeing	9.1	16.5	0.41	----	-0.22
GM	12.1	15.8	0.13	-0.22	----

More information would help

Question: How do we describe the relationship between two random variables? Covariance - Correlation

FINANCIAL ECONOMETRICS

MODULE 3 | LESSON 3

CORRELATION

Reading Time	60 minutes
Prior Knowledge	Basic Statistics
Keywords	Pearson Correlation, Spearman Correlation, Kendall's tau Correlation, Concordance/Discordance, Robust Measure, Agreement, Concordance Correlation, Monotonic relationship

In the last section, we discussed random variables and their probability distributions in a univariate setting. In this section, we are going to introduce some measurements to describe bivariate relationships (relationships for two random variables). Specifically, we are going to look into measures of correlation. The basics of correlation were covered in Module 1. In this lesson, we will go deeper and introduce various measures of correlation and explain why we use them.

In [1]:
```
# Load libraries
import matplotlib.pyplot as plt
import pandas as pd
from scipy.stats import kendalltau, pearsonr, spearmanr

plt.rcParams["figure.figsize"] = (12, 9)  # Figure size and width
```

In [2]:
```
# Download the dataset
data_set = pd.read_csv("M3. bond_and_stock_data.csv")
```

1. Pearson Correlation

The Pearson correlation is the most used correlation measure. The formula for Pearson Correlation is:

$$\rho_{X,Y} = \frac{cov(X,Y)}{\sigma_X \sigma_Y}$$

Where $cov(X,Y)$ is the covariance of X and Y, σ_X is the standard deviation of X and σ_Y is the standard deviation of Y

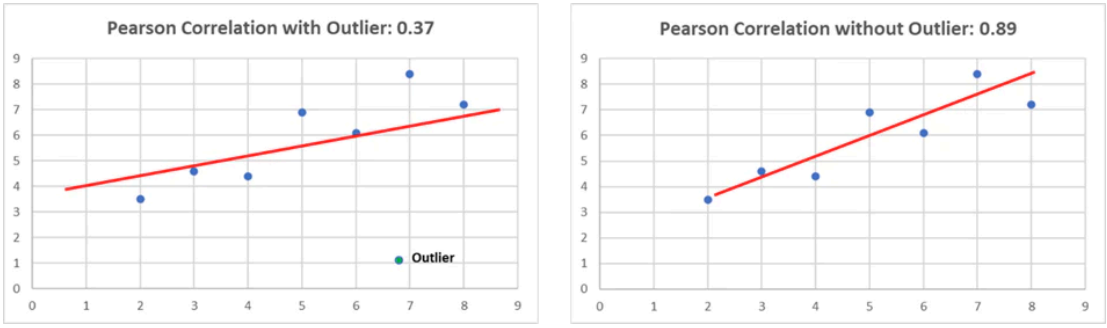
With some math, we can also write the Pearson correlation as follows:

$$\rho_{X,Y} = \frac{E(XY) - E(X)E(Y)}{\sqrt{E(X^2) - (E(X))^2}\sqrt{E(Y^2) - (E(Y))^2}}$$

Pearson correlation is a measure of the linear relationship that exists between two random variables. For example, we can calculate the Pearson correlation of hours of studying and test grades for 50 students to see if there is a linear correlation. In order to calculate Pearson correlation, you must have paired values from each observation. The value of Pearson Correlation is between -1 and 1. In general, if the absolute value of the Pearson correlation for two random variables is over 0.7, we would say the two variables are highly correlated. If the absolute value of the Pearson correlation for two random variables is between 0.5 and 0.7, we would say the two variables are moderately correlated. If the Pearson correlation is between 0.3 and 0.5, we would say the two variables are low correlated. However, we would say the two variables are not correlated when the Pearson correlation is below 0.3. Because the Pearson Correlation only measures the "linear relationship" of two variables, a low Pearson Correlation only says that the two variables are not linearly correlated, but they may be correlated in a non-linear way. The Pearson correlation also assumes that the two variables come from normal distribution and are continuous variables.

One problem with the Pearson correlation is that it can change significantly when outliers are present. A measure whose value does not change significantly with the existence of outliers is called a robust measure. For example, the median is a **robust measure**. In a dataset $\{2, 2, 2, 3, 5, 6, 6, 7, 8, 9, 10, 5000\}$, 5,000 is the outlier. The median is six if you keep the outlier. The median is still six if you drop the outlier. The story is different for the mean. The mean is 421.67 with the outlier. The mean is 5.45 without the outlier. The mean decreases significantly without the outlier while the median does not change without the outlier.

Figure 1: Pearson Correlation with Outlier and without Outlier



From figure 1, you can see that the Pearson Correlation changes significantly if there is an outlier in the dataset. The Pearson Correlation is not resistant to outliers. Due to this issue, we need to find more robust measures to calculate the correlation of two random variables when there are outliers in the dataset. In the following sections, we will introduce the Spearman correlation and Kendall correlation to handle the issue of outliers in calculating correlation.

2. Spearman Correlation

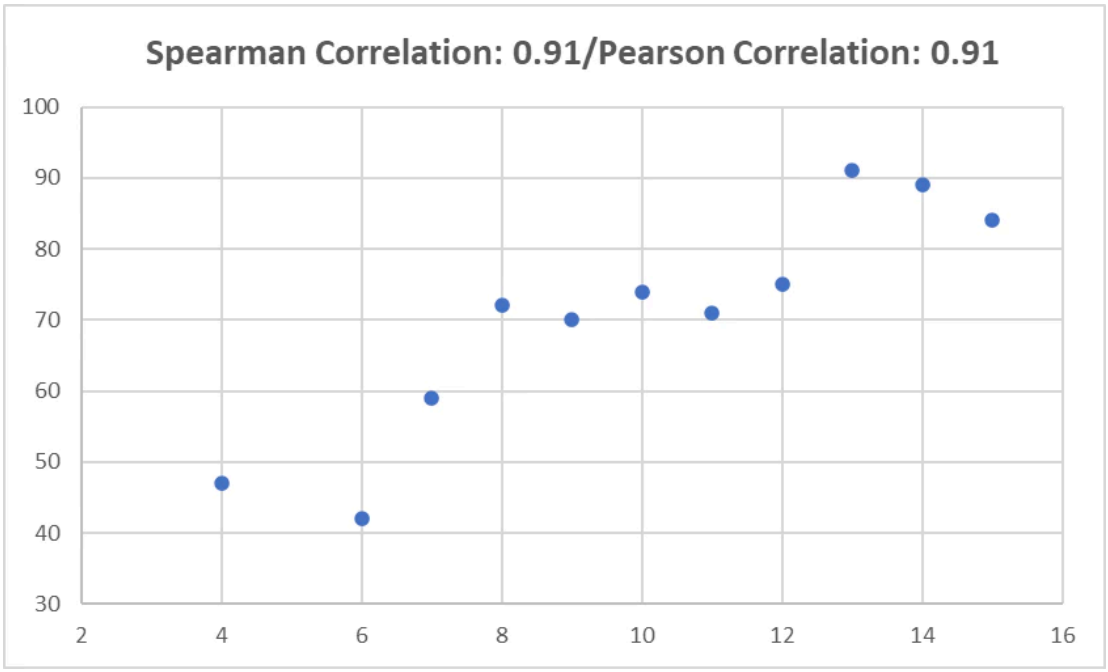
The Spearman correlation ρ is also a measure of the relationship between two random variables. However, instead of using the data values to calculate correlation, the Spearman correlation is a non-parametric method that uses ranks to measure correlation. Let's use the following example to demonstrate how the Spearman correlation is calculated.

Figure 2: Data Example for Calculating Spearman Correlation

Student	Hours of Study	Test Grade	Ranked Hours of Study	Ranked Test Grade
A	14	89	2	2
B	8	72	8	6
C	4	47	11	10
D	13	91	3	1
E	7	59	9	9
F	11	71	5	7
G	9	70	7	8
H	6	42	10	11
I	15	84	1	3
J	12	75	4	4
K	10	74	6	5

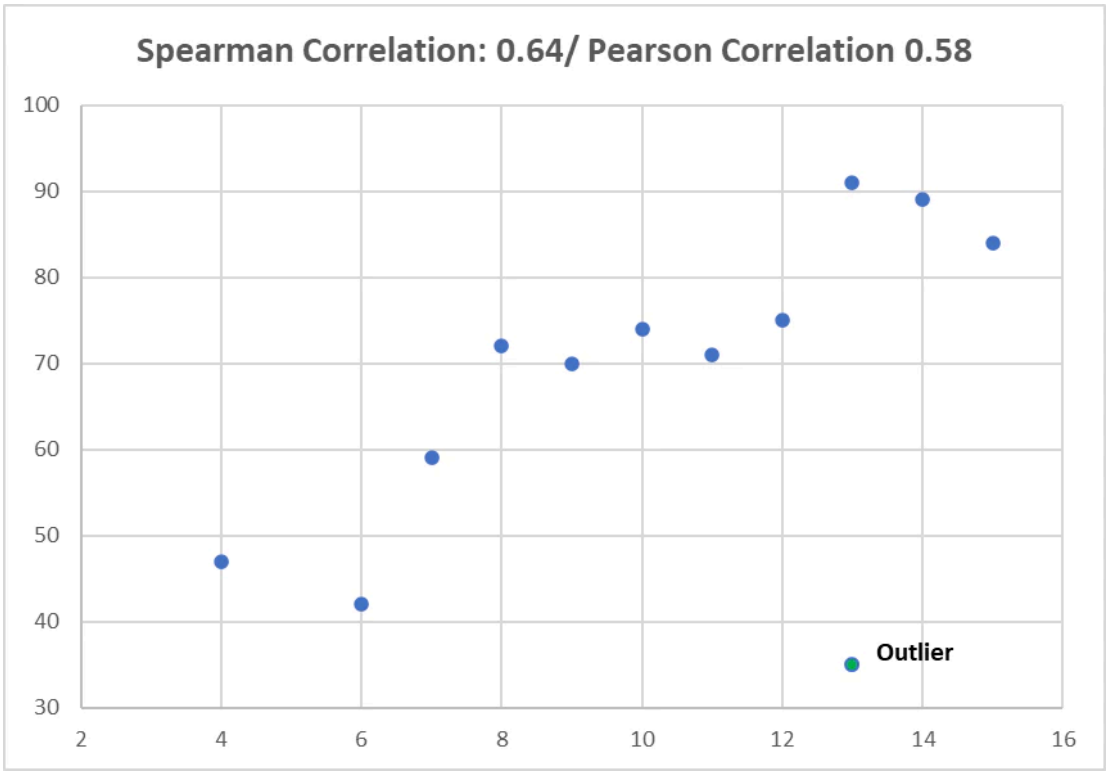
In figure 2, we have paired data from 15 students about how many days they study a subject and the test grade they received. To calculate the Pearson correlation, we use the "Hours of Study" column and "Test Grade" column. However, to calculate the Spearman correlation, we first need to rank each random variable. For the "Hours of Study" random variable, we rank the observations by giving 1 to the highest "Hours of Study" and 2 to the next highest "Hours of Study." Then, we do the same for the "Test Grade" random variable. From figure 2, we can see the two columns on the right are ranked "Hours of Study" and "Test Grade." We then apply these two columns to the Pearson correlation formula to get the Spearman correlation. Hence, we can say that the Spearman correlation is the ranked version of the Pearson correlation. Figure 3 shows the Spearman correlation and the graph.

Figure 3: Graph Example of Spearman Correlation and Pearson Correlation



In figure 3, we've included the values for both the Spearman correlation and Pearson Correlation. Their numbers are the same in this example. However, if we add one outlier to the dataset, the result will change.

Figure 4: Spearman Correlation and Pearson Correlation with Outlier

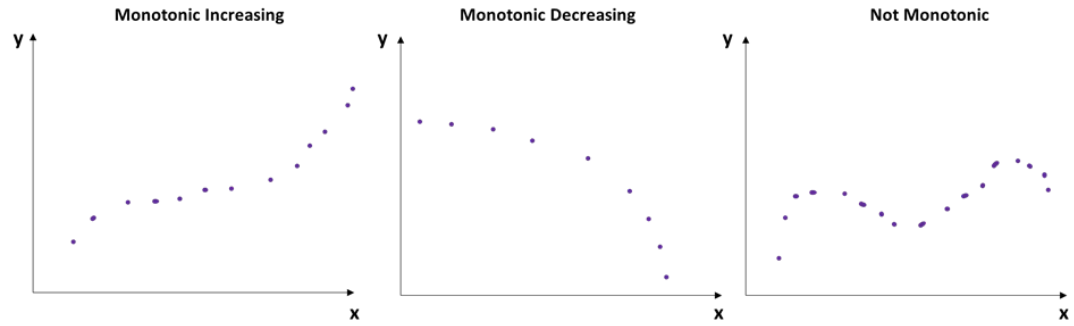


From figure 4, we can see that by adding one outlier, both the Spearman correlation and the Pearson correlation change. However, the Spearman correlation is less sensitive to outliers. This is because the Spearman correlation is calculated on ranked values of the data, not actual values of the data.

Now we know that the Spearman correlation is the ranked version of the Pearson correlation and that the Spearman correlation is less sensitive to outliers. Let's look at other properties of the Spearman correlation.

> **a. The value of the Spearman correlation is between -1 and 1.**
> This conclusion is the result of using the Pearson correlation formula on ranked values of the data.
>
> **b. The Spearman correlation is a ranked data correlation.**
> As demonstrated above, if the data is ranked data, you should use the Spearman correlation.
>
> **c. The Spearman correlation is resistant to outliers.**
> We mentioned this point above.
>
> **d. The Spearman correlation is a non-parametric method.**
> Since Spearman correlation does not require the data to be normal, it is considered a non-parametric method.
>
> **e. The Spearman correlation is a measure of the strength of the monotonic relationship between two random variables.**
> What is a monotonic relationship? We can use the following examples to explain.

Figure 5: Monotonic Relationship



A monotonic function $y = f(x)$ states that when x random variable moves in one direction, y random variable can only move in one direction as well (either the same or the opposite direction) and does not change its direction. For example, in figure 5, the left plot shows when x is increasing, y also is increasing or flat but never is decreasing. The middle plot shows that when x is increasing, y is decreasing all the time and is never increasing. However, the right plot shows that when x is increasing, there are times that y is increasing but also times that y is decreasing. Hence, the plot on the right is not a monotonic relationship.

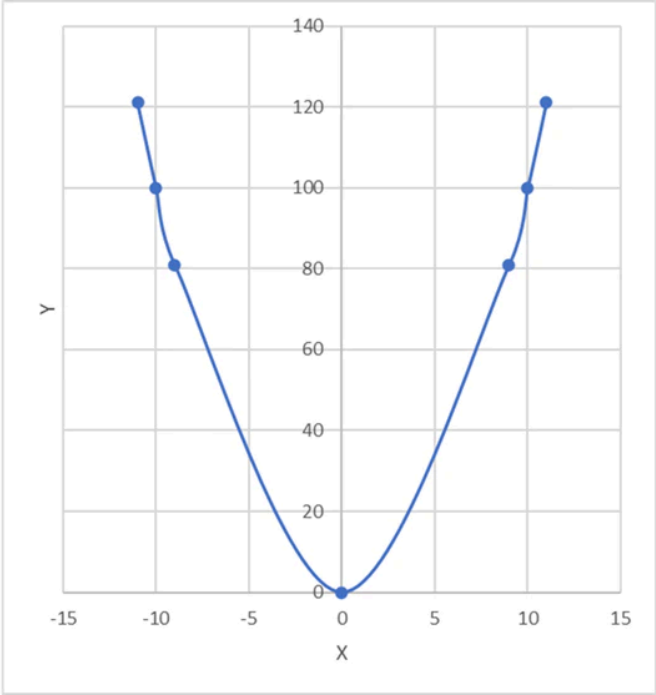
Because Spearman correlation measures the monotonic relationship of two variables, there is no assumption of a linear relationship of two variables. It is the same as using a Pearson correlation; when the Spearman correlation for two variables is 0, it does not

mean that two variables do not have any relationship. They don't have a monotonic relationship, but they might have a quadratic relationship.

Let's look at an example to demonstrate that a quadratic relationship of X and Y can have a Pearson correlation of zero. Figure 6 shows a dataset for two random variables: x and $y = x^2$. The table on the left shows the values of x and y and their quadratic relationship. The plot on the right visualizes their relationship. The Pearson correlation for x and y is 0. This simple example shows that a Pearson correlation cannot identify this particular quadratic relationship (non-linear relationship).

Figure 6: X-Y Quadratic Relationship Example

X	Y=X^2
-11	121
-10	100
-9	81
0	0
9	81
10	100
11	121



3. Kendall Correlation

Kendall Correlation or Kendall's tau Correlation τ is another non-parametric rank-based correlation. To calculate Kendall's tau Correlation, we first need to determine concordant numbers and discordant numbers. Concordant observations mean the values of two variables are moving in the same direction. Discordant observations mean the values of two variables are moving in the opposite direction. It is best to use an example to illustrate how to get the numbers.

Figure 7: Kendall's tau Correlation Data Example

Student	Hours of Study	Test Grade	Ranked Hours	Ranked Grade	Concordant	Discordant
I	15	84	1	3	9	2
A	14	89	2	2	9	1
D	13	91	3	1	9	0

Student	Hours of Study	Test Grade	Ranked Hours	Ranked Grade	Concordant	Discordant
L	13	35	4	12	0	8
J	12	75	5	4	7	0
F	11	71	6	7	4	2
K	10	74	7	5	5	0
G	9	70	8	8	3	1
B	8	72	9	6	3	0
E	7	59	10	9	2	0
H	6	42	11	11	0	1
C	4	47	12	10		
					---	---
					--	--
					51	**15**

In figure 7, we show the same example of a student's study hours and their test grades. We first ordered the table with the student who has the highest "Hours of Study" on top and the least on the bottom. Then, we added a few more columns. For "Ranked Hours," we use 1 for the student who has the highest "Hours of Study" and 2 for the second highest "Hours of Study." We repeat this process with the "Ranked Grade" column. Because the table is ordered by "Hours of Study," you can see "Ranked Hours" has a starting value of 1 and an ending value of 12.

Now we'll examine how to get the value under the "Concordant" column. First, we start with the first row for Student I. Remember the table is already ordered by the first variable "Hours of Study." Let's focus on the "Ranked Grade" column. For student I, the "Ranked Grade" is 3. We need to count how many numbers are larger than 3 below the row where 3 sits. In this case, there are 9 numbers {12, 4, 7, 5, 8, 6, 9, 11, 10} that are larger than 3. We put 9 in the "Concordant" cell in the row where 3 sits. We then move to the next row for student A. The number for "Ranked Grade" is 2. We need to calculate how many numbers are greater than 2 below this row. There are 9 numbers {12, 4, 7, 5, 8, 6, 9, 11, 10}. We put 9 in the "Concordant" cell to the row where 2 sits. We keep doing it until we reach the row of student C. Since there are no numbers under student C, the "Concordant" cell is left empty.

Now let's move to the "Discordant" column. We also start from the first row for student I. We still focus on the "Ranked Grade" column. We need to count how many numbers are smaller than 3. In this case, there are 2 numbers {1, 2}. We put 2 in the "Discordant" cell to the row where 3 sits. Then, we keep doing the same for the following row until the last row. Then, sum up all the numbers in both the "Concordant" column and "Discordant" column.

Kendall's tau Correlation is defined as follows:

$$\tau = \frac{\text{Number of Concordant} - \text{Number of Discordant}}{\binom{n}{2}}$$

Where n is number of observations

Apply our example to the formula:

$$\tau = \frac{51 - 15}{\binom{12}{2}} = \frac{36}{66} = 0.54545$$

The above Kendall's tau is also called Kendall's tau-a. There are alternative definitions, like Kendall's tau-b, which adjusts for tied ranks.

Here are some other properties for Kendall's tau:

> 1. The value of Kendall's tau is between -1 and 1.
> 2. Kendall's tau is also a non-parametric statistic since it does not require the variables to be normally distributed.
> 3. Kendall's tau does not require the two variables to have a linear relationship.
> 4. Kendall's tau is also resistant to outliers.

Both Kendall's tau and the Spearman correlation are rank correlations. In general, the Spearman correlation tends to be larger than Kendall's tau in large sample sizes. The Spearman correlation is also more sensitive to data error or discrepancy. Hence, Kendall's tau is generally preferred over the Spearman correlation because Kendall's tau is more robust and more efficient. Kendall's tau is also easier to explain using the difference between concordance and discordance. Given both methods are in most statistical applications, you can run both and compare the results.

4. Concordant Correlation

In this section, we are going to introduce a correlation metric to evaluate agreement. The correlation metrics we have discussed so far all focus on correlation. But what is the difference between correlation and agreement? We will talk about the difference in this section and then introduce concordant correlation when looking for agreement as the research goal.

4.1 Correlation vs. Agreement

To determine correlation is to measure the strength of association between two variables. These two variables do not need to be the same subjects or on the same scale. For example, we would like to know the association between a student's study hours and their test grades. These are two different variables with different measurement units.

To determine agreement is also to measure the strength of association between two variables. However, it focuses on degree of concordance. For example, two doctors assessing a group of patients might like to see if their assessments agree or not. Another

example is two teachers' critiques on students' writing. The two variables should be the same construct and of course the same measurement. Sometimes, agreement is also related to whether the tests are reproducible (reliable). Hence, agreement is also called reproducibility. For agreement analysis, we not only need to know how two variables are correlated but we also need to know the bias from different opinions or rating systems. For agreement analysis or concordance analysis, we do not need the variables to follow a linear relationship. What we would like to have is for the values of two variables for an observation to be close to each other, which means the opinion or rating is similar. This is why when we analyze the correlation of this kind of assessment/opinion variable, we will need a new correlation. Concordance correlation is useful for this reason.

4.2 Basics of Concordance Correlation

Lawrence Lin proposes how to use the concordance correlation coefficient. Let's see how it works. If we have two random variables x and y, here is the formula for concordance correlation:

$$P_{ccc} = \frac{2\sigma_{xy}}{\sigma_x^2 + \sigma_y^2 + (\mu_x - \mu_y)^2}$$

Where σ_{xy} is the covariance, σ_x^2 and σ_y^2 are variances for x and y, and μ_x and μ_y are expected values (means) for x and y.

We can decompose the above formula as follows:

$$P_{ccc} = \rho \times C_b$$

Where

$$C_b = \frac{2}{\frac{\sigma_1}{\sigma_2} + \frac{\sigma_2}{\sigma_1} + \frac{(\mu_1 - \mu_2)^2}{\sigma_1 \sigma_2}}$$

is bias correction factor and ρ is Pearson Correlation.

Here are some properties of Concordance Correlation:

> 1. If $\rho = 1$, $\mu_x = \mu_y$ and $\sigma_x = \sigma_y$, then $P_{ccc} = 1$ and x and y are perfectly positively in agreement.
> 2. Concordance correlation is between -1 and 1.
> 3. C_b bias correction factor is between 0 and 1.
> 4. The smaller the C_b bias correction factor, the larger the bias is. From the bias correction factor formula, we can see that as the two variables get a higher degree of disagreement, the denominator of the bias correction factor will get bigger, but the numerator will stay the same. Hence, the bias correction factor will get smaller as the disagreement gets worse.
> 5. A low agreement between two variables can stem from low correlation (low ρ) and/or high bias (low bias correction factor).

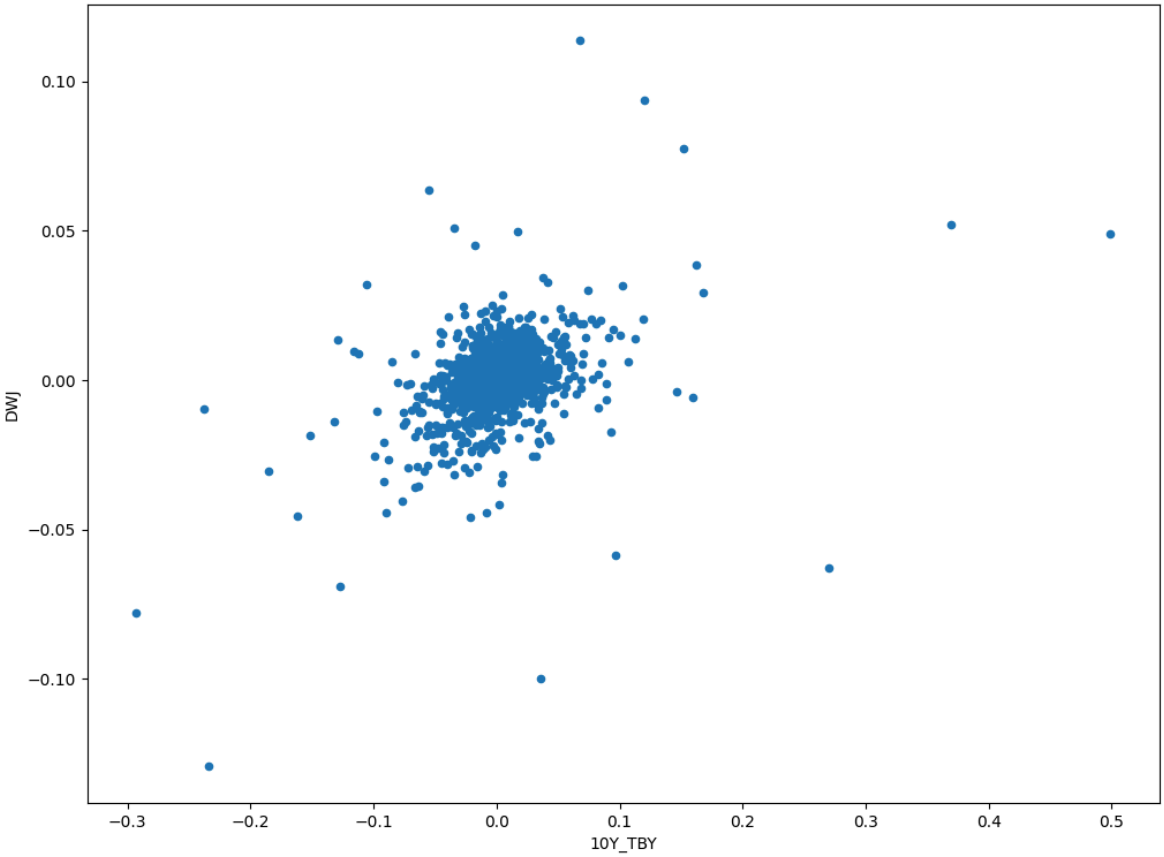
5. Asset Returns and Correlations

Now let's apply the correlation methods we discussed in this lesson to the U.S. 10-Year Treasury Bond Yield daily return and Dow Jones Industrial Average daily return.

Figure 8: Scatter Plot for U.S. 10-Year Treasury Bond Yield Daily Return and Dow Jones Industrial Average Daily Return

In [3]:
```
# Scatter plot for DWJ and 10Y_TBY
scatt = data_set.plot.scatter(x="10Y_TBY", y="DWJ")
scatt.set(xlabel="10Y_TBY", ylabel="DWJ")
plt.show()

# Pearson, Spearman and Kendal tau correlation coefficients
corr_pearson = pearsonr(data_set["10Y_TBY"], data_set["DWJ"])
corr_spearman = spearmanr(data_set["10Y_TBY"], data_set["DWJ"])
corr_kendall = kendalltau(data_set["10Y_TBY"], data_set["DWJ"])
print(
    "Correlation Pearson     : ",
    corr_pearson[0].round(4),
    "\nCorrelation Spearman    : ",
    corr_spearman.correlation.round(4),
    "\nCorrelation kendalltau  : ",
    corr_kendall.correlation.round(4),
)
```




```
Correlation Pearson     :  0.4042
Correlation Spearman    :  0.3386
Correlation kendalltau  :  0.2399
```

From figure 8 and the above correlation results, we can see that there are outliers in the dataset. Hence, the Pearson Correlation is probably not the best choice. As expected, the Spearman correlation has a larger value than Kendall's tau. However, since the correlation is just around 0.3 depending on whether you're using the Spearman correlation or Kendall's tau, the two variables have low correlation.

6. Conclusion

In this lesson, we introduced three commonly used correlation measures: Pearson correlation, Spearman correlation, and Kendall's tau correlation. We also introduced the concept of robust measure: a measurement that is resistant to outliers. We also discussed the difference between correlation and agreement. Lastly, we introduced concordance correlation to evaluate the agreement of two variables. So far in this module, we have been looking at one or two random variables. In the next lesson, we are going to expand to a multivariate space (more than two variables) and discuss both joint probability and marginal probability.

References

- Lin, Lawrence I-Kuei. "A Concordance Correlation Coefficient to Evaluate Reproducibility." Biometrics, vol. 45, no. 1, 1989, pp. 255–268, https://www.jstor.org/stable/2532051.

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FINANCIAL ECONOMETRICS

MODULE 3 | LESSON 4

JOINT PROBABILITY, MARGINAL PROBABILITY AND COPULA

Reading Time	90 minutes
Prior Knowledge	Basic Statistics
Keywords	Joint Probability, Marginal Probability, Inverse Function, Copula, Probability Integral Transformation, Sklar's Theorem, Invariance Under Monotonic, Transformation, Strictly Increasing Function, Comonotonicity, Counter monotonicity, Upper Tail Dependence, Lower Tail Dependence, Elliptical Copula, Normal Copula/Gaussian Copula, Student-t Copula, Archimedean Copula, Clayton Copula, Gumbel Copula

So far, we have talked about probability functions and cumulative probability functions for one random variable (univariate distribution). In this lesson, we are going to explore probability distributions for more than one variable. We will start with two random variables (bivariate distributions). With a solid understanding of bivariate distributions, we can easily extend the theories to multiple random variables (multivariate distributions). At the beginning of the lesson, we will review the concept of joint distribution and marginal distribution. Then, we will move to introduce copulas. A copula is a function to link marginal distributions with joint cumulative distributions. Copulas have been a popular tool in risk management in finance. We will go over the basics of copulas and introduce commonly used copula functions. We will finish the lesson with an example showing how to fit a copula model.

In [1]:
```
# Load libraries
import copulae
import matplotlib.pyplot as plt
import pandas as pd
import pycop
import numpy as np
from copulae import pseudo_obs
from copulae.archimedean import ClaytonCopula, GumbelCopula
from copulae.elliptical import GaussianCopula, StudentCopula
from copulae.gof.goodness_of_fit import gof_copula
from copulas.multivariate import GaussianMultivariate
from scipy.stats import kendalltau


plt.rcParams["figure.figsize"] = (16, 9)  # Figure size and width
```

In [2]:
```
# Download the dataset
data_set = pd.read_csv("M3. bond_and_stock_data.csv")
```

1. Joint Probability Distribution and Marginal Probability Distribution

Oftentimes, we would like to understand the probability of two variables happening at the same time. For example, we would like to know the distribution of both time spent studying and the grades of high school students. Usually, there is a relationship between time spent studying and grades. We would think students who study more may also have higher grades. Hence, the probability of time spent studying and grades is not simply the sum of the probability of time spent studying and the probability of grades.

From the above example, we will need a joint probability distribution for time spent studying and grades. A joint probability distribution is a function to determine the probability of a joint outcome of two random variables. We can write a joint probability distribution function as follows:

Let $X_1, X_2, \cdots, X_d$ be random variables, so their joint probability distribution function would be:

$$f(X_1, X_2, \cdots, X_d) = P(X_1 = x_1, X_2 = x_2, \cdots, X_d = x_d)$$

For example, a joint probability distribution for amount of studying time and grades will give joint probability of the event when study time = 20 hours and grade = 70. For continuous random variables, the joint probability function is called joint probability density function (joint PDF or joint density function). For discrete random variables, the joint probability function is called joint probability mass function (joint PMF or joint mass function). For two random variables, the joint probability distribution is also called bivariate joint probability distribution.

Given a joint distribution of time spent studying and grades, what if we just want to look at the distribution for grades? Then, we will need the marginal probability distribution of grades without study time. A marginal probability distribution is a function to give the probability of grades without study time. Let $X_1, X_2, \cdots, X_d$ be random variables. We can write their marginal probability distribution functions as follows:

$$\begin{cases} f_{X_1}(x_1) = P(X_1 = x_1) \\ f_{X_2}(x_2) = P(X_2 = x_2) \\ \quad \vdots \\ f_{X_d}(x_d) = P(X_d = x_d) \end{cases}$$

For continuous random variables, the marginal probability function is called marginal probability density function (marginal PDF or marginal density function). For discrete random variables, the joint probability function is called marginal probability mass function (marginal PMF or marginal mass function).

We can use the joint probability distribution of studying time and grades to find the marginal probability distribution for grades. The required reading for joint distribution and marginal distribution will provide the detailed process, proof, properties, and examples of joint probability functions and marginal probability functions.

With a joint distribution function, we can also derive the joint cumulative probability distribution function (joint CDF) as follows.

Let $X_1, X_2, \cdots, X_d$ be random variables,

$$F(X_1, X_2, \cdots, X_d) = P(X_1 \leq x_1, X_2 \leq x_2, \cdots, X_d \leq x_d)$$

Sometimes, a distribution function can also mean cumulative probability distribution function (CDF), and a density function means probability distribution function. As such, be careful to understand the term correctly when reading the content.

With a marginal probability distribution, we can also derive the marginal cumulative probability distribution and vice versa. Let $X_1, X_2, \cdots, X_d$ be random variables. We can write their marginal cumulative probability distribution functions as follows:

$$\begin{cases} F_{X_1}(x_1) = P(X_1 \leq x_1) \\ F_{X_2}(x_2) = P(X_2 \leq x_2) \\ \quad \vdots \\ F_{X_d}(x_d) = P(X_d \leq x_d) \end{cases}$$

There is one key property of the joint distribution to emphasize. Joint distribution contains information about the random variables' marginal distributions and their dependence structure. In the following sections, we will explore the relationship among joint distribution, marginal distribution, and random variables' dependence structure more.

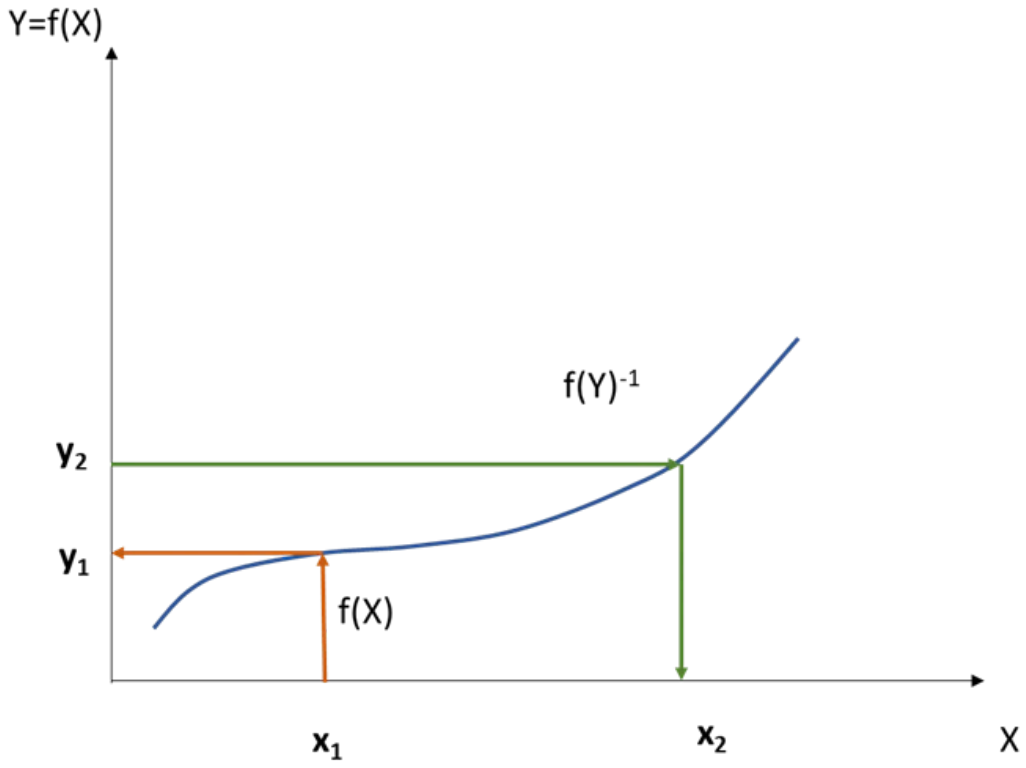
Before we close this section, we'll introduce another concept: inverse function.

When we have a continuous function $f(X) = Y$, it means $f(\cdot)$ maps X to Y. How about if we want to map Y to X? Then, we need an inverse function of $f(\cdot)$. What is an inverse function?

Assume we have a continuous function $f(X) = Y$. If there is a continuous function $g(X)$ such that $g(f(X)) = g(Y) = X$ for all X and Y, then we say $f(\cdot)$ is invertible. The inverse function of $f(X)$ is denoted as $f(Y)^{-1}$. Figure 1 is a graphical depiction of the relationship of a function and its inverse function.

In the above definition, we only look at $f(X)$ and $g(Y)$, which are continuous functions. There will be some variation if one of them is not continuous. We will not cover this topic in this lesson.

Figure 1: Illustration of a Function $f(X)$ and Its Inverse Function $f(Y)^{-1}$



Be careful not to confuse inverse function $f(X)^{-1}$ with the reciprocal of $f(X)$. Inverse Function $f(x)^{-1} eq \frac{1}{f(x)}$. We will use the inverse function concept later in this lesson.

2. Basics of Copulas

2.1 What is a Copula?

In many research projects, we usually need to handle several random variables and find their joint distribution for analysis. For example, in risk management, there's market risk, credit risk, event risk, etc. Usually, we can investigate the market risk and find a distribution that fits the data best. We can do the same thing and find a distribution for each individual risk. However, if we want to analyze all these risks together, how can we find their joint distribution? This is where the concept of a **copula** can help.

A copula is a function that can take the marginal cumulative function of each random variable as input and generate a joint cumulative distribution. Copulas can also estimate the dependence structure of input random variables. You can put in as many marginal distributions from random variables and their correlations as you like. Remember that at the end of the previous section, we mentioned that a joint distribution contains information on the marginal distribution of each random variable and the dependence structure among random variables. However, modeling a joint distribution directly has always been a difficult task in research. With a copula function, we can fit the joint distribution function and estimate the dependence structure of input random variables. That is why copulas have been very popular recently in high-dimensional data analysis. We can summarize what a copula is in the following way:

Assuming X and Y are two continuous random variables, the bivariate joint cumulative distribution of X and Y can be written as follows:

$$H(X,Y) = C(F(X), F(Y))$$

> $H(\cdot)$: joint cumulative function
> $F(\cdot)$: marginal distribution function
> $C(\cdot)$: Copula function

2.2 Definition and Theorems of Copulas

In the last section, we described what a copula is. In this section, we are getting a bit more technical. We will go over a more precise definition for a copula and its theorems.

2.2.1 Copula Definition

A copula is a d-dimensional joint cumulative distribution function of uniform random variables $(u_1, u_2, \cdots, u_d)$ with a range between 0 and 1. Here are the propositions that define a copula function:

> 1. $C(U) = C(u_1, u_2, \cdots, u_d)$ where u_i is a uniform marginal distribution.
> 2. $C(u_1, u_2, \cdots, u_d) : [0,1]^d \rightarrow [0,1]$. This means $C(u_1, u_2, \cdots, u_d)$ is defined in a d-dimensional hypercube space between 0 and 1. $C(u_1, u_2, \cdots, u_d)$ maps the d-dimensional cubic space to a range between 0 and 1.
> 3. $C(u_1, u_2, \cdots, u_{i-1}, 0, u_{i+1}, \cdots, u_d) = 0$ if any component in $C(\cdot)$ is 0.
> 4. $C(1, 1, \cdots, , u_i, \cdots, 1, 1) = u_i$ if all other components except u_i are 1.
> 5. $C(\cdot)$ must increase or stay the same when any of the components increases. $C(\cdot)$ is a non-decreasing function for each component.
> 6. $C(\cdot)$ must increase or stay the same when all components increase or stay the same (none of them decrease) at the same time. This is also called rectangular inequality or n-increasing.

Propositions 1 and 2 tell us that a copula is a joint cumulative function for uniform random variables with a range between 0 and 1. Propositions 3–6 are all rules to set up a copula function so that the copula function behaves like a joint uniform cumulative function.

Please read the required reading for uniform distribution to learn more details about this distribution.

Before moving on to the next topic, we would like to introduce another concept: probability integral transformation.

2.2.2 Probability Integral Transformation

Probability integral transformation states that the cumulative distribution function for a continuous random variable is a uniformly distributed random variable with a range between 0 and 1. We can write in math presentation as follows:

$$F(X) \sim U[0,1]$$

where $F(\cdot)$ is the cumulative distribution function of continuous random variable X.

Please study section 1.4 of the required reading on probability integral transformation for proof and details. We will use this result in the following Sklar's theorem. Sklar's theorem is the single most important theorem for copula functions because it sets the foundation for how we can apply copulas to data analysis.

2.2.3 Sklar's Theorem

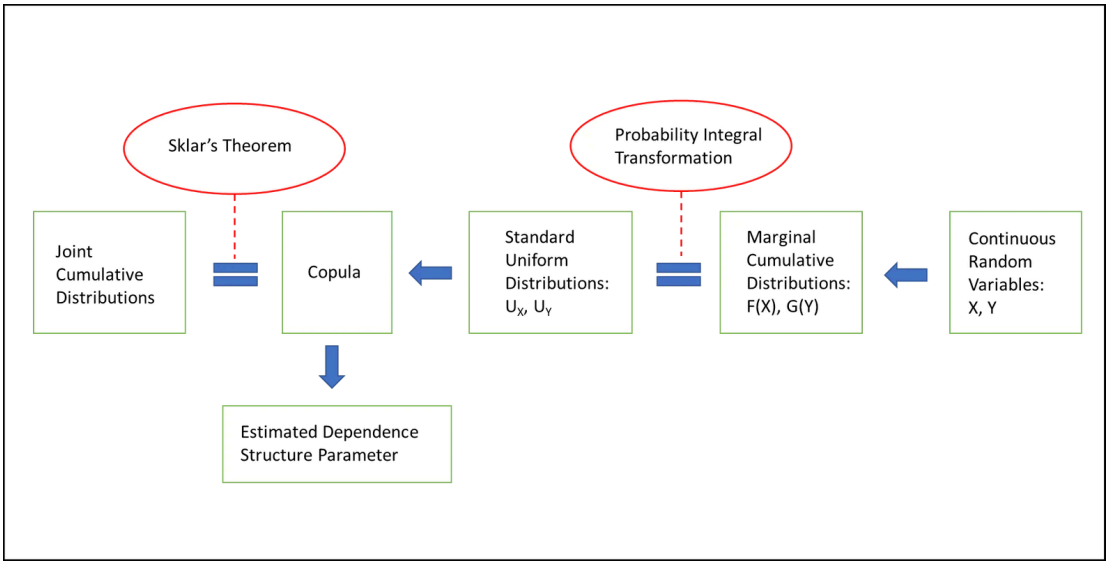
Sklar's theorem is defined with the following two propositions:

> 1. Let $H(X,Y)$ be the joint cumulative distribution function of continuous random variables X and Y. Also, $F(X)$ and $G(Y)$ are marginal cumulative distribution functions of X and Y. Then, there exists a unique copula function C such that
>
> $$H(X,Y) = C(F(X), G(Y))$$
>
> 2. On the other hand, if we have marginal cumulative distribution functions $F(X)$ and $G(Y)$ for continuous random variables X and Y and we also have a copula function C, then $H(X,Y)$ defined in (1) is a bivariate joint cumulative function with marginal cumulative distributions $F(X)$ and $G(Y)$.

By applying probability integral transformation as described in the last section, we know $F(X)$ and $G(Y)$ are uniform distributions with a range between 0 and 1. Hence, we can use $F(X)$ and $G(Y)$ as inputs for $C(\cdot)$.

Our description of a copula function in section 2.1 is based on part 2 of Sklar's theorem. According to Sklar's theorem, we can construct a joint cumulative distribution by using a copula function and marginal cumulative functions. By estimating a copula function with the marginal cumulative functions, we can also get the dependence structure among random variables from estimation. Figure 2 connects all the information we have presented above.

Figure 2: Flow Chart to Use a Copula Function and Marginal CDFs of X and Y to Build Joint CDF of X and Y



From the above flow chart, we can also see that while estimating a copula model for joint CDF, we also estimate the dependence structure parameter for X and Y.

2.2.4 Invariance under Strictly Monotonic Transformations

Invariance under strictly monotonic transformation states that:

Let random variables X and Y have continuous marginal CDFs and a copula C_{XY}. We also have strictly increasing functions T_X and T_Y and we transform X and Y to $X' = T_X(X)$ and $Y' = T_Y(Y)$. The dependence structure of X and Y and the dependence structure of X' and Y' are the same and (X, Y) and (X', Y') share the same copula C_{XY}.

This is a statement about the dependence structure property for a copula. Here is one example:

$$C_{XY}(X, Y) = C_{XY}(ln(X), exp(Y))$$

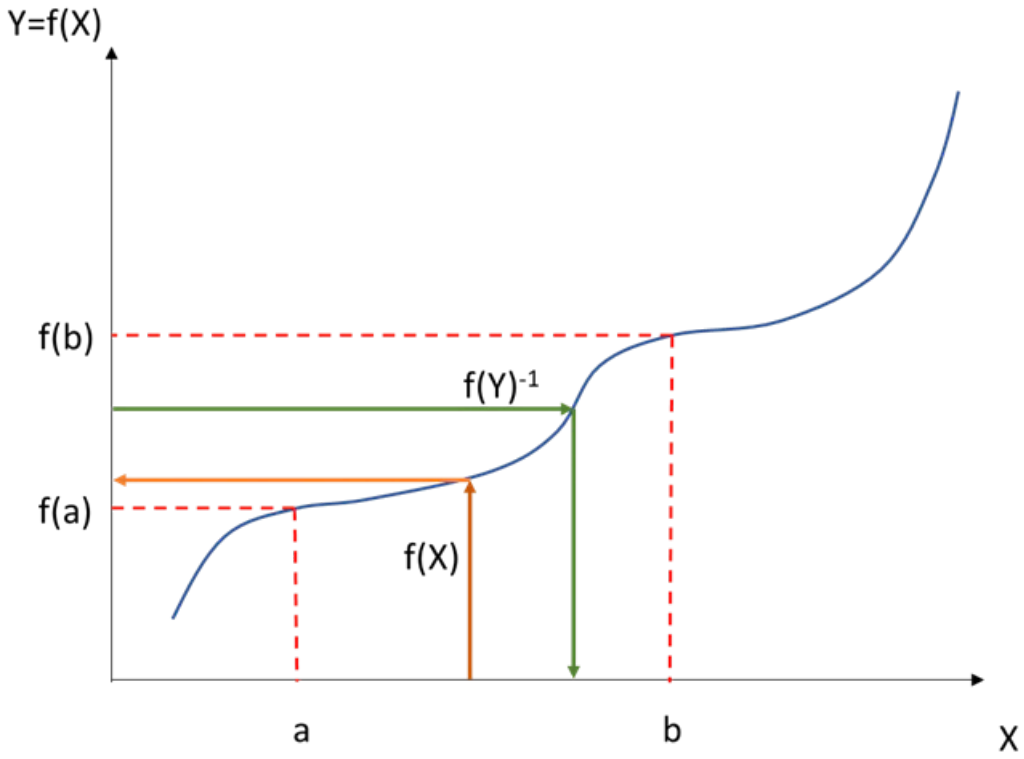
We can see that $ln(\cdot)$ and $exp(\cdot)$ are strictly increasing functions; (X, Y) and $(ln(X), exp(Y))$ share the same copula. Because of this property of copulas, we can also say that a copula is scale invariant under strictly increasing functions.

So what is a strictly increasing function? Let's use the following graphs to explain.

In figure 3 below, we show one example of strictly increasing functions. As we can see, as x increases, $f(x)$ increases too. We can use the following math to present the idea:

A function $f(x)$ is a strictly increasing function if $f(b) > f(a)$ for all $b > a$.

Figure 3: A Strictly Increasing Function Example

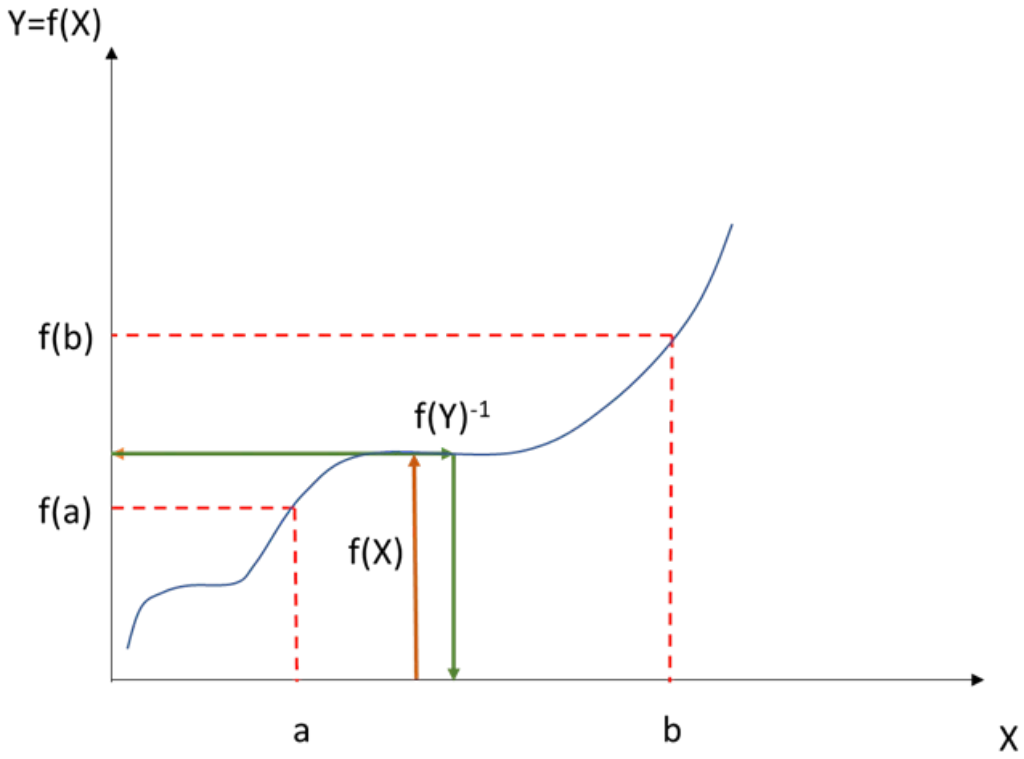


One key point to pay attention to is that there is not $=$ for $f(a)$ and $f(b)$ in the definition. We can see from figure 3, when X increases, $f(X)$ always increases too. $f(X)$ never stays at the same value as X increases.

Why do we need this specific restriction? It is because we want a one-to-one mapping between X and $Y = f(X)$. Only when we have a one-to-one mapping can we preserve the order of X after the transformation. By using a strictly increasing function to transform data, we can transform the data back and forth, but the order of the data always stays the same.

We also demonstrate the relationship with brown arrows and green arrows in figure 3. The graph shows that X and $f(X)$ have a one-to-one relationship. For any X, you can get a unique $f(X) = Y$. For any Y, you can get a unique $f(Y)^{-1} = X$. In figure 4 below, we show an example of a non-strictly increasing function. From figure 4, for a specific Y, we cannot get a unique X. Actually, for a specific Y, we may get multiple Xs. This is a one-to-many relationship, and a function does not apply for this relationship. Hence, $f(Y)^{-1}$ does not exist in this case.

Figure 4: A Non-Strictly Increasing Function Example



Also, remember that although we have only used two random variables in our lesson so far, we can always extend copula analysis by using more random variables. All the results from the theorems will apply. In the next section, we are going to go over some popular copula properties.

2.2.5 Copula Density Functions

A copula density function is defined as follows:

$$c(u,u) = \frac{\partial^2}{\partial u\, \partial u} C(u,u)$$

$f(x,y)$ is the joint density for X and Y, $F(x)$ and $G(y)$ are marginals for X and Y. And $f(x)$ and $g(y)$ are marginal density functions for X and Y.

By using calculus and chain rules, we can obtain the following relationship:

$$f(x,y) = c(F(x), G(y)) f(x) g(y)$$

The above result shows us that a joint density function can be expressed as its copula and marginals.

2.3 Properties of Copulas

In this section, we are going to introduce properties of copulas. We will continue to use our bivariate example X and Y in this section.

2.3.1 Bounds of copulas

Hoeffding and Frechet found that a copula always falls between an upper bound and a lower bound.

$$W(u_1, \cdots, u_d) = max\{1 - d + \sum_{i=1}^{d} u_i, 0\} \leq C(\mathbf{u}) \leq min\{u_1, \ldots, u_d\} = M(u_1, \cdots, u_d$$

W is the Hoeffding and Frechet lower bound, and M is the Hoeffding and Frechet upper bound.

2.3.2 Measures of Dependence

In last lesson, we discussed Pearson's correlation, the Spearman correlation and Kendall's tau correlation. Since we know that a copula tries to estimate scale-invariant association between random variables, we will restrict out correlation measures to rank correlations: Spearman correlation and Kendall's tau correlation. The reasons we use rank correlations are:

> a. The rank correlation = 0 when X and Y are monotonically independent
> b. The rank correlation = 1 when X and Y are comonotonic
> c. The rank correlation = -1 when X and Y are countermonotonic

2.3.3 X and Y are Perfectly Positive Dependent

When X and Y are increasing functions of each other, or we can say when X and Y are moving monotonically in the same direction, then X and Y are perfectly positive dependent (comonotonicity). X and Y have Kendall's tau = 1. According to Hoeffding and Frechet bounds, their copula is as follows:

$$C(U_X, U_Y) = M(U_X, U_Y) = \min(U_X, U_Y)$$

2.3.4 X and Y are Perfectly Negative Dependent

When X and Y are decreasing functions of each other, or we can say when X and Y are moving monotonically in the opposite direction, then X and Y are perfectly negative dependent (countermonotonicity). X and Y have Kendall's tau = -1. According to Hoeffding and Frechet bounds, their Copula is as follows:

$$C(U_X, U_Y) = W(U_X, U_Y) = \max(U_X + U_Y - 1, 0)$$

2.3.5 X and Y are Perfectly Independent

When X and Y are perfectly independent, U_X and U_Y are perfectly independent too due to invariance under monotonic transformation. Copula of U_X and U_Y can be as follows:

$$C(U_X, U_Y) = U_X \cdot U_Y$$

2.3.6 Tail Dependence

Tail dependence is to investigate the correlation of random variables when one random variable is approaching to the extreme level. There are upper-tail dependence and lower-tail dependence, which we can analyze. Let's use two continuous random variables X and Y as an example to define **upper-tail dependence** and **lower-tail dependence**.

> **Upper-Tail Dependence Coefficient:** λ_u: The probability that X will approach its extreme high values given that Y is already approaching its extreme high values
>
> **Lower-Tail Dependence Coefficient:** λ_l: The probability that X will approach its extreme low values given that Y is already approaching its extreme low values.

The coefficients are between 0 and 1 since they are probabilities. When they are 0, it means X and Y are upper tail asymptotically independent or lower tail asymptotically independent. The term asymptotically independent comes from extreme value analysis, which is beyond the scope of this course.

3. Commonly Used Copulas

In this section, we are going to introduce commonly used copulas. We will divide these copulas into two groups: One group is **elliptical copulas**, which includes the **normal copula** and **Student's t-copula**. The other group is **Archimedean copulas**, which includes **Clayton** and **Gumbel** copulas. We will use bivariate copulas with continuous random variables X and Y as an example.

3.1 Elliptical Copulas

The **Normal (Gaussian) copula** and **Student's t-copula** are two copulas in the elliptical copula family.

Gaussian Copula:

> Bivariate Copula: $C_\rho(U_x, U_y) = \Phi_\Sigma(\Phi^{-1}(U_x), \Phi^{-1}(U_y))$
>
> Parameter: ρ = Pearson Correlation where $\rho \in [-1, 1]$

Σ is a 2-by-2 matrix with 1 on the diagonal and ρ off the diagonal. Φ^{-1} is the inverse of the cumulative function of standard normal distribution. Φ_Σ is the cumulative function of a bivariate normal distribution with mean = 0 and correlation = Σ.

The relationship between the Gaussian copula's parameter ρ and Kendall's tau is as follows:

$$\rho(X, Y) = \sin\left(\frac{\pi}{2}\tau\right)$$

Student's t-Copula:

> Bivariate Copula: $C_{u,\Sigma}(U_x, U_y) = t_{u,\Sigma}(t_u^{-1}(U_x), t_u^{-1}(U_y))$
>
> Parameter: ρ = Pearson Correlation where $\rho \in [-1, 1]$ and u = Degree of Freedom

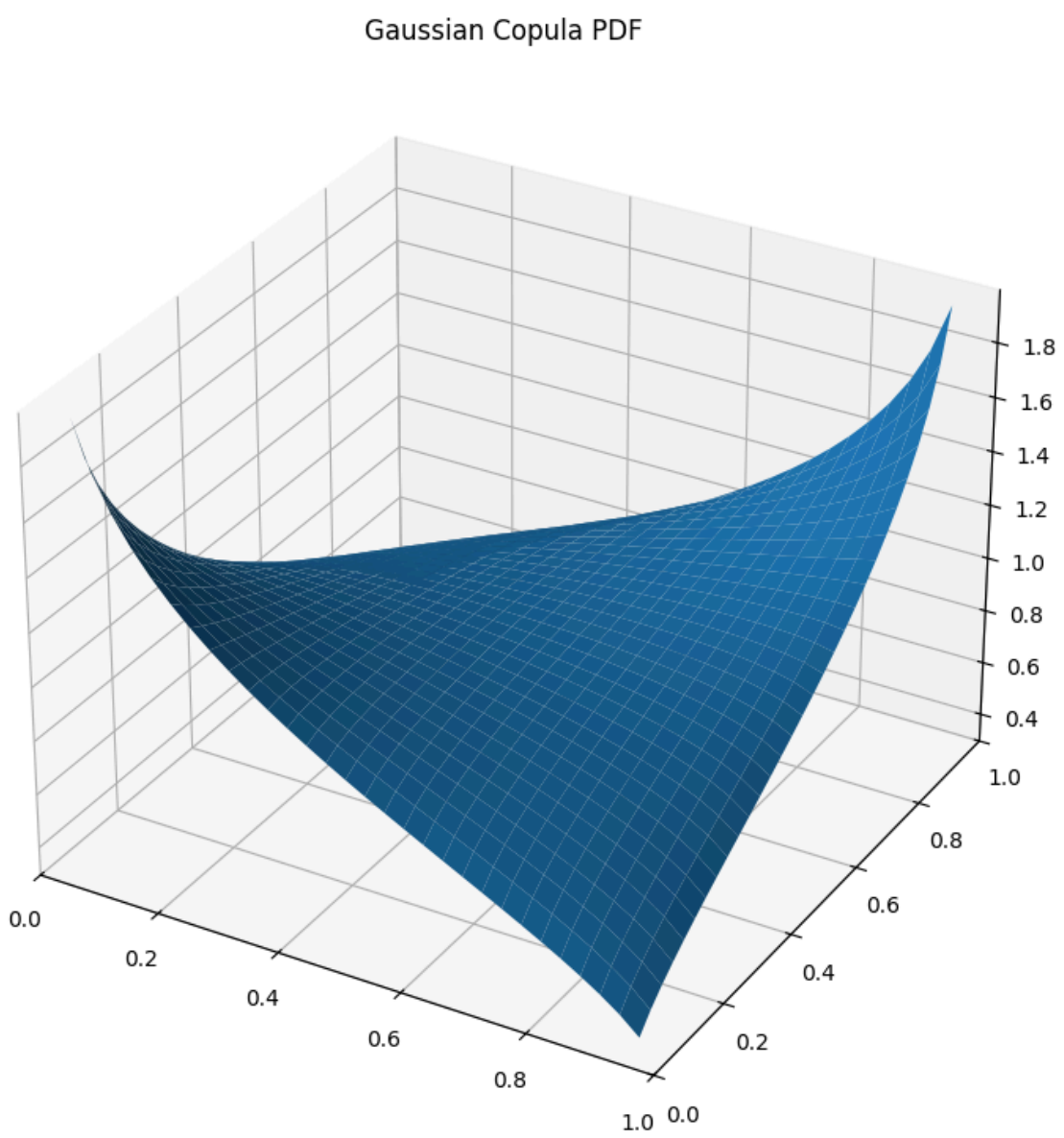
Σ is a 2-by-2 matrix with 1 on the diagonal and ρ off the diagonal. t^{-1} is the inverse of the cumulative function of univariate Student's t-distribution. $t_{u,\Sigma}$ is the cumulative function of a bivariate Student's t-distribution with degree of freedom = u and correlation = Σ.

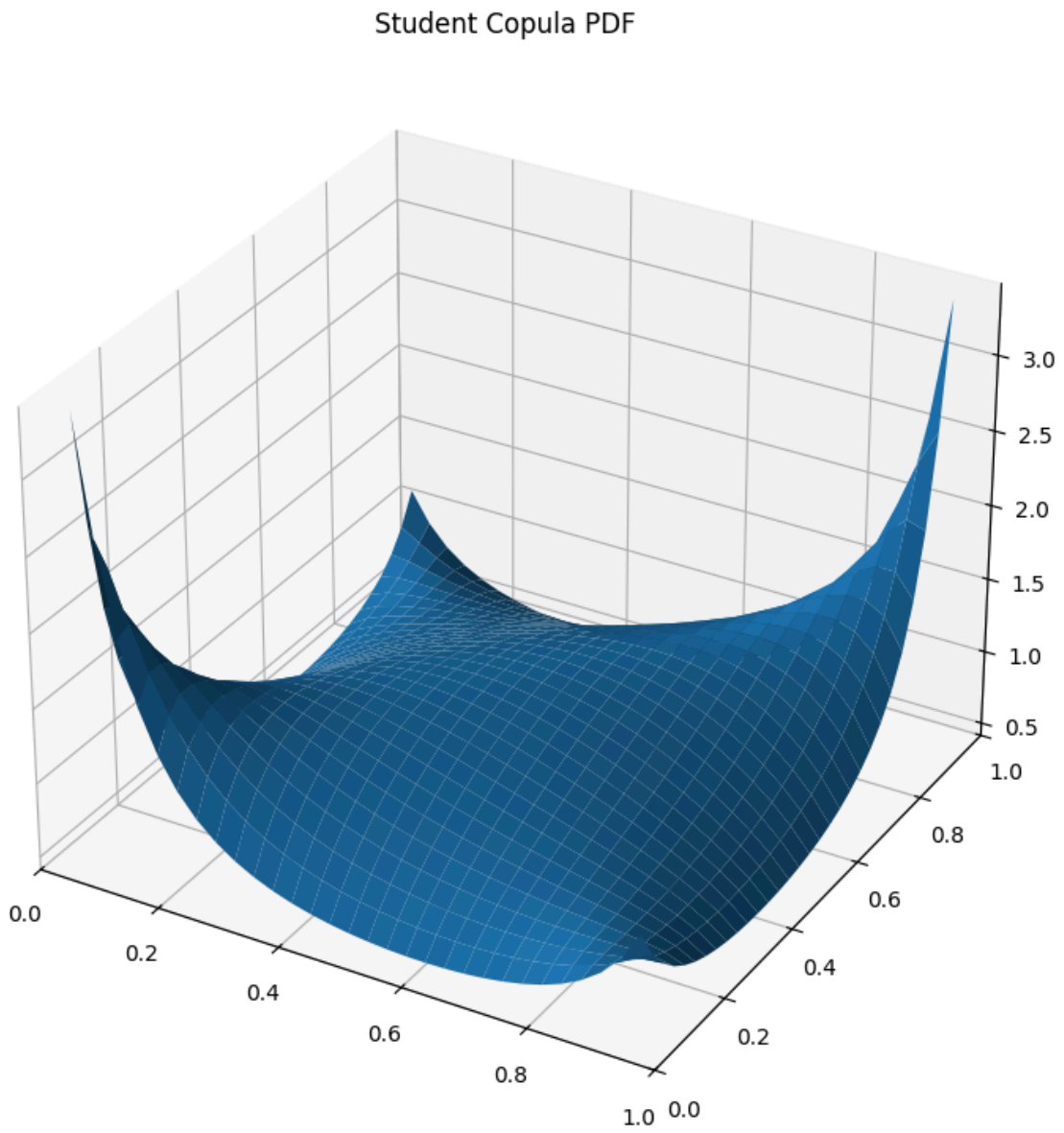
When the degree of freedom u gets large, the Student's t-distribution will converge to Gaussian distribution. The relationship between the Student's t-copula parameter ρ and Kendall's tau is as follows:

$$\rho(X, Y) = \sin\left(\frac{\pi}{2}\tau\right)$$

Figure 5: Copula Densities for Gaussian Copula vs. Student's t-Copula with $\rho = 0.3$ and df $= 2$

In [3]:
```
# Gaussian Copula vs. Student's t-Copula
pycop.gaussian().plot_pdf([0.3, 2], plot_type="3d", Nsplit=30)
pycop.student().plot_pdf([0.3, 2], plot_type="3d", Nsplit=30)
```


Gaussian Copula PDF
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.2
0.4
0.6
0.8
1.0



From figure 5, we draw the copula density functions for both a Gaussian copula and Student's t-copula with a Pearson correlation = 0.3 and degrees of freedom = 2 (for Student's t-copula). We can see that both copulas have a similar pattern in the middle, but in the Student's t-copula, the four corners are higher. It means that the Student's t-copula has more extreme cases than the Gaussian copula. It shows that the Student's t-copula has both upper-tail dependence and lower-tail dependence whereas the Gaussian copula is asymptotically independent for both upper- and lower-tail dependence.

3.2 Archimedean Copulas

Another popular Copula group among researchers is **Archimedean copulas**. Archimedean copulas have an explicitly defined copula formula and math properties that we can manipulate. An Archimedean copula is defined as follows:

$$C(u_x, u_y) = \varphi^{-1}(\varphi(u_x) + \varphi(u_y))$$

Where φ is a decreasing function mapping $[0, 1]$ to $[0, \infty)$

φ is called the generator of Archimedean Copula. With different φ, we can create different copulas. We will go through two common ones: Clayton and Gumbel copulas.

Clayton Copula:

> Bivariate Copula: $C_\theta(u_x, u_y) = [\max(u_x^{-\theta} + u_y^{-\theta} - 1; 0)]^{\frac{-1}{\theta}}$
>
> Parameter: θ where $\theta \in [-1, \infty)$ without 0
>
> Generator: $\phi(u) = \frac{1}{\theta}(\theta^{-1} - 1)$

The relationship between Clayton Copula's parameter θ and Kendall's-tau is as follows:

$$\theta = \frac{2\tau}{1-\tau}$$

From the above formula, we know as $\theta \to 0$, a Clayton copula turns into an independence copula. As $\theta \to \infty$, a Clayton copula turns into a comonotonic copula. As $\theta = -1$, a Clayton copula reaches the Hoeffding and Frechet lower bound.

Gumbel Copula:

> Bivariate Copula: $C_\theta(u_x, u_y) = \exp[-((-\ln u_x)^\theta + (-\ln u_y)^\theta)^{\frac{1}{\theta}}]$
>
> Parameter: θ where $\theta \in [1, \infty)$
>
> Generator: $\phi(u) = (-\ln u)^\theta$

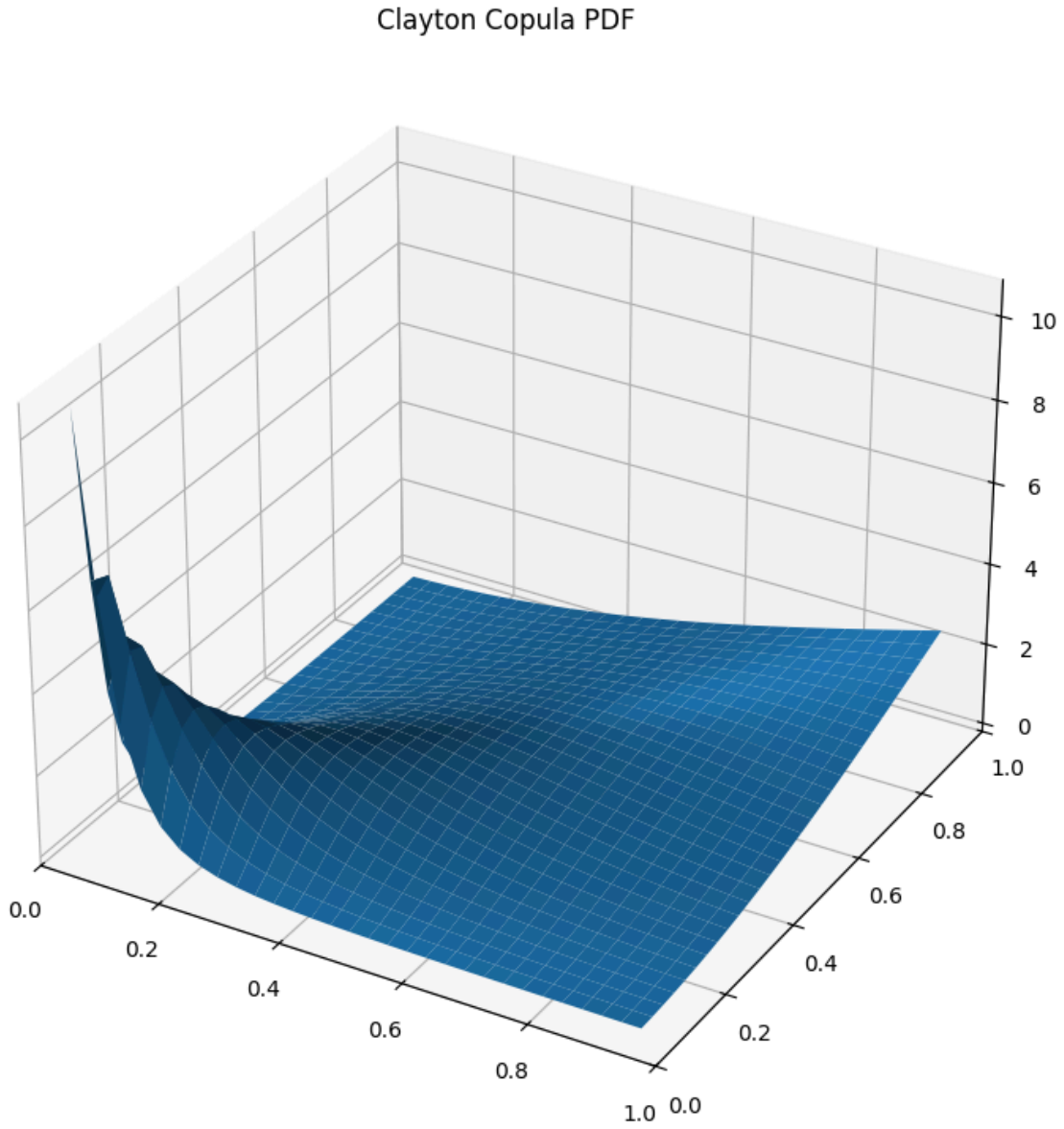
The relationship between a Gumbel copula's parameter θ and Kendall's tau is as follows:

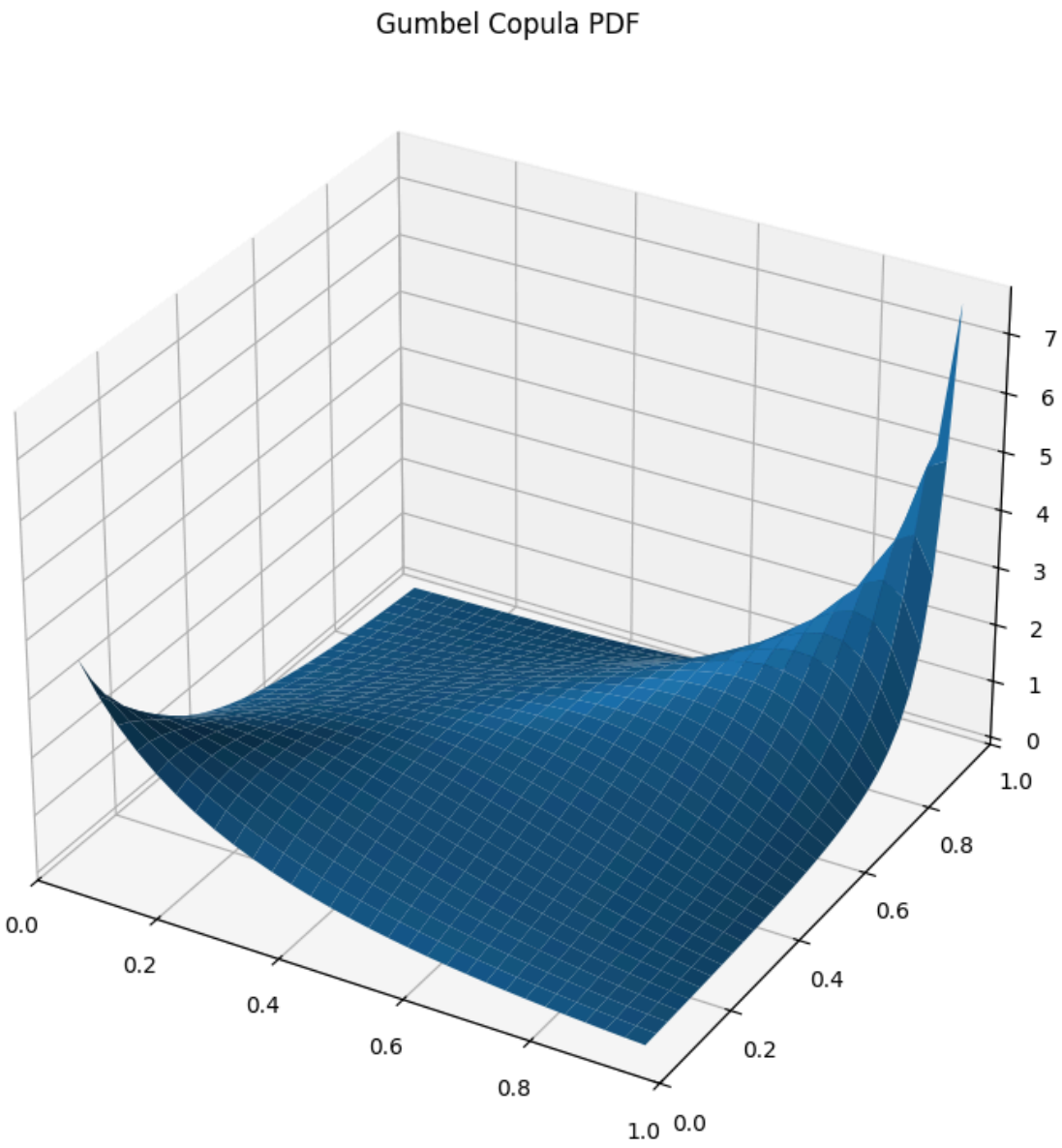
$$\theta = \frac{1}{1-\tau}$$

When $\theta = 1$, a Gumbel copula becomes an independence copula. As $\theta \to \infty$ a Gumbel copula turns into a comonotonic copula.

Figure 6: Copula Densities for Clayton Copula and Gumbel Copula with $\theta = 2$

In [4]:
```
# Clayton Copula and Gumbel Copula
pycop.archimedean(family="clayton").plot_pdf([2], plot_type="3d", Nsplit=30)
pycop.archimedean(family="gumbel").plot_pdf([2], plot_type="3d", Nsplit=30)
```


Clayton Copula PDF
10
8
6
4
2
0
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.2
0.4
0.6
0.8
1.0



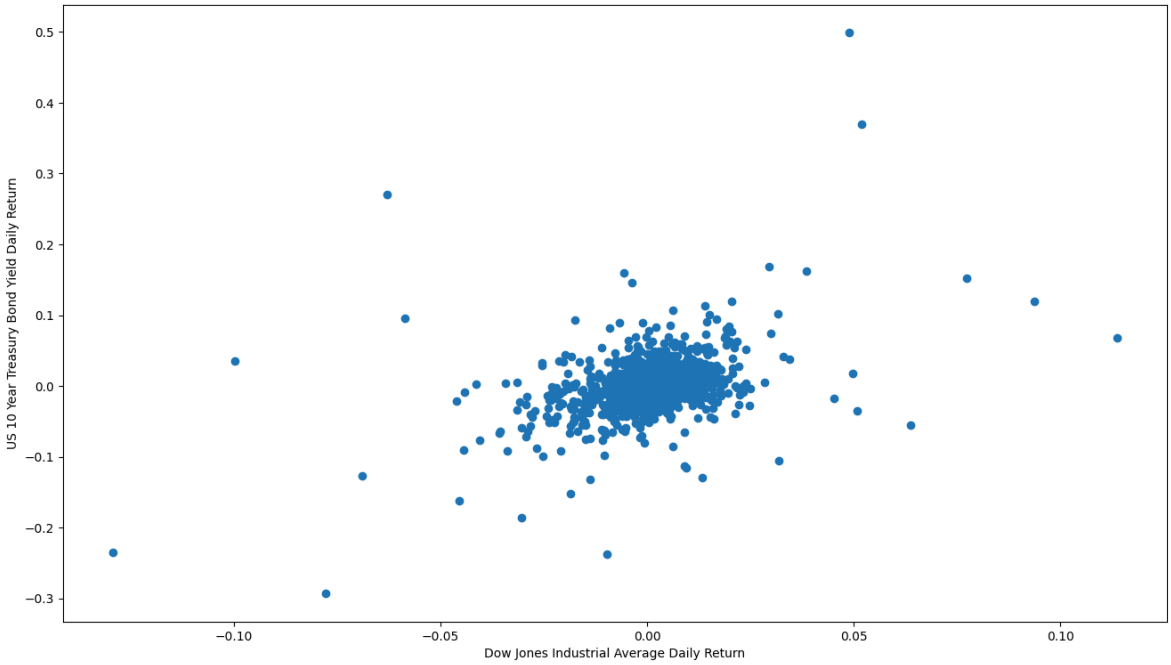
In figure 6, we have copula densities for Clayton and Gumbel copulas. We can see Clayton has a high tail near $(0, 0)$ while Gumbel has a high tail at $(1, 1)$. According to the graphs, Clayton has lower-tail dependence, and Gumbel has upper-tail dependence. Both Clayton and Gumbel are one-tail dependence Copulas. This is important in finance because there are asset returns that exhibit many negative extreme value cases. Clayton copulas can be a good choice for modeling this phenomenon.

4. Example: Joint Distribution for Stock Market Index Returns and Treasury Yield Return

In this section, we are going to use a copula to find a joint distribution function for Dow Jones Industrial Average daily returns and U.S. 10-Year Treasury Bond Yield daily return. Let's look at the scatter plot first.

Figure 7: Scatter Plot for Dow Jones Industrial Average Daily Return and U.S. 10-Year Treasury Bond Yield Daily Return

In [5]:
```
# Scatter Plot for DWJ and 10Y_TBY
plt.scatter(data_set["DWJ"], data_set["10Y_TBY"])
plt.xlabel("Dow Jones Industrial Average Daily Return")
plt.ylabel("US 10 Year Treasury Bond Yield Daily Return")
plt.show()
```



From figure 7, we can see the two variables are mildly positive correlated. Their Kendall's tau is 0.24.

From previous lessons, we know that daily returns do not exhibit normal distribution characteristics. Our data exhibits some heavy tails on both the lower and upper sides. A Student's t-copula seems to be a good candidate since this copula also has heavy tails on both sides. But we need a more formal method for selecting the copula function. Fortunately, the statistical application used here can conduct copula model selection automatically as shown in next figure. The result suggested by the application is to use Student's t-copula for our data.

Our next step before fitting a Student's t-copula model is to transform the data. We know from the theory that we need to transform our data to become standard and uniformly distributed. After having transformed the data, we can now fit the Student's t-copula model. Here is the summary of the model fit.

Figure 8: Student's t Model Result for Dow Jones Industrial Average Daily Return and U.S. 10-Year Treasury Bond Yield Daily Return

In [6]:
```
# Identifying copula subclass from family of univariate copulas

# Fit GaussianMultivariate copula and estimate parameters
data = pd.DataFrame([data_set["DWJ"], data_set["10Y_TBY"]]).T
dist = GaussianMultivariate()
dist.fit(data)
parameters = dist.to_dict()

# print covariance matrix
```

```
print(
    "correlation : ",
    pd.DataFrame(
        parameters["correlation"],
        index=parameters["columns"],
        columns=parameters["columns"],
    ),
)

# print df/loc/scale parameters and copula subclass type
print(
    "- - - - - - - - - - - - - - -Parameters : ",
    pd.DataFrame(parameters["univariates"], index=parameters["columns"]),
)
```

```
correlation :                DWJ   10Y_TBY
DWJ      1.000000  0.369284
10Y_TBY  0.369284  1.000000
- - - - - - - - - - - - - - -Parameters :                df       loc     scale
\
DWJ      2.071275  0.000959  0.005320
10Y_TBY  2.476938 -0.000266  0.017414

                                                      type
DWJ      copulas.univariate.student_t.StudentTUnivariate
10Y_TBY  copulas.univariate.student_t.StudentTUnivariate
```

In [7]:
```
print(parameters.keys())
```

```
dict_keys(['correlation', 'univariates', 'columns', 'type'])
```

The above results suggest Student's t-copula is best fit so we proceed with fitting Student's t-copula model.

In [8]:
```
# Fit student copula and estimate df and rho parameters
result = copulae.elliptical.student.StudentCopula(df=2).fit(
    data, method="ml", fix_df=True
)
print("- - - - - - - - - - - - - - -df and rho parameters : ", result.params)
```

```
              DWJ      10Y_TBY
count  1507.000000  1507.000000
mean      0.500000     0.500000
std       0.288579     0.288578
min       0.000663     0.000663
25%       0.250332     0.250332
50%       0.500000     0.499005
75%       0.749668     0.749668
max       0.999337     0.999337
- - - - - - - - - - - - - - -df and rho parameters :  StudentParams(df=np.float64
(4.669), rho=array([0.3957575]))
```

From figure 8, we can see the two estimated parameters. The First one is the degree of freedom. The second one 'rho' is the dependence parameter. In this model, the estimated rho is 0.3957575. We can use the parameter–rho conversion formula for Student's t-copula in section 3.1 to obtain Kendall's tau. The estimated Kendall's tau is 0.24, the same as we obtained from the original data.

Next, we are going to do some tests to check for model goodness-of-fit. We selected a normal copula, Student's t-copula, Clayton copula, and Gumbel copula to run the test. Figure 9 shows the results.

The goodness-of-fit statistic for the class of copula compares the empirical copula against a parametric estimate of the copula derived under the null hypothesis. Given the pseudo-observations $U_{ij} \forall i = 1, \cdots, n; j = 1, \cdots, d$ and the empirical copula given by

$$C_n(\mathbf{u}) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{I}(U_{i1} \leq u_1, \cdots, U_{id} \leq u_d)$$

where $\mathbf{u} \in [0,1]^d$, the null hypothesis, H_0 thus tests if

$$C \in C_0$$

where C_0 is the true class of the copulae under H_0. The test statistic T is defined as

$$T = n \int_{[0,1]^d} [C_n(\mathbf{u}) - C_{\theta_n}(\mathbf{u})]^2 dC_n(\mathbf{u})$$

where $C_{\theta_n}(\mathbf{u})$ is the estimation of C under H_0.

The approximate p-value is then given by:

$$\frac{\sum_{k=1}^{M} \mathbf{I}(|T_k| \geq |T|)}{M}$$

Figure 9: Kendal-tau and Goodness-of-Fit for Normal Copula, Student's t-Copula, Clayton Copula, and Gumbel Copula

In [9]:
```
# Kendal tau comparisons

# Obtain Transformed (pseudo observations) sample
var_dwj = pseudo_obs(data_set["DWJ"])
var_tb10y = pseudo_obs(data_set["10Y_TBY"])
data = pd.DataFrame([var_dwj, var_tb10y]).T
data.columns = ["var_dwj", "var_tb10y"]

# Obtain Simulated Data
result = copulae.elliptical.student.StudentCopula(df=2).fit(
    data, method="ml", fix_df=True
)
u = result.random(1507, seed=12345)

# Kendal tau - original data
kendal_o = kendalltau(data_set["DWJ"], data_set["10Y_TBY"])
print("Kendal tau for original data")
print(
    "Kendall correlation : ",
    kendal_o.correlation,
    "Kendall p value     : ",
    kendal_o.pvalue,
)

# Kendal tau - synthesized sample
```

```
kendal_t = kendalltau(var_dwj, var_tb10y)
print("Kendal tau for synthesized sample")
print(
    "Kendall correlation : ",
    kendal_t.correlation,
    "Kendall p value     : ",
    kendal_t.pvalue,
)

# Kendal tau - simulated data
kendal_s = kendalltau(u.iloc[:, 0], u.iloc[:, 1])
print("Kendal tau for simulated data")
print(
    "Kendall correlation : ",
    kendal_s.correlation,
    "Kendall p value     : ",
    kendal_s.pvalue,
)
```

```
           var_dwj    var_tb10y
count  1507.000000  1507.000000
mean      0.500000     0.500000
std       0.288579     0.288578
min       0.000663     0.000663
25%       0.250332     0.250332
50%       0.500000     0.499005
75%       0.749668     0.749668
max       0.999337     0.999337
Kendal tau for original data
Kendall correlation :  0.23986070670687398 Kendall p value     :  3.2258360735212
43e-44
Kendal tau for synthesized sample
Kendall correlation :  0.23986070670687398 Kendall p value     :  3.2258360735212
43e-44
Kendal tau for simulated data
Kendall correlation :  0.27498499697295753 Kendall p value     :  1.4105590652835
42e-57
```

In [10]:
```
# GaussianCopula fitmetrics
gaus_cop = GaussianCopula().fit(
    pd.DataFrame([data_set["DWJ"], data_set["10Y_TBY"]]).T, method="ml"
)
gaus_cop_ll = gaus_cop.log_lik(pd.DataFrame([data_set["DWJ"], data_set["10Y_TBY"
gaus_cop_gof = gof_copula(
    copula=GaussianCopula,
    data=pd.DataFrame([data_set["DWJ"], data_set["10Y_TBY"]]).T,
    reps=1,
)

print(gaus_cop)
print("The covariance parameters : ", gaus_cop.params)
print("The covariance matrix :", gaus_cop.sigma)
print("Log likelihood (LL) : ", gaus_cop_ll)
print("", gaus_cop_gof)
```

```
                DWJ      10Y_TBY
count  1507.000000  1507.000000
mean      0.500000     0.500000
std       0.288579     0.288578
min       0.000663     0.000663
25%       0.250332     0.250332
50%       0.500000     0.499005
75%       0.749668     0.749668
max       0.999337     0.999337
<copulae.elliptical.gaussian.GaussianCopula object at 0x779b87553cd0>
The covariance parameters :  [0.36971151]
The covariance matrix : [[1.         0.36971151]
 [0.36971151 1.        ]]
Log likelihood (LL) :  109.21611542923105
 Goodness-of-Fit statistics
==========================
Method     : Parametric bootstrap-based goodness-of-fit of Gaussian with Sn
Parameter  : [0.36971151]
Statistic  : 0.067799045
P-Value    : 0.25
```

In [11]:
```
# StudentCopula fitmetrics
st_cop = StudentCopula().fit(
    pd.DataFrame([data_set["DWJ"], data_set["10Y_TBY"]]).T, method="ml"
)
st_cop_ll = st_cop.log_lik(pd.DataFrame([data_set["DWJ"], data_set["10Y_TBY"]]).
# st_cop_gof = gof_copula(copula=StudentCopula, data=pd.DataFrame([data_set["DWJ

print(st_cop)
print("The covariance parameters : ", st_cop.params)
print("The covariance matrix :", st_cop.sigma)
print("Log likelihood (LL) : ", st_cop_ll)
# print('', st_cop_gof) ##Not implemented
```

```
Warning: Possible convergence problem with copula fitting
                DWJ      10Y_TBY
count  1507.000000  1507.000000
mean      0.500000     0.500000
std       0.288579     0.288578
min       0.000663     0.000663
25%       0.250332     0.250332
50%       0.500000     0.499005
75%       0.749668     0.749668
max       0.999337     0.999337
<copulae.elliptical.student.StudentCopula object at 0x779b875cc350>
The covariance parameters :  StudentParams(df=np.float64(3.040590002001409e-11),
rho=array([0.37029274]))
The covariance matrix : [[1.         0.37029274]
 [0.37029274 1.        ]]
Log likelihood (LL) :  34675.74334744192
```

In [12]:
```
# ClaytonCopula fitmetrics
cl_cop = ClaytonCopula().fit(
    pd.DataFrame([data_set["DWJ"], data_set["10Y_TBY"]]).T, method="ml"
)
cl_cop_ll = cl_cop.log_lik(pd.DataFrame([data_set["DWJ"], data_set["10Y_TBY"]]).
cl_cop_gof = gof_copula(
    copula=ClaytonCopula,
    data=pd.DataFrame([data_set["DWJ"], data_set["10Y_TBY"]]).T,
    reps=1,
```

```
)

print(cl_cop)
print("The covariance parameters : ", cl_cop.params)
# print('The covariance matrix :', cl_cop.sigma) ##Not implemented
print("Log likelihood (LL) : ", cl_cop_ll)
print("", cl_cop_gof)
```

```
                DWJ      10Y_TBY
count  1507.000000  1507.000000
mean      0.500000     0.500000
std       0.288579     0.288578
min       0.000663     0.000663
25%       0.250332     0.250332
50%       0.500000     0.499005
75%       0.749668     0.749668
max       0.999337     0.999337
<copulae.archimedean.clayton.ClaytonCopula object at 0x779b87551210>
The covariance parameters :  0.5879553410302689
Log likelihood (LL) :  125.11848541324596
 Goodness-of-Fit statistics
==========================
Method     : Parametric bootstrap-based goodness-of-fit of Clayton with Sn
Parameter  : 0.5879553410302689
Statistic  : 0.106061625
P-Value    : 0.75
```

In [13]:
```
# GumbelCopula fitmetrics
gumb_cop = GumbelCopula().fit(
    pd.DataFrame([data_set["DWJ"], data_set["10Y_TBY"]]).T, method="ml"
)
new_data = np.array(pd.DataFrame([data_set["DWJ"], data_set["10Y_TBY"]]).T)
gumb_cop_ll = gumb_cop.log_lik(new_data,to_pobs=True, ties='average')
gumb_cop_gof = gof_copula(
    copula=GumbelCopula,
    data=pd.DataFrame([data_set["DWJ"], data_set["10Y_TBY"]]).T,
    reps=1,
)

print(gumb_cop)
print("The covariance parameters : ", gumb_cop.params)
# print('The covariance matrix :', gumb_cop.sigma) ##Not implemented
print("Log likelihood (LL) : ", gumb_cop_ll)
print("", gumb_cop_gof)
```

```
                DWJ      10Y_TBY
count  1507.000000  1507.000000
mean      0.500000     0.500000
std       0.288579     0.288578
min       0.000663     0.000663
25%       0.250332     0.250332
50%       0.500000     0.499005
75%       0.749668     0.749668
max       0.999337     0.999337
<copulae.archimedean.gumbel.GumbelCopula object at 0x779b877812d0>
The covariance parameters :  1.3192097974315677
Log likelihood (LL) :  124.49942782189672
 Goodness-of-Fit statistics
==========================
Method     : Parametric bootstrap-based goodness-of-fit of Gumbel with Sn
Parameter  : 1.3192097974315677
Statistic  : 0.136252685
P-Value    : 0.75
```

Let's observe the log likelihood (LL) of the selected copulas for our given data. The greater the LL (closer to ∞) the better. Based on our results LL of 188.438 for StudentCopula outperforms other copulas and indicates that StudentCopula is the best option.

Now let's random sample or simulate data points from a new joint distribution derived from a copula and compare them to our observed data.

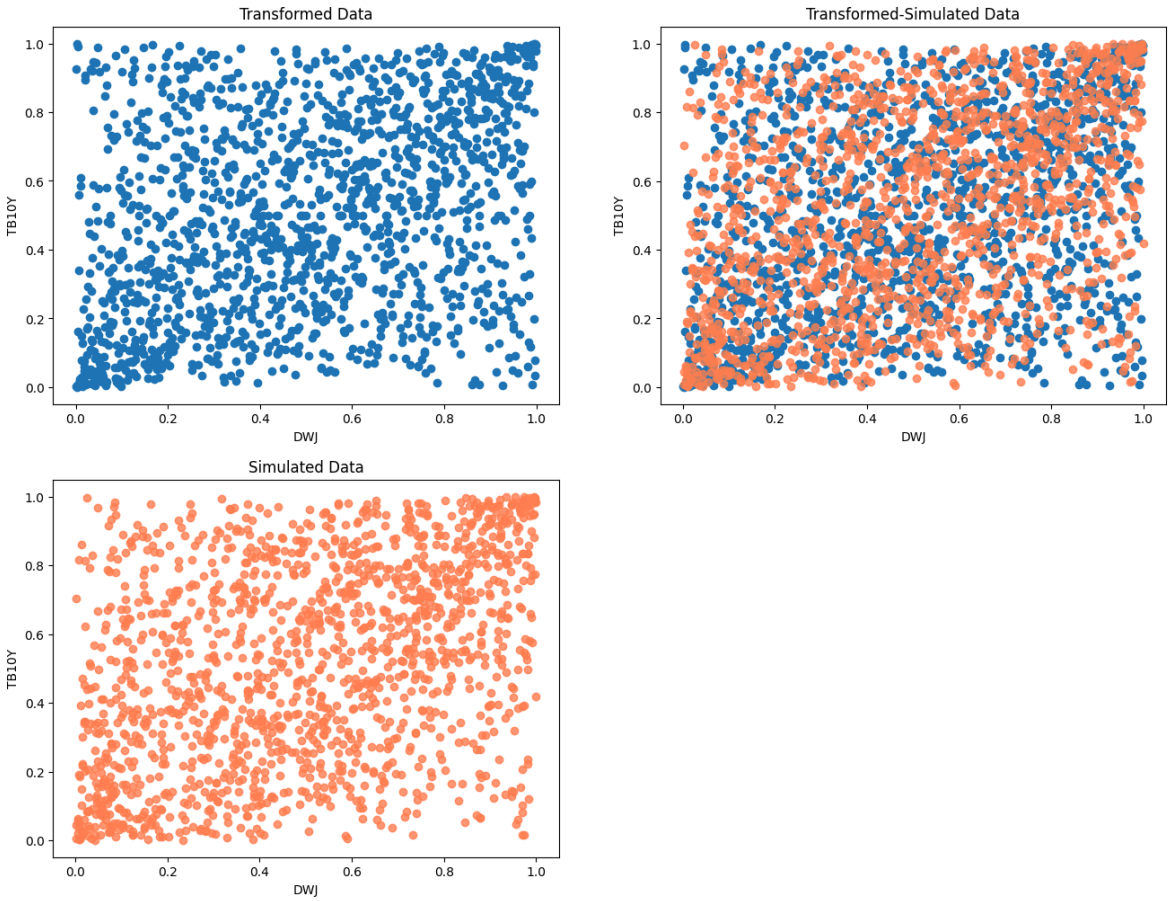
Figure 10: Comparison of Transformed Observed Data and Simulated Data from Copula

In [14]:
```
# Comparison of Transformed Observed Data and Simulated Data from Copula
plt.figure(figsize=(16, 12))

# Plot Transformed Data
ax1 = plt.subplot(221)
plt.scatter(data.var_dwj, data.var_tb10y)
plt.xlabel("DWJ")
plt.ylabel("TB10Y")
plt.title("Transformed Data")

# Plot Simulated Data
ax3 = plt.subplot(223)
plt.scatter(u.var_dwj, u.var_tb10y, c="coral", alpha=0.8)
plt.xlabel("DWJ")
plt.ylabel("TB10Y")
plt.title("Simulated Data")

# Plot Transformed-Simulated Data
ax2 = plt.subplot(222)
plt.scatter(data.var_dwj, data.var_tb10y)
plt.scatter(u.var_dwj, u.var_tb10y, c="coral", alpha=0.8)
plt.xlabel("DWJ")
plt.ylabel("TB10Y")
plt.title("Transformed-Simulated Data")
plt.show()
```



From figure 10, we drew scatter plots from transformed observed data and from simulated data and overlaid them. From the scatter plots on the left, we can see that transformed data or simulated data alone have a slightly higher concentration of data on the lower left corner and upper right corner. It indicates that the Student's t-copula captures the observed data pattern around the tails. The overlaid scatter plot on the right gives us some idea of how the two sets of data points match.

5. Caution on Using Copula

We have demonstrated how powerful a copula is for data analysis. Copulas are a very general technique for modeling random variable dependence structure. This kind of generality also creates some issues. Now, let's look at the issues we need to pay attention to when using copulas.

5.1 Static Dependence

The measure of dependence applied to copula is static, meaning it does not change as time or leadership changes. Hence, it is important to conduct exploratory data analysis and check the correlation in different time periods before running any copula model.

5.2 Questionable Marginal CDF Selection

Because of the invariant property of copulas under monotonic transformation, marginal CDFs can change while the dependence structure stays the same for a copula. We know

the joint distribution of normal random variables is a multivariate normal distribution from basic statistics. However, from a copula, we can create non-normal marginals for a multivariate normal distribution with the same dependence structure. For example, by using a copula, we can obtain a gamma marginal distribution and beta marginal distribution from a multivariate normal distribution. This practice is questionable, so we should check and have a good reason for using this practice.

5.3 Convenient Copula Function Selection Without Data Support

Many people blame the use of normal copula as the cause of the 2008 financial crisis. The model didn't predict the domino effect on mortgage market default. However, we now know that the normal copula does not model tail dependence. The Student's t-copula or Clayton copula would be a better choice to incorporate heavy tail modeling. We understand that normal copula have nice mathematical tractability, but this reason alone is not sufficient for selecting a normal copula function for modeling when it does not model the heavy tails. That's why it is important to conduct a goodness-of-fit process when using copulas.

6. Shiny Application for Copula Model

In this module, we provide a Shiny application to build a Student's t copula model. The application also gives a data summary to let you know what variables are available in the application.

Click here to access the application. You first choose two variables (X variable and Y variable) to build the Student's t copula model. The X and Y data summary tab will show the scatter plot for the two variables and their correlation metrics. The X and Y normality test tab will show the QQ plots for the two variables. The last two tabs give us the copula model result and comparison graph for both the original data and simulated data.

7. Conclusion

In this lesson, we first talked about joint probability distribution and marginal probability distribution. We then introduced copula functions as a link between joint probability distribution and marginal probability distribution. We discussed the definition, properties, and common types of copula functions. We demonstrated how to use a copula to find the joint probability distribution for two asset returns. We finished up with a brief discussion of what areas to pay attention to when using copulas.

This lesson also concludes Module 3. In Module 3, we took a break from regression analysis and went over several topics on random variables and their probability distribution functions. In the next module, we will go back to regression analysis and explore the situation where data points from the same variable are correlated.

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FINANCIAL ECONOMETRICS

MODULE 3 | LESSON 1

RANDOM VARIABLES AND DISTRIBUTIONS

Reading Time	90 minutes
Prior Knowledge	Basic Statistics, Linear Algebra
Keywords	Random Variable,Discrete Random Variable, Continues Random Variable, Probability Distribution, Cumulative Probability Distribution, Expected Value, Variance, Standard Deviation, Probability Mass Function, Probability Density Function, Bernoulli Distribution, Binomial Distribution, Normal Distribution, QQ Plot, Shapiro Wilk Test, Empirical Distribution

In the last module, we discussed many regression methods for different scenarios. We went through which regression method we should apply when the data shows certain characteristics. In this module, we are switching our focus to random variables. We are going to review what a random variable is, some of the probability distributions associated with random variables, how to investigate the relationship among random variables, and several applications for random variables. In this lesson, we are going to discuss the basics of random variables and some of their probability distributions. We will end this lesson with the introduction of empirical probability distribution. This is a method to find a probability distribution from the dataset you have without assuming what the data's distribution is.

In [1]:
```
# Load libraries
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import pylab
import scipy.stats as stats
from statsmodels.distributions.empirical_distribution import ECDF

plt.rcParams["figure.figsize"] = (16, 9)  # Figure size and width
```

In [2]:
```
# Import dataset
data_set = pd.read_csv("M3. bond_and_stock_data.csv")
```

1. Random Variables, Discrete Random Variables, and Continuous Random Variables

A random variable is a variable whose possible outcome comes from a specific experiment. The chance of an outcome happening from this specific experiment is determined by a probability distribution function. The set of all possible outcomes is called a **sample space**. Let's look at an example. Rolling a fair dice once and observing what face comes up on top is an experiment. There are six possible outcomes from rolling a dice: 1, 2, 3, 4, 5, and 6. The top face from rolling a dice is a random variable. Let's call this random variable X. The random variable X can take any of the values in the sample space, i.e., {1, 2, 3, 4, 5, 6}. The probability distribution of each possible outcome for X would be as follows:

$$Pr[X = i] = 1/6, \quad i = 1, 2, .., 6$$

The above relationship between the outcome and the associated probability of the outcome forms a **probability distribution function**. The first key property of a probability distribution function is that the probability of each outcome is a number between 0 and 1. The second key property of a probability distribution function is that the sum of the probabilities of all possible outcomes equals 1:

For X, the sum of probabilities is:

$$\sum_{n=1}^{6} Pr[X = i] = \sum_{n=1}^{6} \frac{1}{6} = 1$$

Sometimes, we would like to know the probability of outcomes over a certain range instead of a certain value. For example, we would like to know the probability that X is, say less than or equal to 3. A cumulative probability distribution of X can provide this information. A **cumulative probability distribution** is a function that provides the probability of the outcome(s) of a random variable to be equal or less than a certain value, x. For X, the cumulative probability function is as follows:

x	Pr[X <= x]
1	1/6
2	2/6
3	3/6
4	4/6
5	5/6
6	6/6=1

How do we read the above relationship? On the left, you have all the possible outcomes. On the right, you have the cumulative probability associated with each outcome on the left. Take outcome 1 as an example. It means the cumulative probability of getting a face that is equal to or less than 1 is $1/6$. When the outcome is 2, it means the cumulative probability of getting a top face that is equal to or less than 2 is $2/6$. For outcome 3, the cumulative probability that a top face is equal to or less than 3 is $3/6$.

Next, we would like to introduce two measurements for a random variable: expected value and variance.

Expected value is a metric to measure where the center of the random variable distribution is. Here is the formula for expected value:

$$E(Y) = \sum_{i=1}^{n} y_i f(y_i)$$

Where $f(y_i)$ is the probability function of a random variable Y and X has $y_1, y_2, \ldots, y_n$ possible outcomes

Variance is a metric to measure how spread out the random variable distribution is. Here is the formula for variance:

$$V(Y) = \sigma_Y^2 = E(Y - E(Y))^2 = \sum_{i=1}^{n} (y_i - E(Y))^2 f(y_i) = \sum_{i=1}^{n} y_i^2 f(y_i) - [E(Y)]^2$$

The scale of variance is the square of the scale of the underlying random variable. In order to be able to use the spread metric directly with the random variable, we would take the square root of the variance. The square root of the variance is called **standard deviation**. The formula for standard deviation is as follows:

$$SD(Y) = \sigma_Y = \sqrt{V(Y)}$$

Figure 1: Expected Value and Variance Calculation for Random Variable X, X = Top Face of Rolling a Dice

x_i	$f(x_i)$	$x_i f(x_i)$	$(x_i - E(X))^2$	$(x_i - E(X))^2 f(x_i)$
1	1/6	1/6	6.25	1.04
2	1/6	1/3	2.25	0.38
3	1/6	1/2	0.25	0.04
4	1/6	2/3	0.25	0.04
5	1/6	5/6	2.25	0.38
6	1/6	1	6.25	1.04
	--------	--------	--------	--------
	$E(X)$	3.5	$V(X)$	2.92

Figure 1 shows the expected value and variance calculation steps for our dice-rolling random variable. From the above table, we can see the average outcome from rolling a dice is 3.5. The standard deviation is $\sqrt{2.92} = 1.71$. We can use the expected value and standard deviation to form an interval and describe the probability as well. If we set an

interval as follows $(3.5 - 1.71, 3.5 + 1.71) = (1.79, 5.21)$, then when we roll a dice, there is a 4/6 chance that an outcome will fall into this interval $(1.79, 5.21)$.

There are two types of random variables: discrete random variables and continuous random variables.

A discrete random variable has a sample space whose outcome is countable. In our dice throwing example, the possible outcomes are countable: 1, 2, 3, 4, 5, and 6. Hence, it is a discrete random variable. There can be finite outcomes or infinite outcomes for a discrete random variable. In our example, we have six outcomes, so it is a finite discrete random variable. If we are looking at the possible number of defective tires in a tire production line, we can count how many tires are defective, and the number can be infinite. Hence, the number of defective tires from a production line is an infinite discrete random variable. The probability distribution function for a discrete random variable is called a **probability mass function**.

On the other hand, a **continuous random variable** has a sample space whose outcome is a real number in a range but whose outcome is not countable. For example, if we want to investigate the height of students aged eight to 15 in New York City, our sample space will be bounded by 0 and infinity (theoretically). But the outcome can be any real number between 0 and infinity if we pick one student. There will be unlimited outcomes. Hence, the height is a continuous random variable. The probability distribution function for a continuous random variable is called a **probability density function** or **density function**.

In the next sections, we are going to review two discrete random variables and their distributions: the Bernoulli random variable and binomial random variables. After looking at two discrete random variables, we will switch to a continuous random variable. We will discuss two continuous random variables: a uniform random variable and normal random variable. We will use these random variables and distributions frequently in the following modules. That's why we are going to cover them in this lesson.

2. Bernoulli Distribution

Bernoulli distribution is the probability function for a Bernoulli random variable. There are many situations in real life that have Bernoulli distribution. For example, when you buy a lottery ticket, you can either win or lose. Say the probability of winning is 0.00001, and the probability of losing is 0.99999. Another popular example is to toss a coin. When you toss a coin, you either get heads or tails. If the coin is fair, there is a 50% chance that you get heads and a 50% chance that you get tails. Bernoulli distribution has only two possible outcomes from an experiment, so it is a discrete distribution. This kind of experiment is called a Bernoulli process or Bernoulli trial. In general, we would call one outcome success (S) and the other outcome failure (F). It does not mean one outcome is better than the other; it is just to differentiate between the two outcomes.

When we assign the outcome of success as 1 and the outcome of failure as 0, we come up with a special Bernoulli distribution: **indicator distribution**. Indicator distribution can

be used to indicate if an event happens or not. When an event happens, the outcome is 1. When an event does not happen, the outcome is 0. Here are some properties of an indicator distribution:

Probability Mass Function:

$$f(x) \begin{cases} p & \text{if } x = 1 \\ 1 - p = q & \text{if } x = 0 \end{cases}$$

Expected Value:

$$E(X) = P$$

Variance:

$$V(X) = P(1 - p)$$

3. Binomial Distribution

Sometimes, you will run a trial several times instead of once. We know one trial with the outcomes of success or failure is a Bernoulli distribution. What distribution can we use if we conduct a series of Bernoulli trials? Binomial distribution is used to describe this experiment. Let's use an example to demonstrate how binomial distribution works.

Assume we are tossing a coin three times and we want to understand the distribution of getting heads during this experiment. Since the outcome of the first trial will not affect the outcome of the second trial, we can safely say the three trials are independent of each other, which means the outcome of the first trial will not affect the outcome of the second trial. If the coin is fair, the probability of getting heads is equal to the probability of getting tails. Hence, the probability of getting either heads or tails is 0.5. Let's say random variable X is the outcome of the number of times we get heads during this experiment. We can write the probability mass function for X as follows:

$$f(X) = \begin{cases} \binom{3}{0}(0.5)^3(0.5)^0 = 0.125 & \text{if } X = 0 \\ \binom{3}{1}(0.5)^2(0.5)^1 = 0.375 & \text{if } X = 1 \\ \binom{3}{2}(0.5)^1(0.5)^2 = 0.375 & \text{if } X = 2 \\ \binom{3}{3}(0.5)^0(0.5)^3 = 0.125 & \text{if } X = 3 \end{cases}$$

Where $\binom{3}{0}, \binom{3}{1}, \binom{3}{2}, \binom{3}{3}$ are binomial coefficients

Expected Value: $E(X) = 0 \times 0.125 + 1 \times 0.375 + 2 \times 0.375 + 3 \times 0.125 = 1.5$

Variance: $V(X) = 3 \times 0.5 \times 0.5 = 0.75$

We can generalize the above example to write the probability mass function, expected value, and variance for a binomial Distribution in which all trials are independent:

Probability Mass Function:

$$B(X;n,p) = \binom{n}{x} p^x q^{n-x}$$

where $q = 1 - p$ and $x = 0, 1, 2, \ldots, n$

Cumulative Distribution Function:

$$F(x \leq m) = \sum_{i=0}^{m} \binom{n}{i} p^i q^{n-i}$$

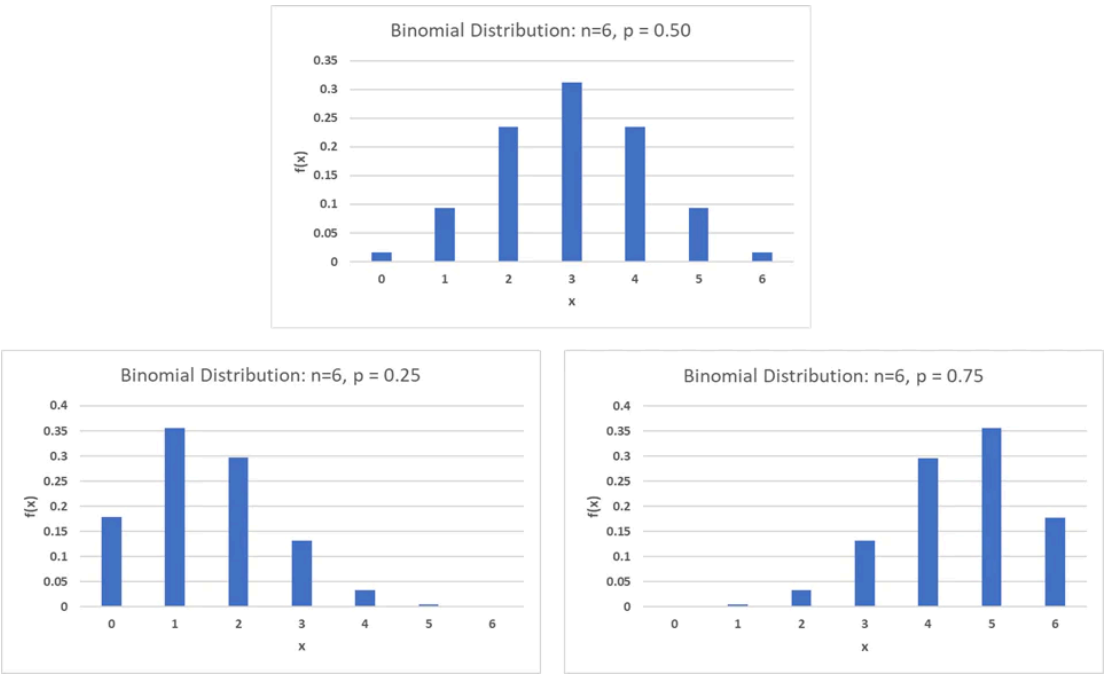
Expected Value:

$$E(X) = np$$

Variance:

$$V(X) = npq$$

Figure 2: Binomial Distribution with Different p-Values



Binomial distributions would have different distribution shapes depending on the value of p. In Figure 2, we can see when $p = 0.5$, binomial distribution is a symmetric distribution. When $p < 0.5$, binomial distribution is a right-skewed distribution. When $p > 0.5$, binomial distribution is a left-skewed distribution.

4. Normal Distribution

4.1 Basics of Normal Distribution

Now we are moving to talk about continuous random variables. In this section, we will review the most popular continuous distribution in data analysis: normal random variable or normal distribution. Since we are all familiar with normal distribution from basic

statistics coursework, we will only quickly review the key properties of normal distribution and discuss how to test a random variable for normality in this section.

Normal distribution is also called Gaussian distribution. For a normal random variable X with the expected value μ_X and variance σ_X^2, the probability density function is as follows:

$$f(X) = \frac{1}{\sigma_X \sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{X-\mu_X}{\sigma_X}\right)^2}$$

Where $-\infty < X < \infty$

This is also written in the following compact manner:

$$X \sim N(\mu_X, \sigma_X^2)$$

Here are some of the properties for Normal Distribution:

a. Bell-Shaped Normal Distribution Curve

Figure 3: Normal Distribution Curve

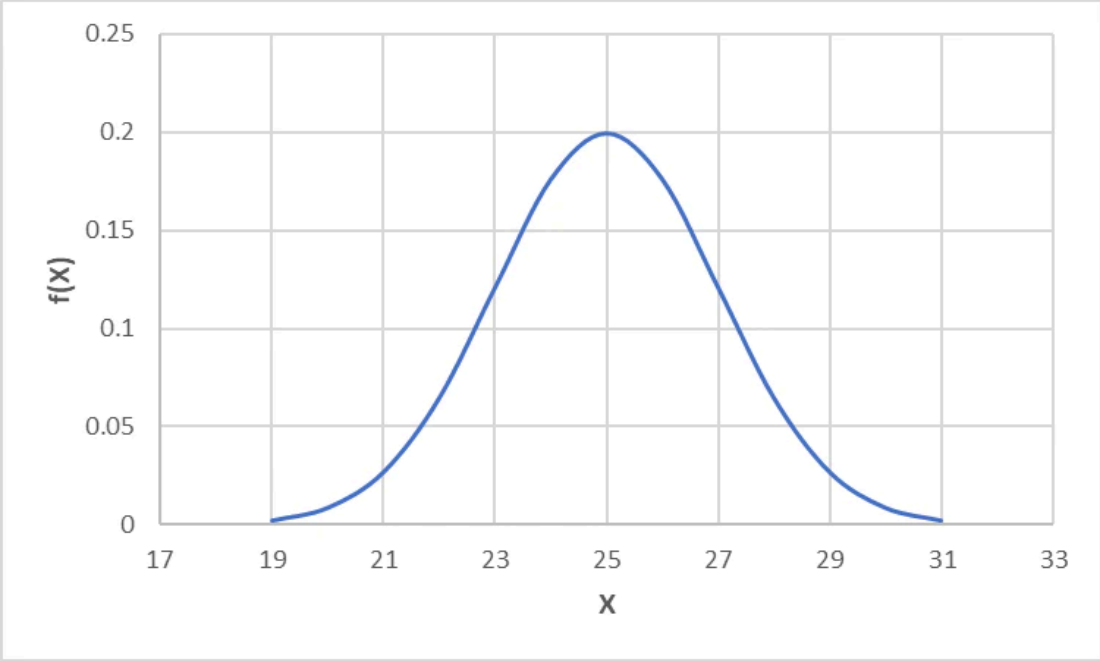


Figure 3 is the probability density curve for a normal distribution with the expected value = 25 and standard deviation = 2. We can see the curve is a symmetric, bell-shaped curve. The curve is centered around the expected value. It means the probability density function has a maximum at the expected value.

b. Empirical Rule for Normal Distribution

The empirical rule for normal distribution is a rule that helps us quickly know what the probability is when given a range of values from a normally distributed variable. Say X is a normally distributed random variable. The empirical rule provides the following rules:

> 68% of the data from X will fall into this interval:
>
> $(\text{expected value} - 1 \times \text{standard deviation},\ \text{expected value} + 1 \times \text{standard dev}$
>
> 95% of the data from X will fall into this interval:
>
> $(\text{expected value} - 2 \times \text{standard deviation},\ \text{expected value} + 2 \times \text{standard dev}$
>
> 99.7% of the data from X will fall into this interval:
>
> $(\text{expected value} - 3 \times \text{standard deviation},\ \text{expected value} + 3 \times \text{standard dev}$

In our example in Figure 3, 68% of the data points will come from the interval $(23, 27)$. 95% of the data points will come from the interval $(21, 29)$. 99.7% of the data points will come from the interval $(19, 31)$. This piece of information will be very helpful for us going forward when conducting statistical inferences.

c. Standard Normal Distribution

Standard normal distribution is a special case for normal distribution. It has the expected value = 0 and variance = 1. If Z has standard normal distribution, its probability density function is as follows:

$$f(Z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}Z^2}$$

where $-\infty \leq Z \leq \infty$. We can write Z as follows: $Z \sim N(0, 1)$

Now, standard normal distribution has a bell-shaped curve centered around 0.

If X is a normally distributed random variable, you can use the following formula to convert X into a standard normal variable.

$$\frac{X - \mu_X}{\sigma_X}$$

This is basically to standardize a random variable. In previous modules, there are several regression analyses that require standardizing the variable before running the model. By standardizing a variable, we can see that it will convert a normal variable into a standard normal variable with the expected value of 0 and variance of 1. Then, all standardized variables will have the same expected values and variances.

4.2 Test of Normality

a. QQ Plot

In order to test if a random variable is a normally distributed random variable, we usually use a graphic method as a starting point. The most popular graphic method is **QQ plot** or **quantile-quantile plot**.

QQ plots show if two sets of data come from the same distribution. In our case, a QQ plot will compare an observed dataset with a normal distribution to see if they match in distribution. If they do have the same distribution, then a QQ plot should show a 45-degree line pattern formed by the observed data points.

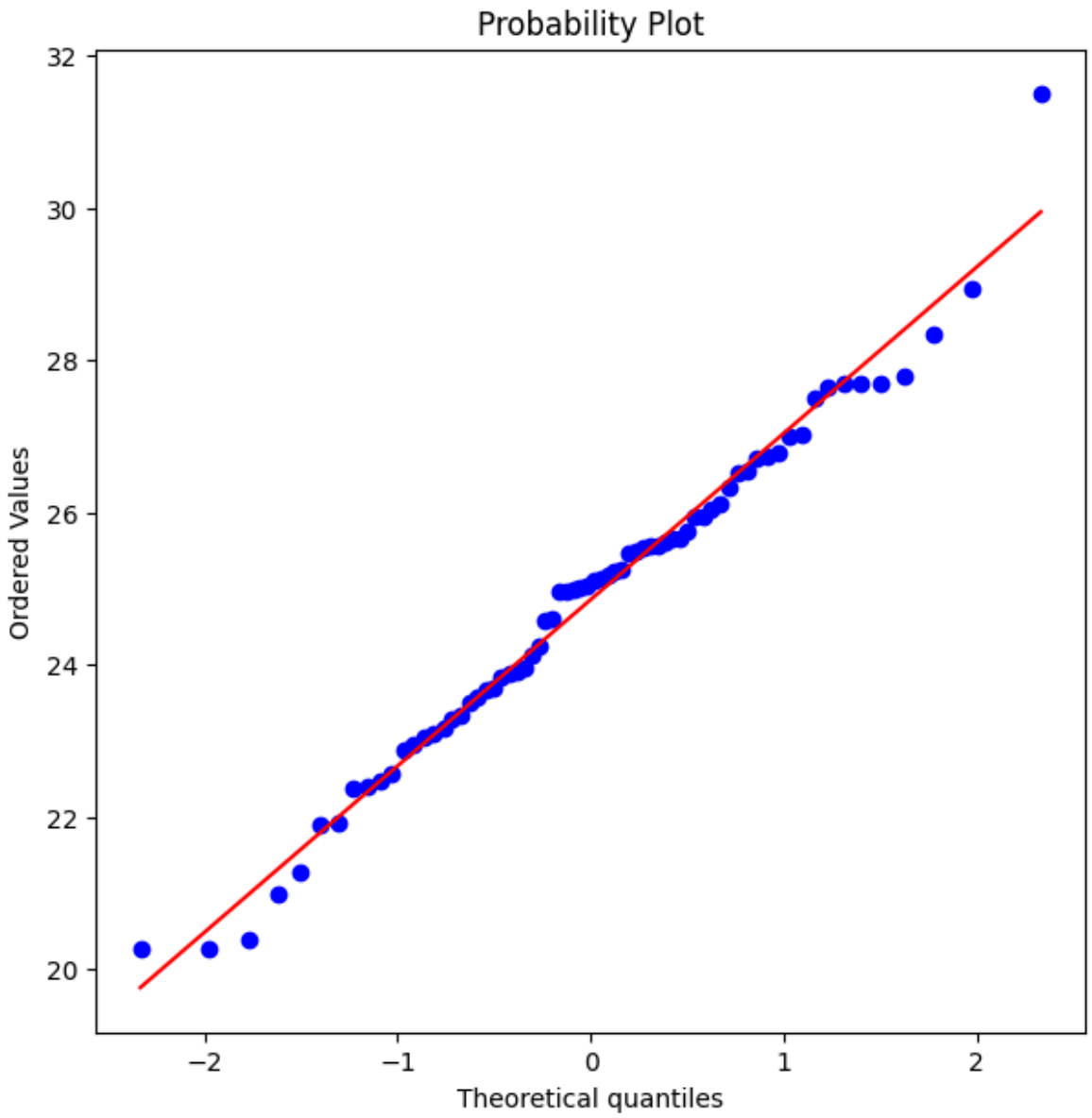
Let's use an example to understand more about QQ plots. We randomly sampled 70 data points from a normal random variable $N(25, 2)$. We first calculated the mean and variance of the 70 data points. Then, we standardized these data points. We then ordered 70 standardized data points from smallest to largest. We also used the mean and the variance calculated from 70 data points to construct a normal distribution. Then, for each data point from smallest to largest, we used the constructed normal distribution to determine the percentile location of the data point. This percentile location on the constructed normal distribution is the X-axis of the QQ plot. The Y-axis is the original value of the data point. If the sampled data points follow normal distribution, all the ordered data points should form a straight line in the plot as shown in Figure 4.

Figure 4: QQ Plot for Normal Distribution

In [3]:
```
# QQ Plot for Normal Distribution

# Create a normal distribution
np.random.seed(12345)
y = np.random.normal(loc=25, scale=2, size=70)  # np.random.normal(mean, std, na

# Plot
plt.figure(figsize=(7, 7))
stats.probplot(y, dist="norm", plot=pylab)
pylab.show()
```



b. Shapiro–Wilk Test

However, sometimes the graph is not very accurate. We need a formal normality test. There are several formal tests for normality. We are going to introduce one that works very well: the Shapiro–Wilk test. The Shapiro Wilk test uses the W statistic. The W statistic is the correlation between the ordered observed data points and the theoretical data points on normal distribution. If the W statistic is too small, the Shapiro–Wilk test rejects the normality hypothesis.

Figure 5: Shapiro–Wilk Test for 70 Data Points

In [4]:
```
# Shapiro Wilk normality test
shapiro_test = stats.shapiro(y)
print(
    "Shapiro W: {0} \nShapiro p-value {1}".format(
        shapiro_test.statistic, shapiro_test.pvalue
    )
)
```

```
Shapiro W: 0.9852498693782097
Shapiro p-value 0.5832105563259571
```

In figure 5, we have the Shapiro–Wilk test for our example of 70 data points. The W is 0.98 and the p-value is 0.58. The p-value is greater than 0.05 so we cannot reject that this series of 70 data points is normally distributed. Hence, they are normally distributed.

5. Empirical Distribution

In the previous sections, we assumed that we knew the theoretical distribution underlying our data. In application, many times we don't know what the theoretical distribution should be. In this section, we will use the observed data points to construct a distribution. This distribution derived from the observed dataset is called an empirical distribution function or empirical cumulative distribution function. We will go through how to construct an empirical distribution from the dataset, discuss empirical distribution's properties, and also show how to overlay theoretical distribution onto empirical distribution to check if the dataset comes from the theoretical distribution. As usual, let's use an example to understand empirical distribution.

Assume we have a data series X with 10 observations as follows: X: 8, 9, 9, 0, 2, 4, 5, 5, 5, 9

First, we are going to reorder X data from the smallest to the largest as follows: X: 0, 2, 4, 5, 5, 5, 8, 9, 9, 9

Then, for each unique outcome, we calculate the number of times it appears divided by the total number of the data series. The result is as follows: (Make sure the unique outcomes are still in ascending order, smallest to largest)

0:	1/10
2:	1/10
4:	1/10
5:	3/10
8:	1/10
9:	3/10

Now we can define the following empirical distribution function $F_n(X)$:

$$F_n(X) = \begin{cases} 0, & \text{for } X < 0 \\ \frac{1}{10}, & \text{for } 0 \leq X < 2 \\ \frac{1}{10} + \frac{1}{10} = \frac{2}{10}, & \text{for } 2 \leq X < 4 \\ \frac{1}{10} + \frac{1}{10} + \frac{1}{10} = \frac{3}{10}, & \text{for } 4 \leq X < 5 \\ \frac{1}{10} + \frac{1}{10} + \frac{1}{10} + \frac{3}{10} = \frac{6}{10}, & \text{for } 5 \leq X < 8 \\ \frac{1}{10} + \frac{1}{10} + \frac{1}{10} + \frac{3}{10} + \frac{1}{10} = \frac{7}{10}, & \text{for } 8 \leq X < 9 \\ \frac{1}{10} + \frac{1}{10} + \frac{1}{10} + \frac{3}{10} + \frac{1}{10} + \frac{3}{10} = \frac{10}{10}, & \text{for } X \geq 9 \end{cases}$$

From the above empirical distribution function, we can see that $F_n(X)$ is the cumulative probability for X equal to or less than certain value C. The empirical distribution jumps when X is an observed outcome because there is a new probability value to add to the existing cumulative probability at each observed outcome. Between observed outcomes, the cumulative probability is flat because there is no observed outcome between two consecutive observed outcomes. We can also draw a step graph to demonstrate the jump of the empirical distribution at observed outcomes.

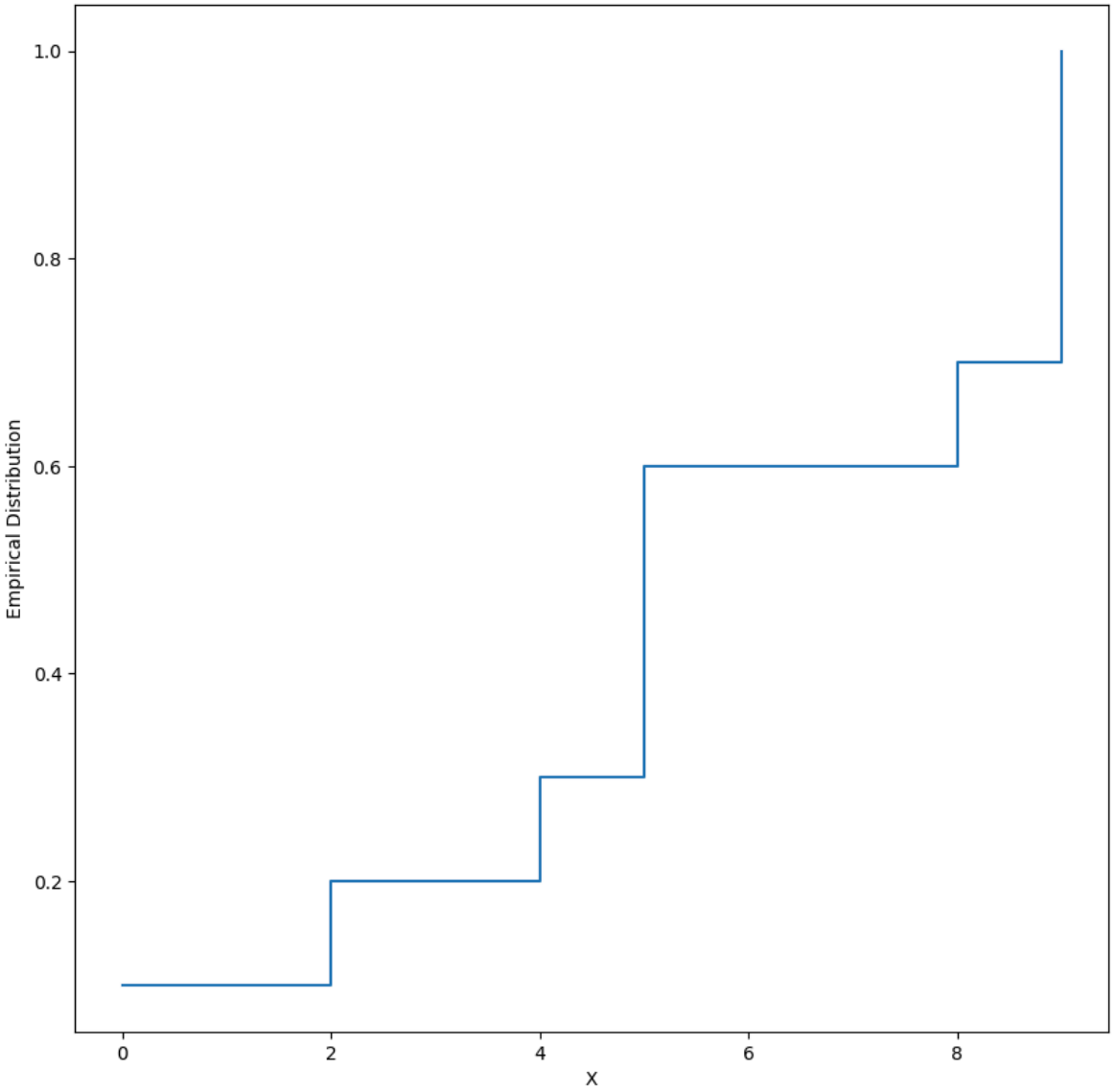
Figure 6: Empirical Distribution Function

In [5]:
```
# Empirical Distribution function

M = [8, 9, 9, 0, 2, 4, 5, 5, 5, 9]

# Empirical Cumulative distribution function
ecd_function = ECDF(M)

# plot
plt.figure(figsize=(10, 10))
plt.plot(ecd_function.x[1:], ecd_function.y[1:], drawstyle="steps-post")
plt.xlabel("X")
plt.ylabel("Empirical Distribution")
plt.show()
```



From figure 6, we can see the function jumps on each observed outcome as we described before. We also see the cumulative probability is flat between two consecutive outcomes in the graph.

Let's formalize the formula for the empirical distribution function.

Assume we have a sample of n observations: $X_1, X_2, \ldots, X_n$. They are ordered from smallest to largest. They are also identical and independently distributed (i.i.d.). The empirical cumulative distribution function is:

$$F_n(t) = \frac{1}{n} \sum_{j=1}^{n} 1_{(X_j \leq t)}$$

Where $1_{(X_j \leq t)}$ is an indicator function and t is defined as a threshold to decide the count of $X_1, X_2, \ldots, X_n$ that are less than or equal to t.

We can write this indicator function as follows:

$$1_{(X_j \leq t)} = \begin{cases} 1 & \text{if } X_j \leq t \\ 0 & \text{if } X_j > t \end{cases}$$

The expected value of the empirical cumulative distribution function is:

$$E_n(X) = \frac{1}{n}\sum_{j=1}^{n} X_j$$

The variance of the empirical cumulative distribution function is:

$$V_n(X) = \frac{1}{n}\sum_{j=1}^{n}(X_j - E_n(X))^2$$

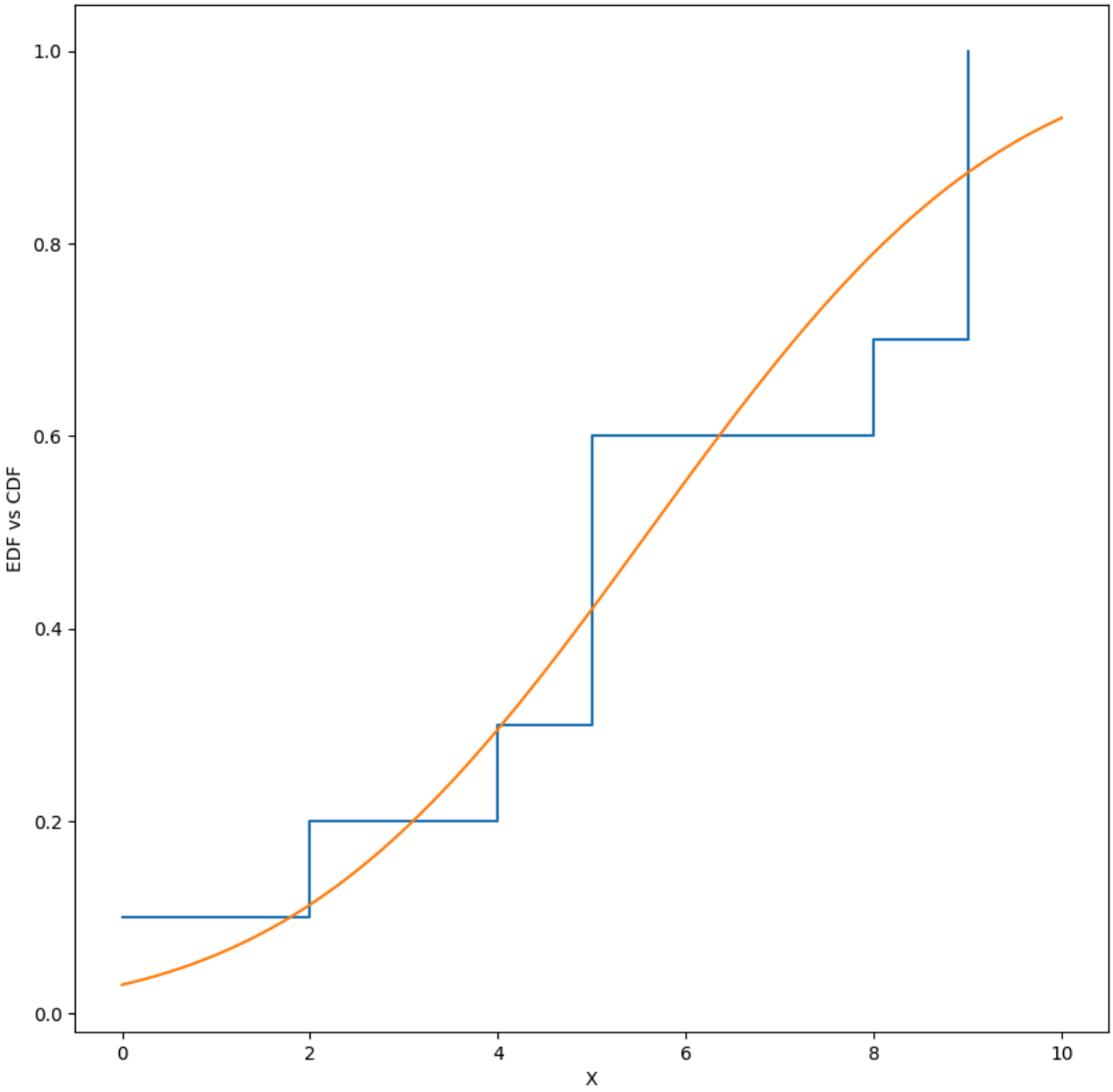
Figure 7: Empirical Distribution Function Overlays with Normal Cumulative Distribution Function

In [6]:
```
# EDF overlay with NCDF
plt.figure(figsize=(10, 10))

# EDF plot
plt.plot(ecd_function.x[1:], ecd_function.y[1:], drawstyle="steps-post")

# CDF plot
x = np.linspace(0, 10, 100)
y = stats.norm.cdf(x, np.mean(M), np.std(M))
plt.plot(x, y)

plt.xlabel("X")
plt.ylabel("EDF vs CDF")
plt.show()
```



We can overlap a theoretical distribution onto our empirical distribution plot to check if the data fits the theoretical distribution. In figure 7, a normal cumulative distribution is overlapped with the empirical distribution plot. From the plot, it seems the data fits the normal distribution. Let's do a Shapiro–Wilk test to see if the data also passes the normality test.

Figure 8: Shapiro–Wilk Test for Observed Data

In [7]:
```
# Shapiro Wilk Test for Observed Data
shapiro_testM = stats.shapiro(M)
print(
    "Shapiro W: {0} \nShapiro pvalue {1}".format(
        shapiro_testM.statistic, shapiro_testM.pvalue
    )
)
```

```
Shapiro W: 0.8973177145071203
Shapiro pvalue 0.20467842718237345
```

Figure 8 shows the Shapiro–Wilk test result. The result shows the data passed the normality test. Hence, we can use normal distribution to describe our 10-observation data series.

6. Asset Return Distributions and Normality Tests

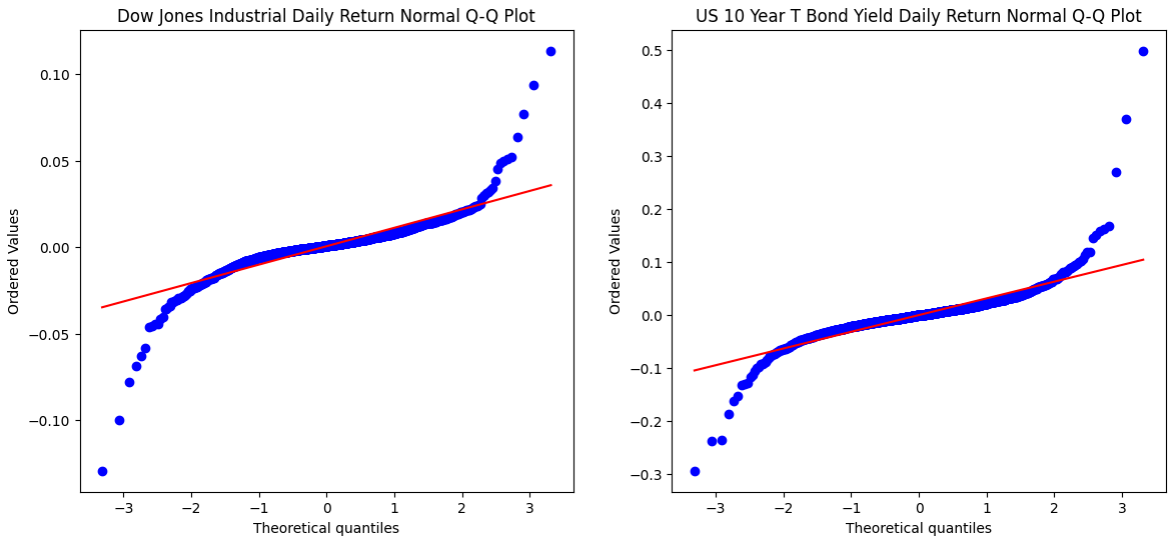
Now, let's use what we learned in this lesson to test the following two popular financial market returns for normality: U.S. 10-Year Treasury Bond yield daily returns and the Dow Jones Industrial Index daily returns to check if they are normally distributed.

We first draw the QQ plot for both daily returns.

Figure 9: Normal QQ Plot for U.S. 10-Year Treasury Bond Yield and Dow Jones Industrial Average Daily Returns

In [8]:
```
# Normal QQ plots
plt.figure(figsize=(14, 6))
ax1 = plt.subplot(121)
stats.probplot(data_set.DWJ, dist="norm", plot=pylab)
plt.title("Dow Jones Industrial Daily Return Normal Q-Q Plot")

ax2 = plt.subplot(122)
stats.probplot(data_set["10Y_TBY"], dist="norm", plot=pylab)
plt.title("US 10 Year T Bond Yield Daily Return Normal Q-Q Plot")
pylab.show()
```



From figure 9, we can see the daily returns for both assets are not normally distributed. They have a similar curvy look in the plots. These curves indicate they have more extreme values than a normal distribution.

Figure 10: Normality Test for U.S. 10-Year Treasury Bond Yield Daily Returns and Dow Jones Industrial Average Daily Returns

In [9]:
```
# Shapiro Wilk Tests
shapiro_test10Y_TBY = stats.shapiro(data_set["10Y_TBY"])
print("10Y_TBY")
print(
    "Shapiro W: {0} \nShapiro pvalue {1}".format(
        shapiro_test10Y_TBY.statistic, shapiro_test10Y_TBY.pvalue
    )
```

```
)

shapiro_testDWJ = stats.shapiro(data_set.DWJ)
print("\nDWJ")
print(
    "Shapiro W: {0} \nShapiro pvalue {1}".format(
        shapiro_testDWJ.statistic, shapiro_testDWJ.pvalue
    )
)
```

```
10Y_TBY
Shapiro W: 0.7625240202833228
Shapiro pvalue 4.045607123564828e-42

DWJ
Shapiro W: 0.7877278994598755
Shapiro pvalue 1.8210701203587463e-40
```

From figure 10, we see that both the U.S. 10 Year Treasury Bond Yield daily returns and DJ Industrial Average daily returns did not pass the Shapiro–Wilk Test. We cannot use normal distribution to describe these two asset daily returns.

7. Conclusion

In this lesson, we first reviewed random variables, including discrete and continuous random variables. Then, we went through two popular discrete distributions: Bernoulli distribution, binomial distribution. After that, we reviewed normal distribution and two methods to test normality for a random variable: the QQ plot and Shapiro–Wilk test. Lastly, we talked about empirical distribution and its properties. In the next lesson, we will talk about some distributions that are similar to normal distribution but not exactly the same.

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